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Boyd

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(54) **AIR COOLING SYSTEM FOR AN ELECTRIC VEHICLE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 151 days.

This patent is subject to a terminal disclaimer.

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(51) **Int. Cl.**
B60H 1/00 (2006.01)
B60H 1/32 (2006.01)

(52) **U.S. Cl.**
CPC **B60H 1/00278** (2013.01); **B60H 1/00378**
(2013.01); **B60H 1/00392** (2013.01); **B60H**
1/3227 (2013.01)

(58) **Field of Classification Search**

CPC B60H 1/00378; B60H 1/00278; B60H
1/00392

See application file for complete search history.

(56) **References Cited**

PUBLICATIONS

Wright et al., "Air Cooling System for an Electric Vehicle", U.S. Appl. No. 18/379,764, filed Oct. 13, 2023.
Boyd et al., "Air Cooling System for an Electric Vehicle", U.S. Appl. No. 18/379,767, filed Oct. 13, 2023.
Boyd et al., "Air Cooling System for an Electric Vehicle", U.S. Appl. No. 18/379,769, filed Oct. 13, 2023.
Nadiadi, "Air Cooling System for an Electric Vehicle", U.S. Appl. No. 18/379,779, filed Oct. 13, 2023.

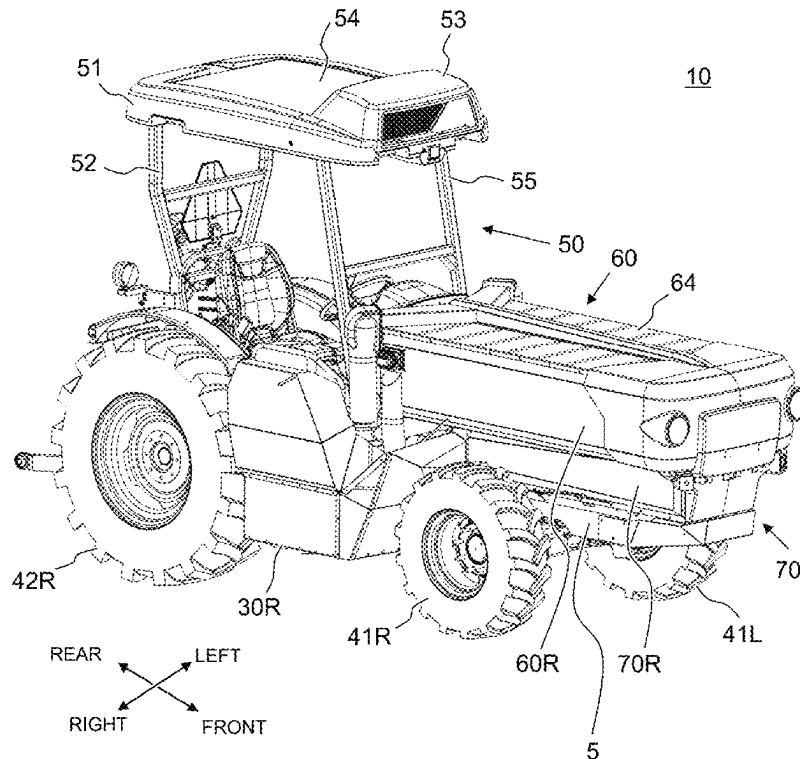
Primary Examiner — Nael N Babaa

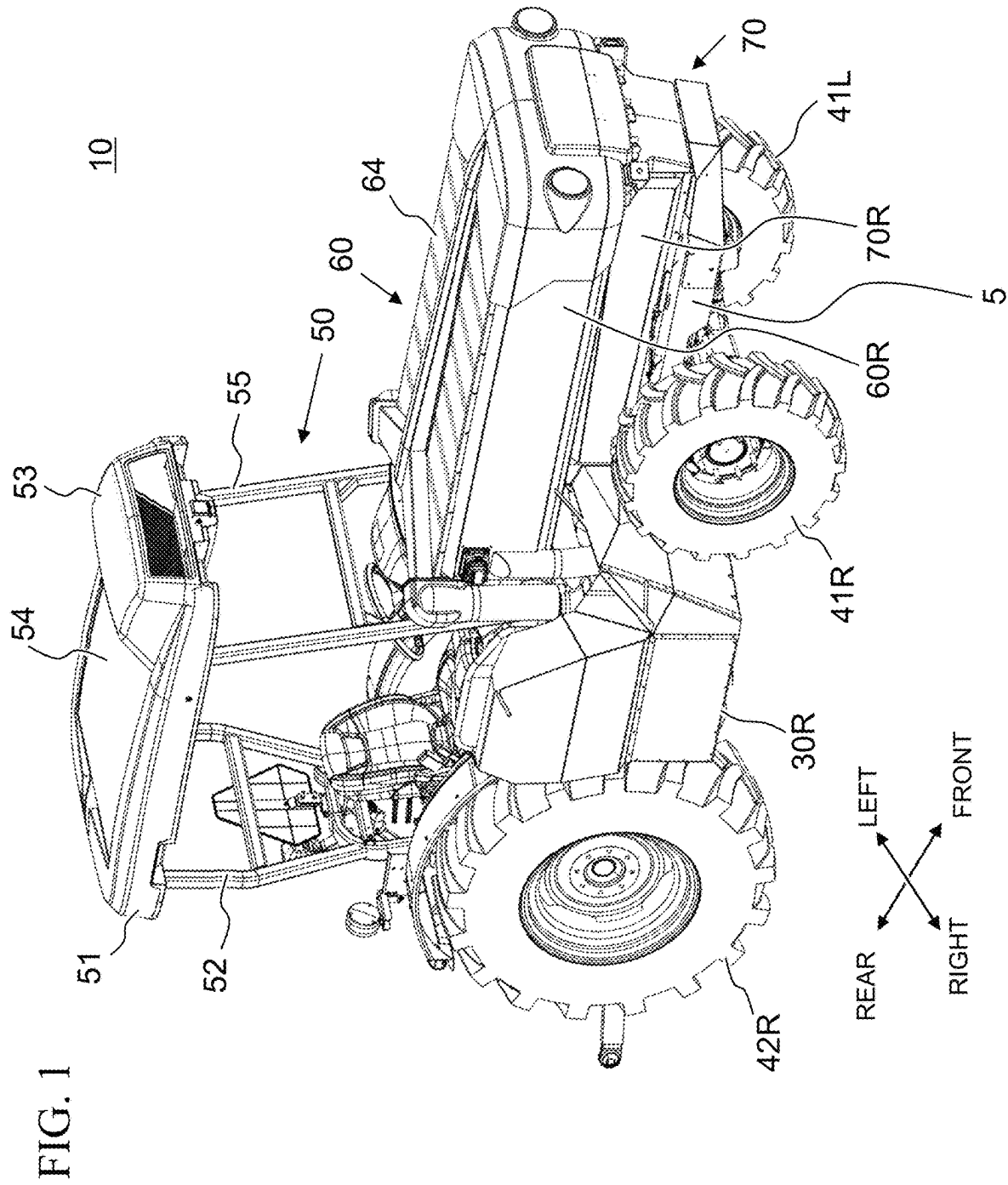
(74) *Attorney, Agent, or Firm* — Keating & Bennett, LLP

(57) **ABSTRACT**

A vehicle includes a first battery housing to house a plurality of first battery modules, a second battery housing to house a plurality of second battery modules, and an air cooling system to cool the plurality of first battery modules and the plurality of second battery modules. The air cooling system includes a compressor, a condenser, a first evaporator coil that includes a first side at which warm air is received and a second side at which cool air exits after having passed through the first evaporator coil, and a second evaporator coil that includes a first side at which warm air is received and a second side at which cool air exits after having passed through the second evaporator coil.

20 Claims, 44 Drawing Sheets





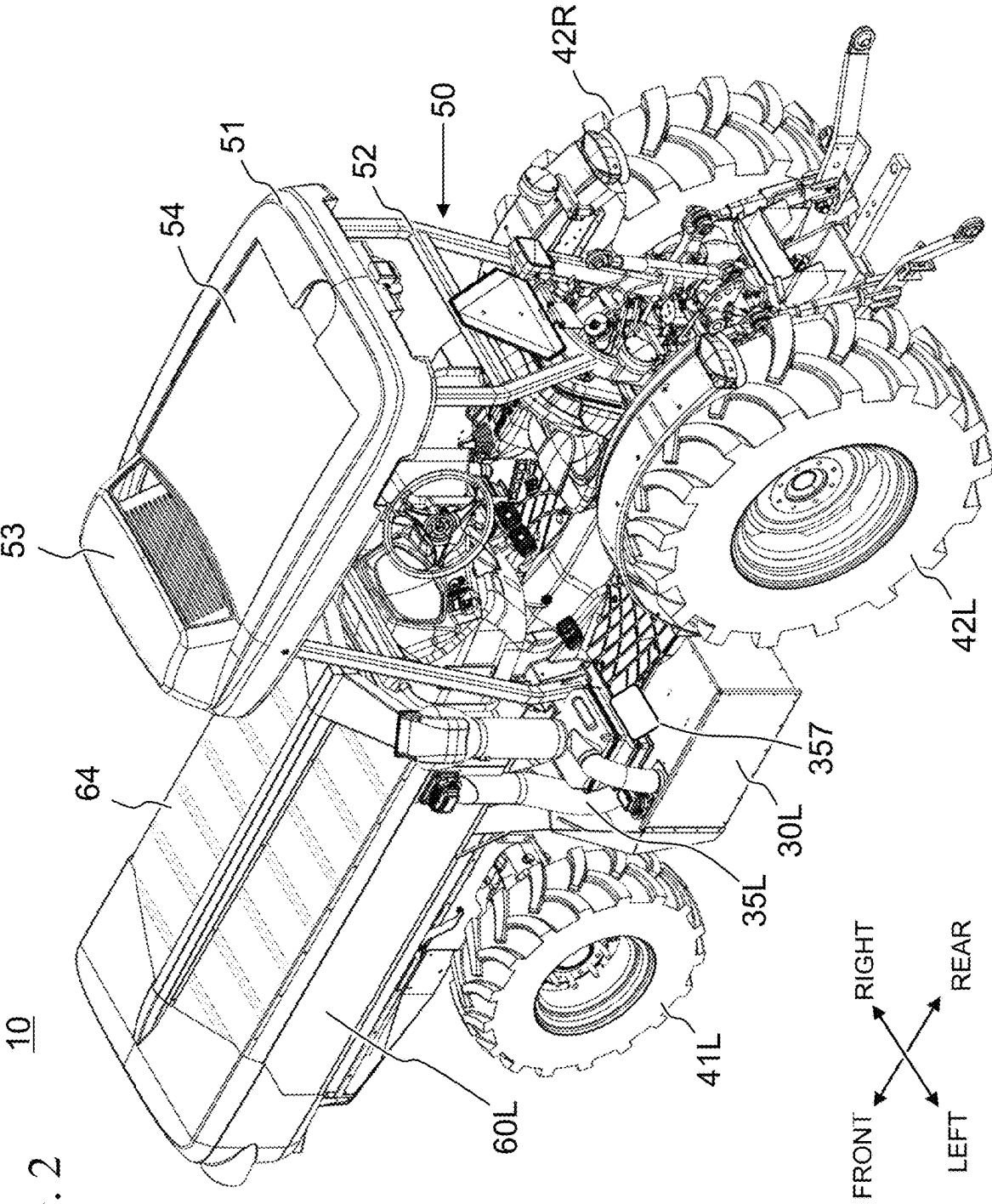


FIG. 3

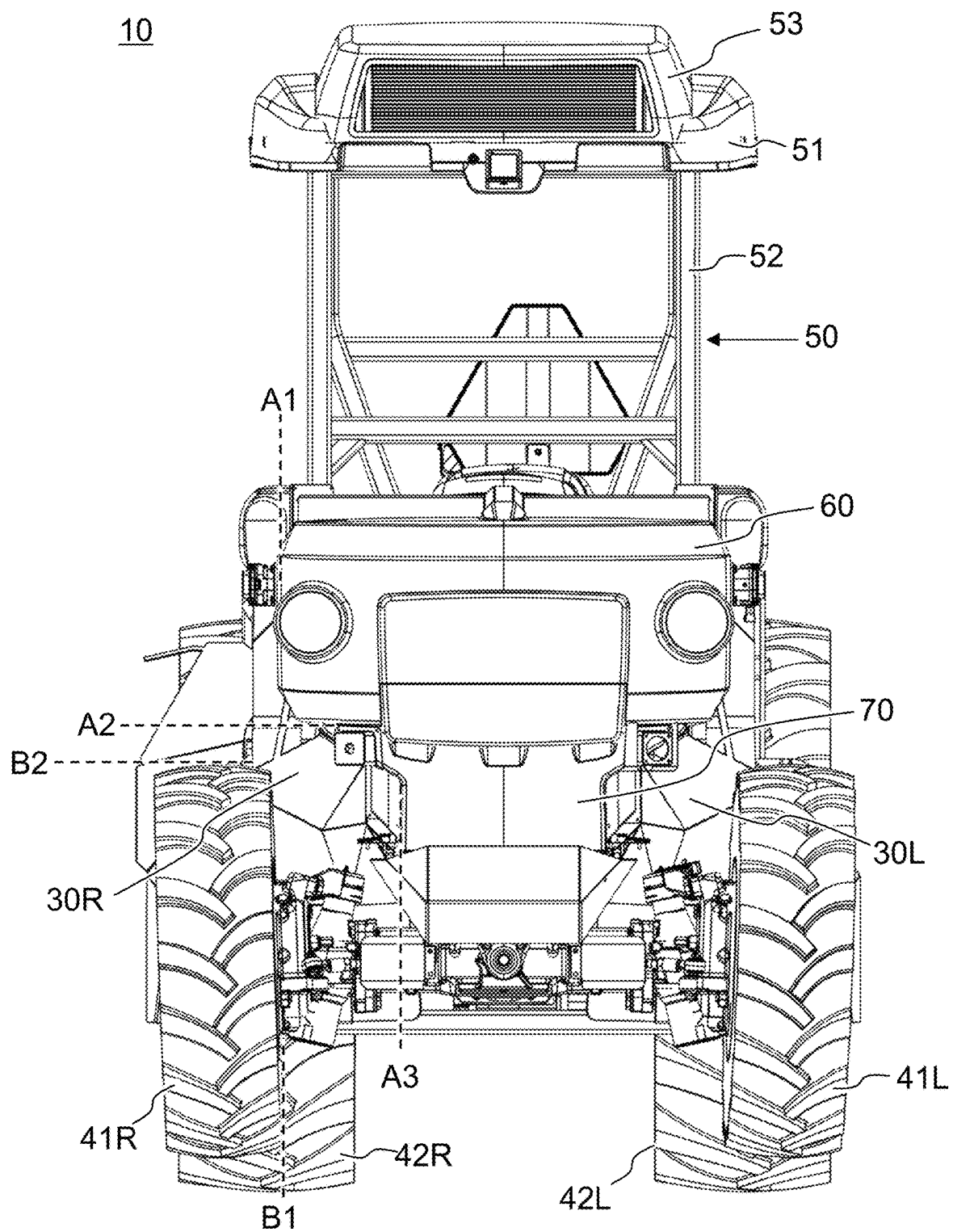


FIG. 4

10

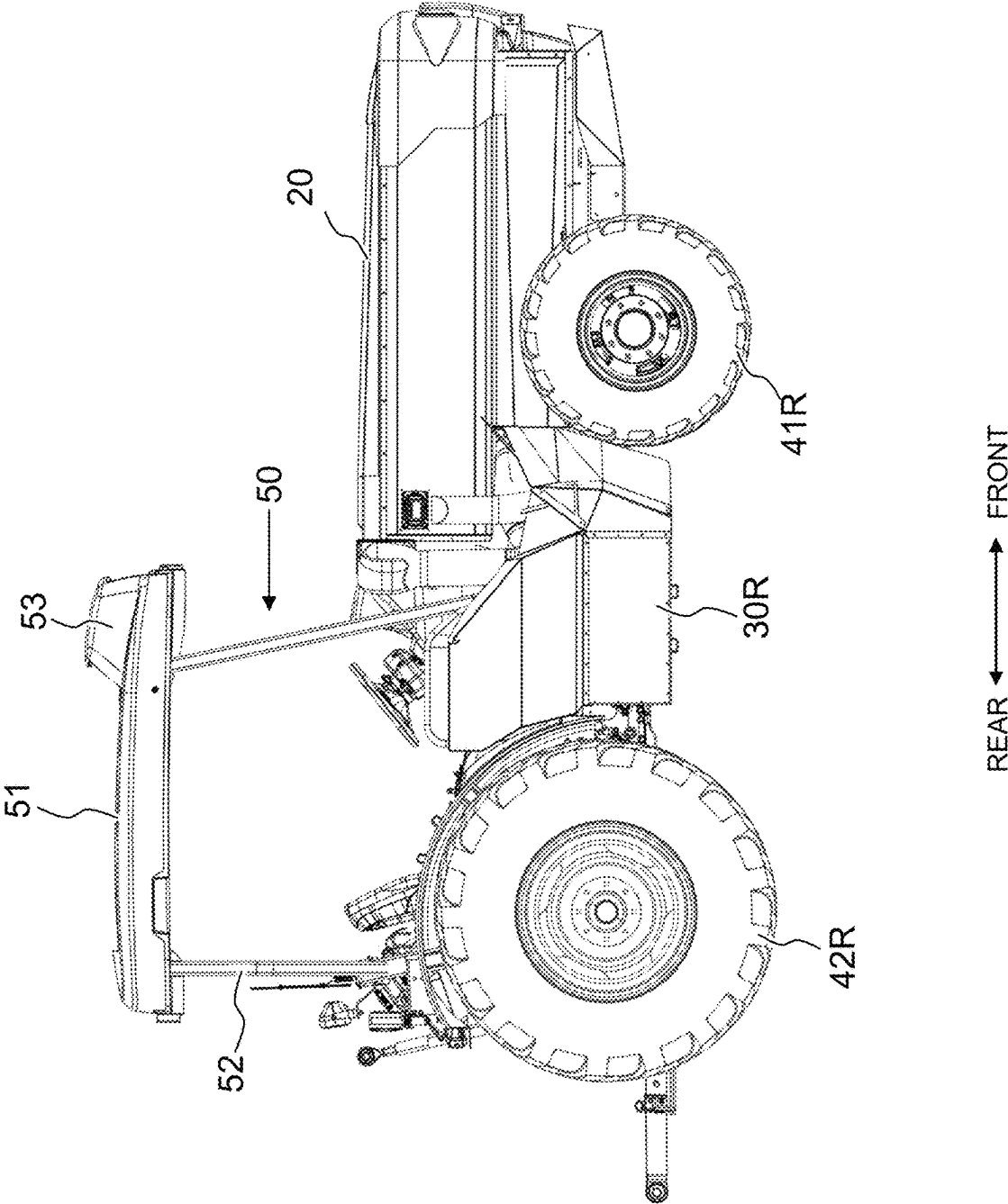


Fig. 5.

10

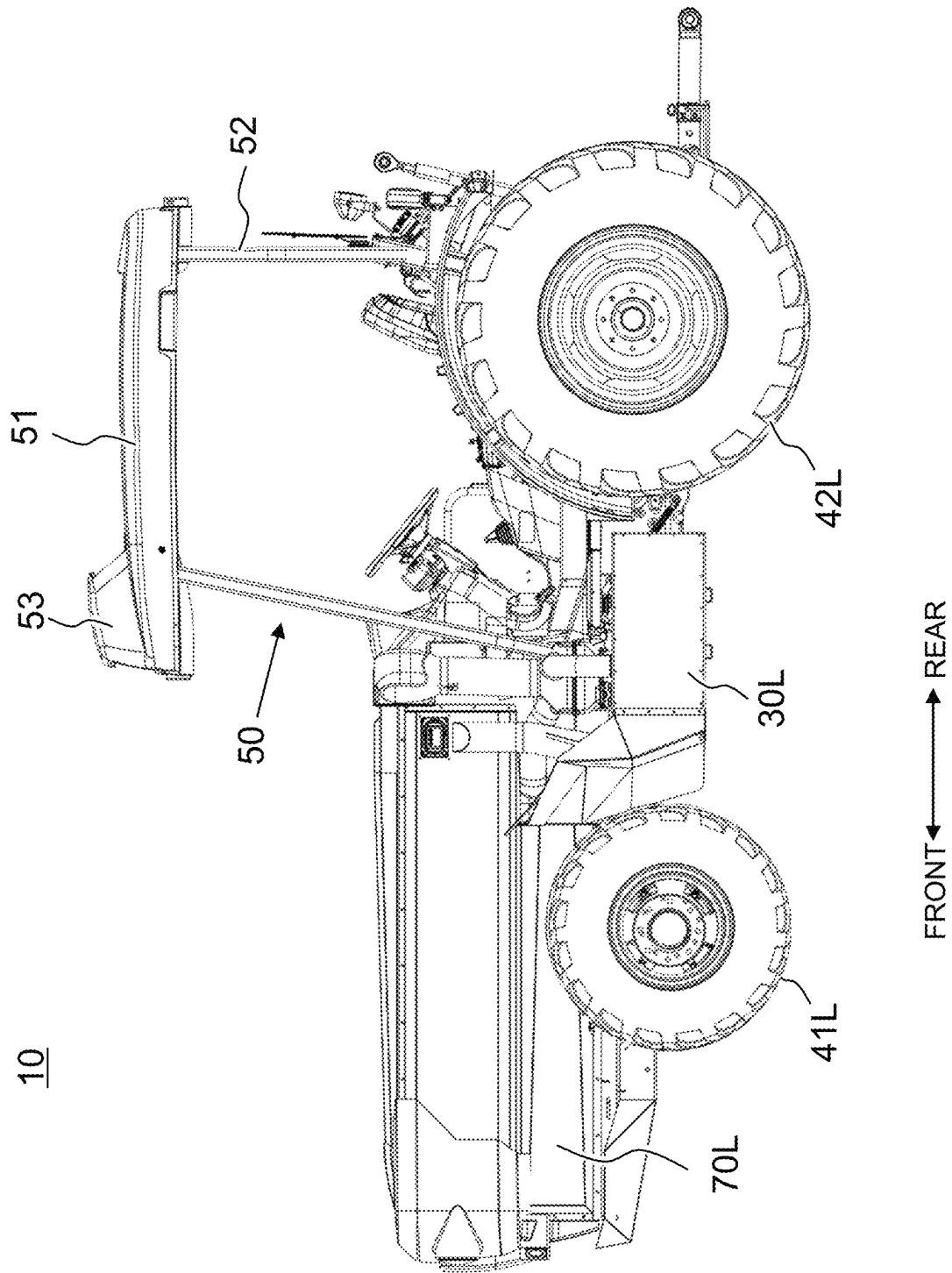


FIG. 6

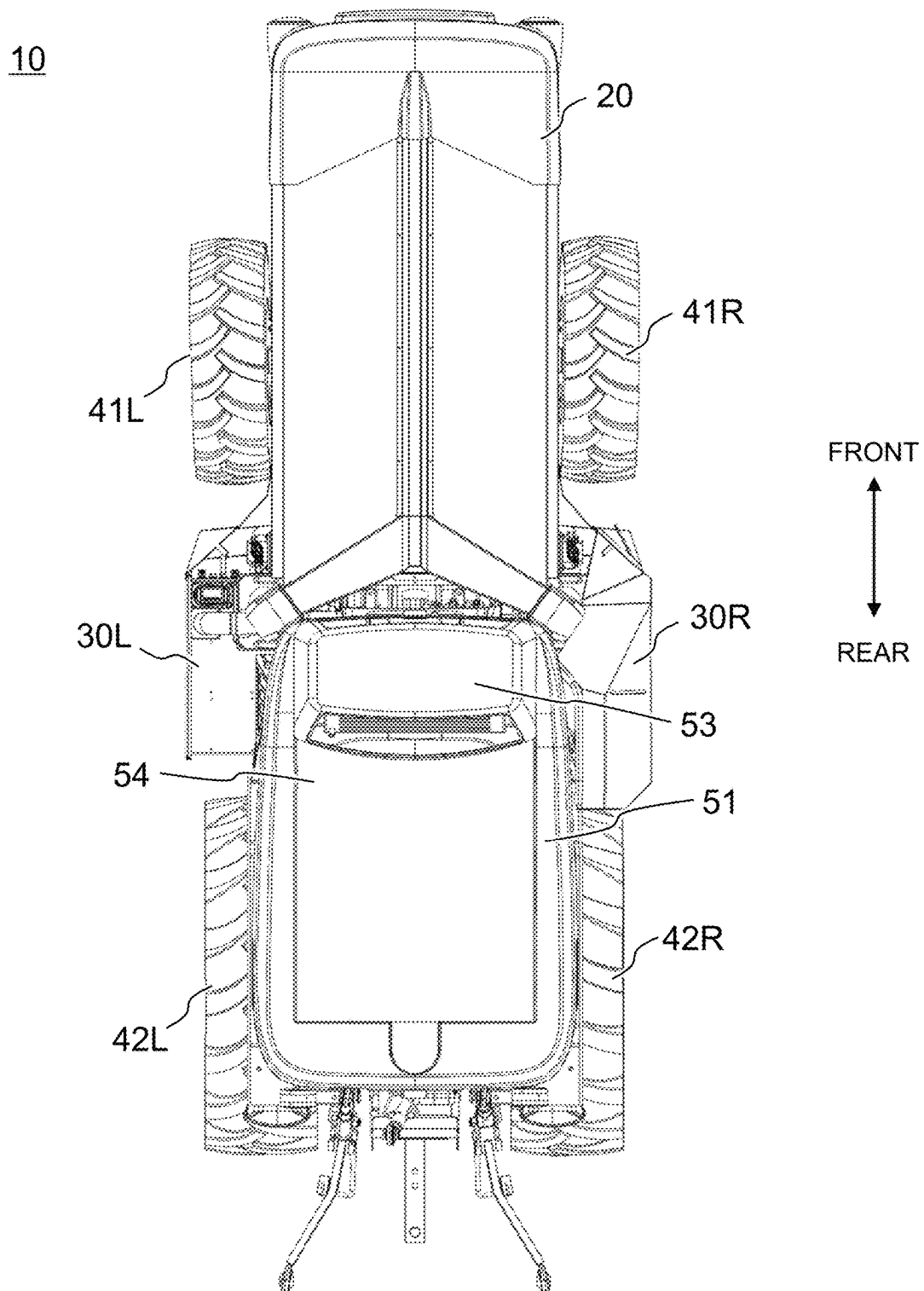


FIG. 7

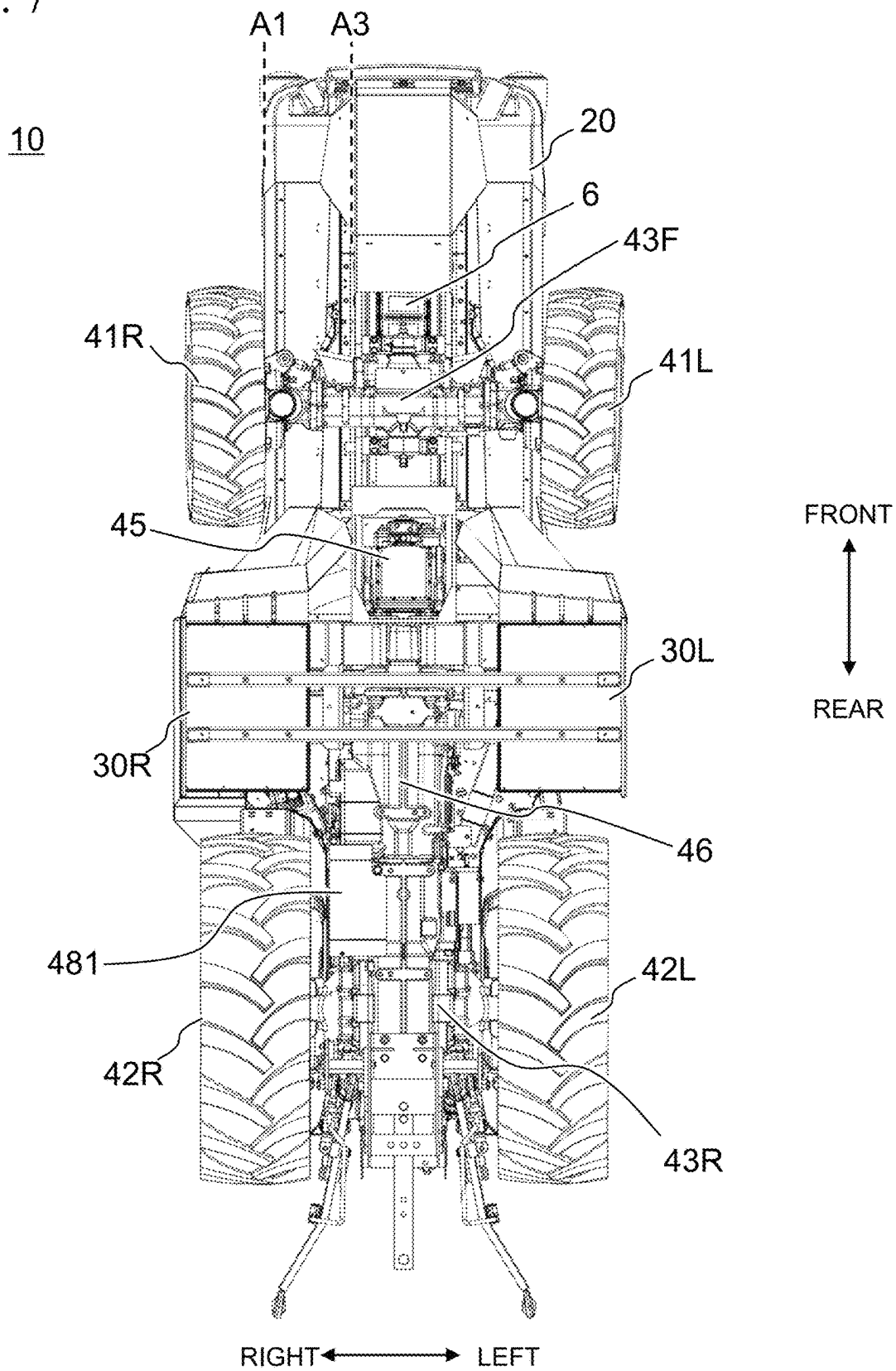


FIG. 8

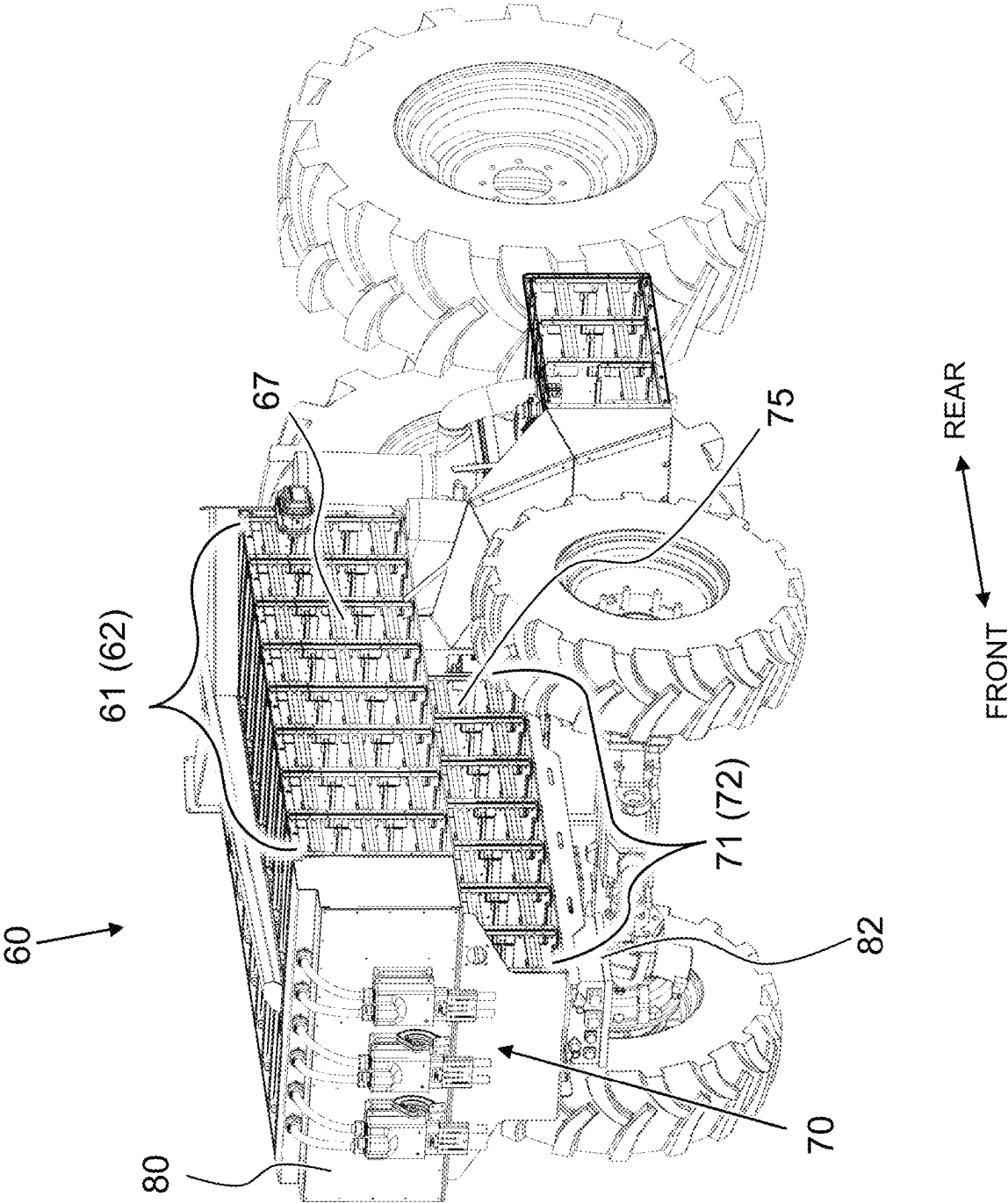


FIG. 9

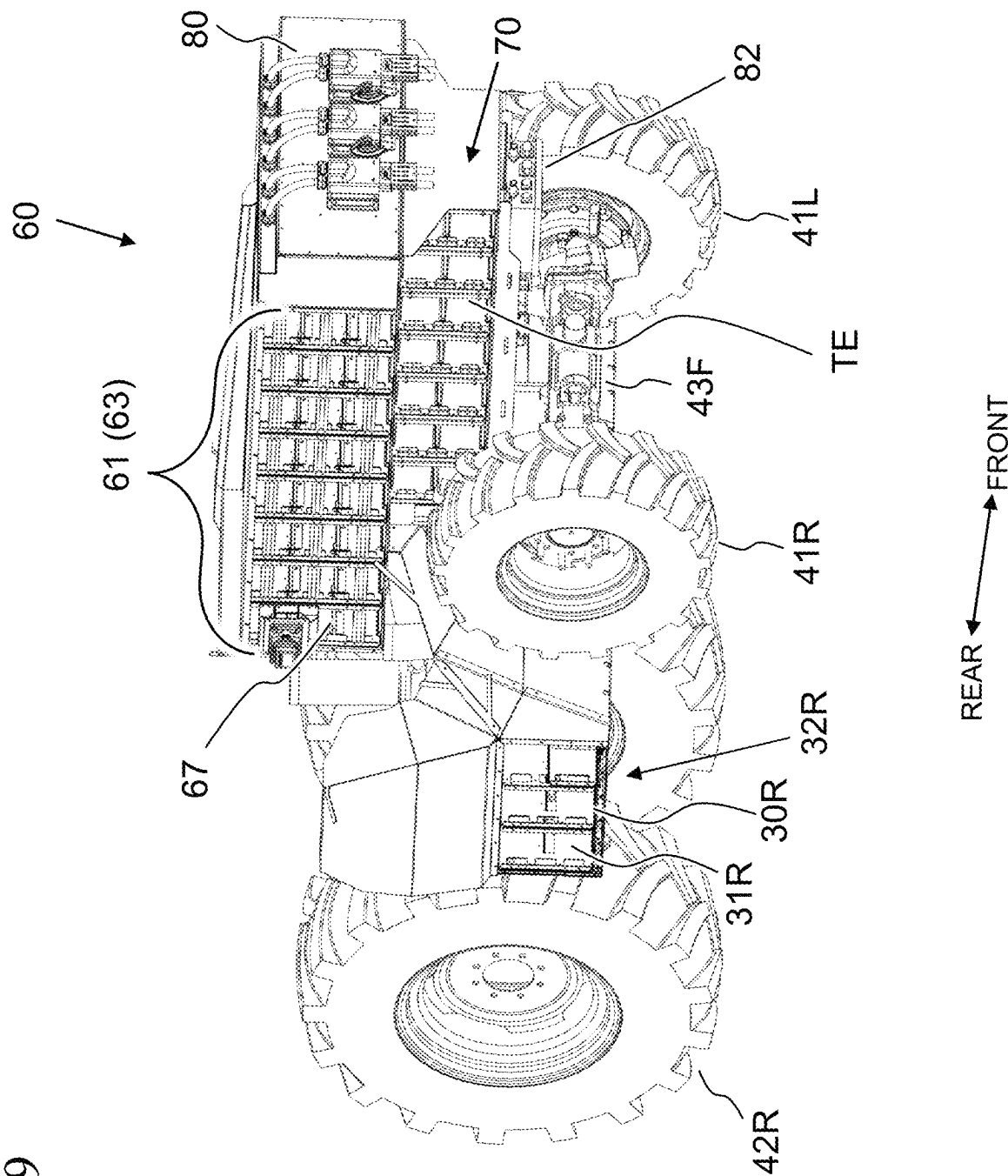


FIG. 10

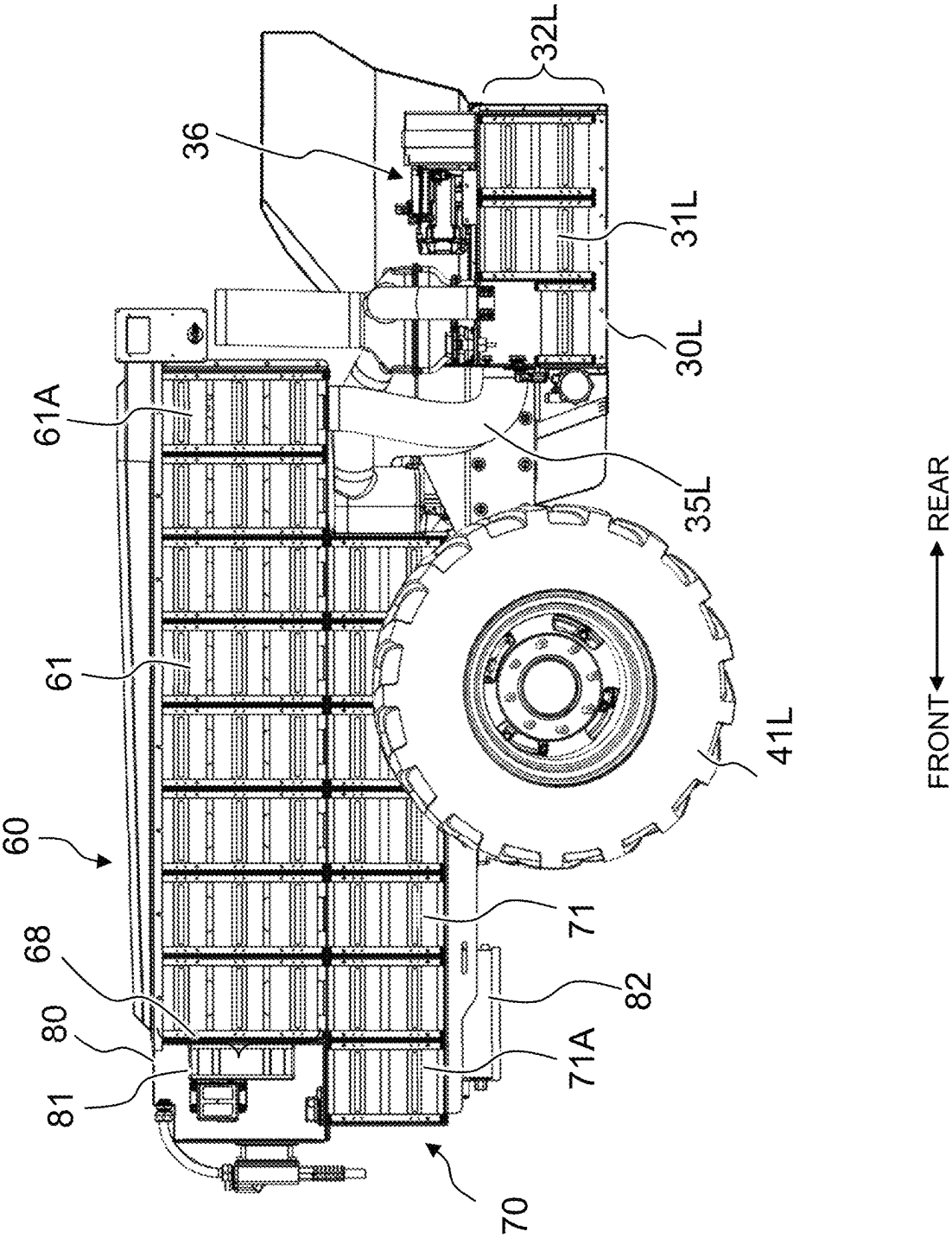
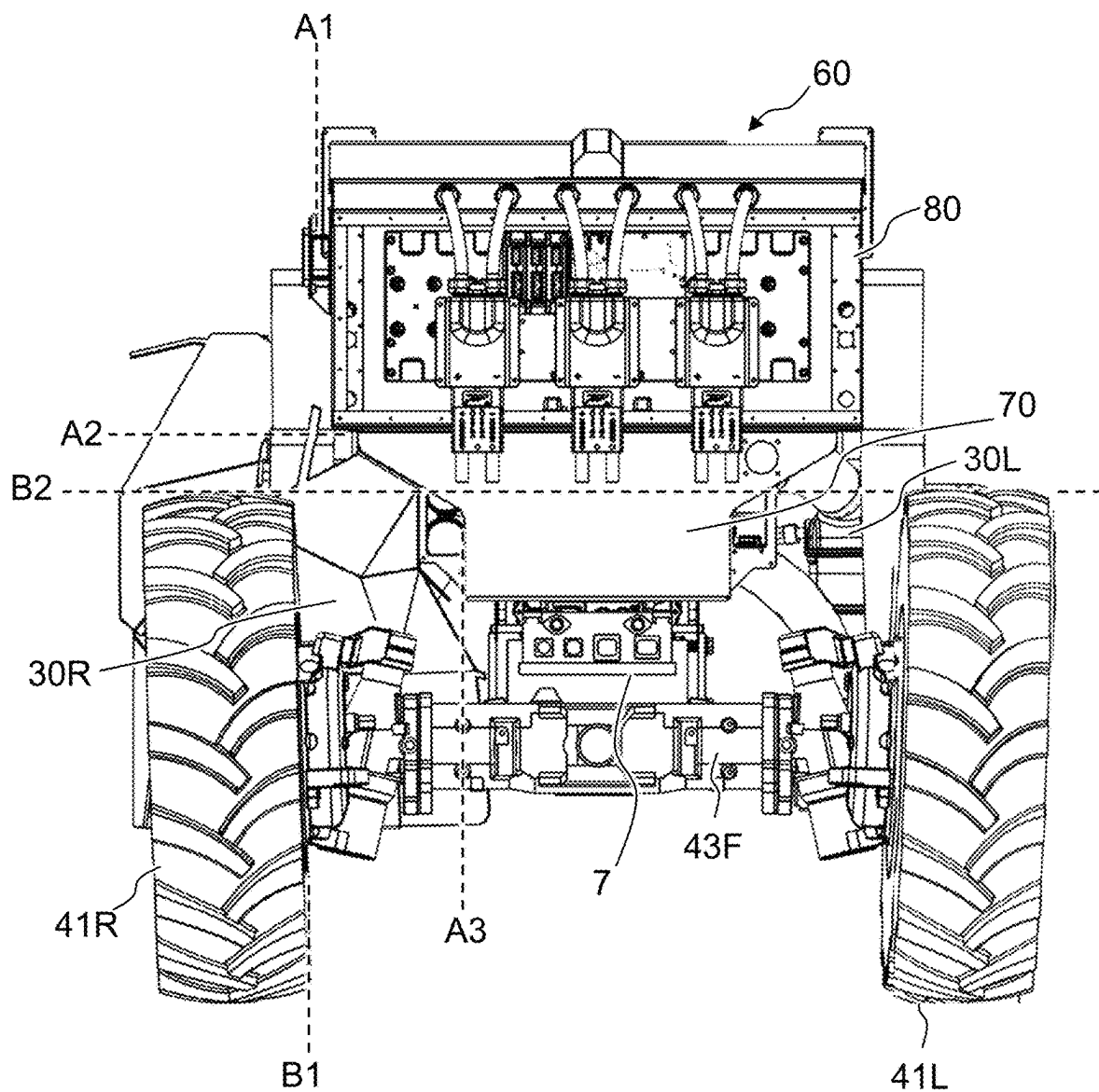


FIG. 11



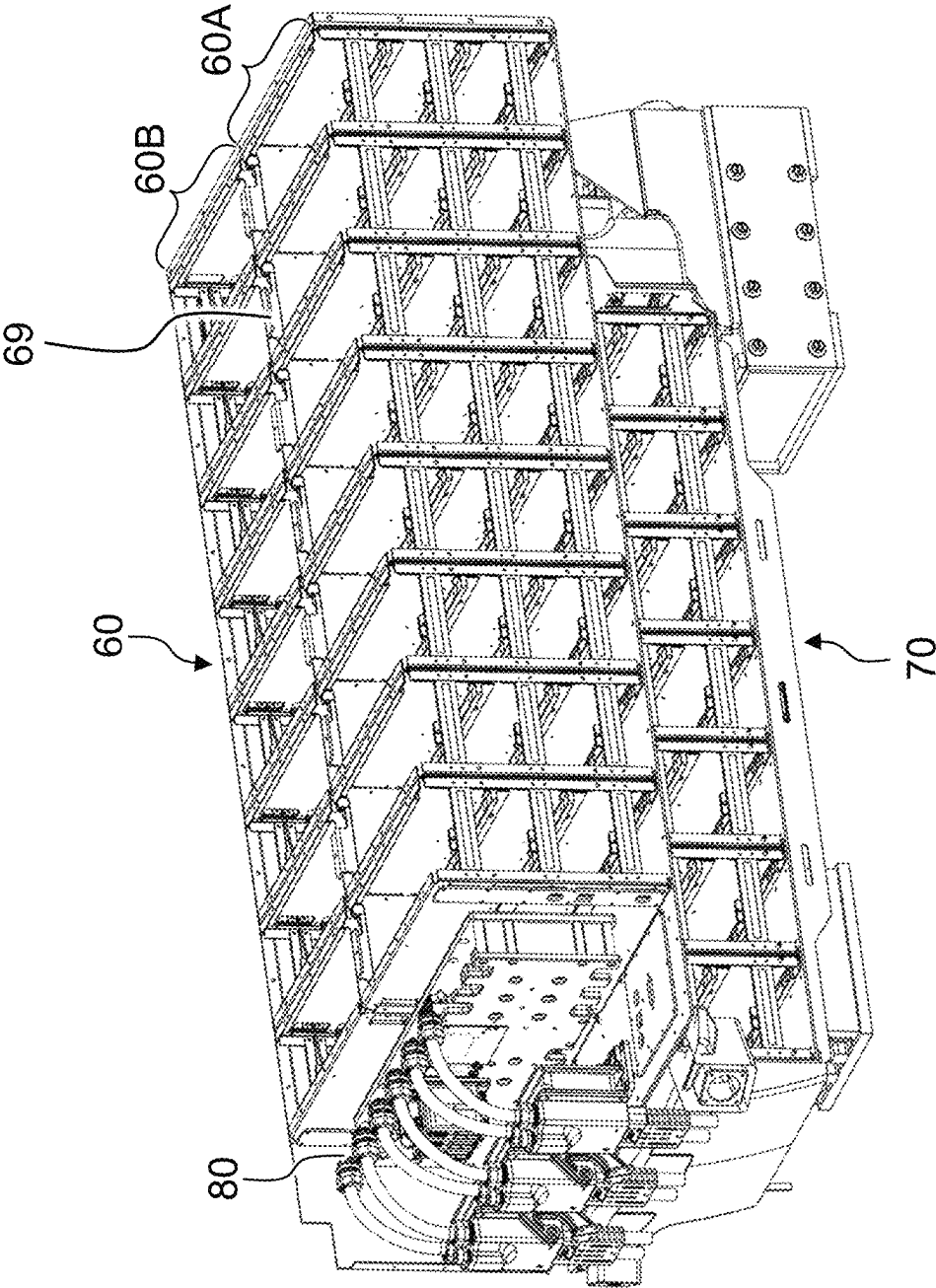


FIG. 12

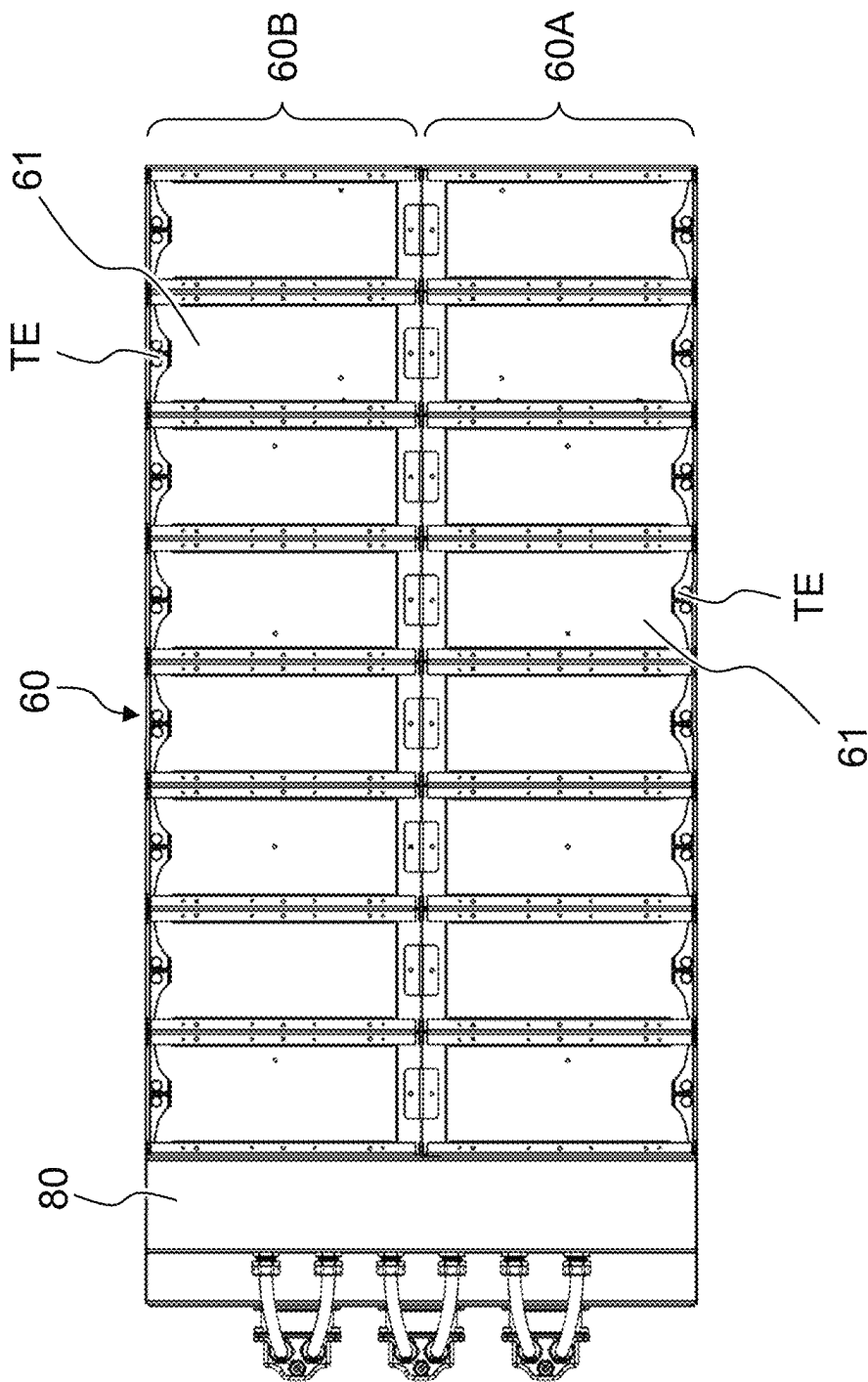


FIG. 13

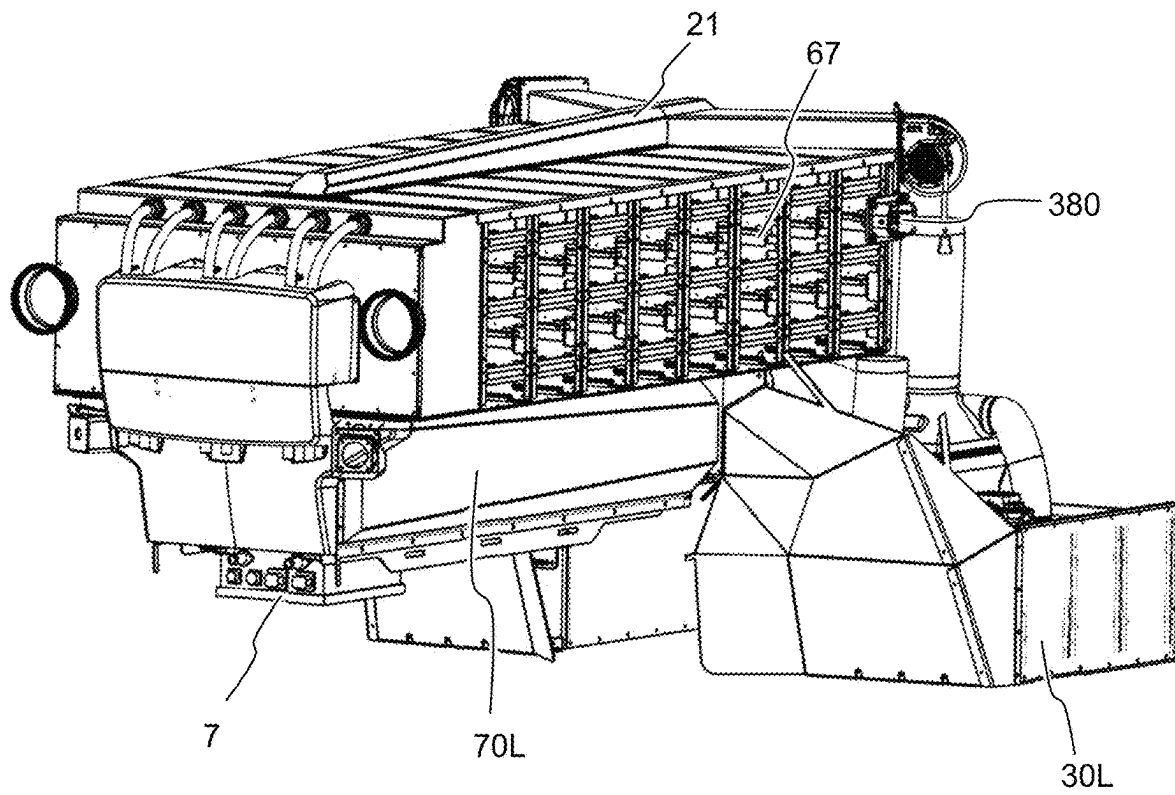


FIG. 14

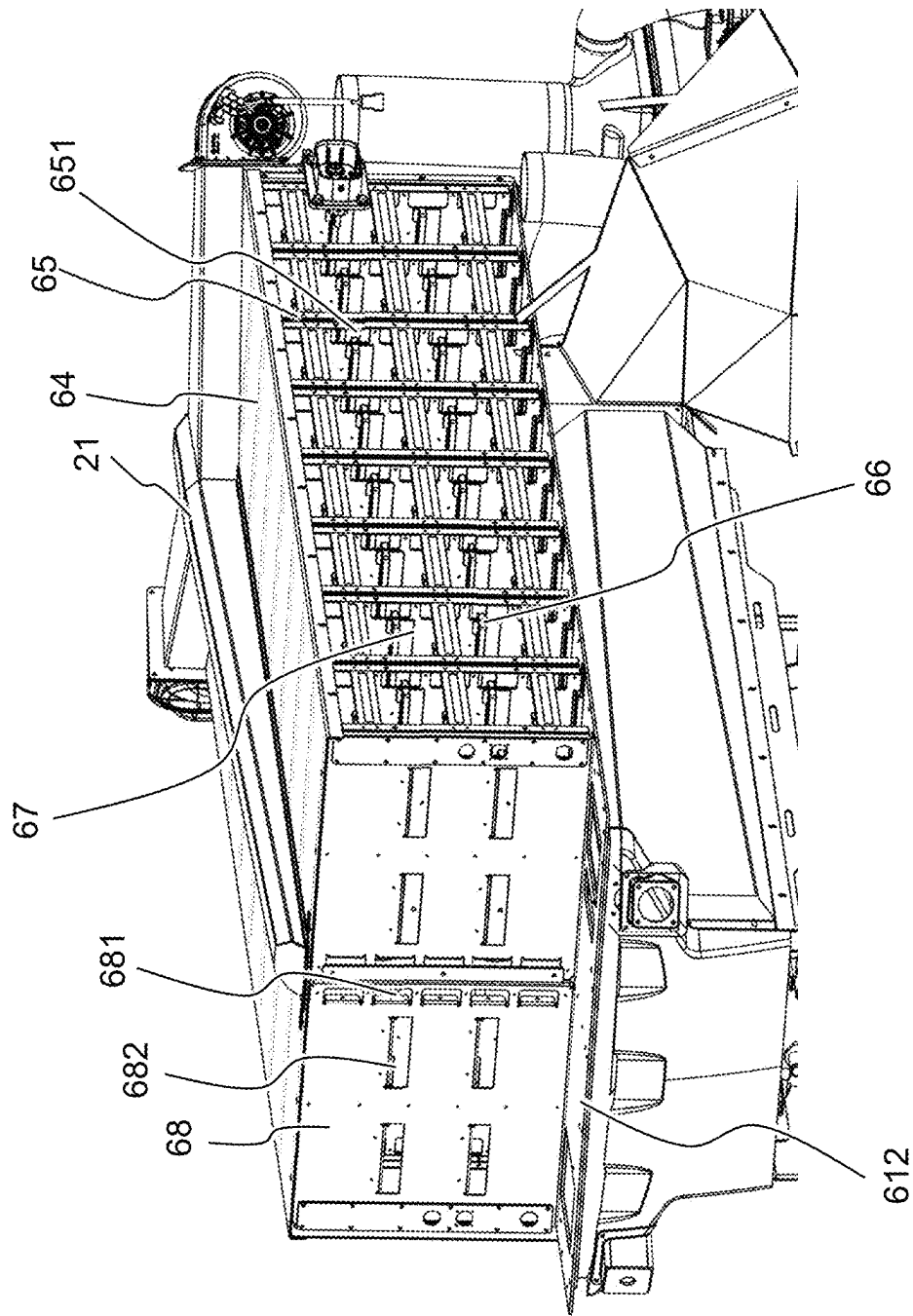


FIG. 15

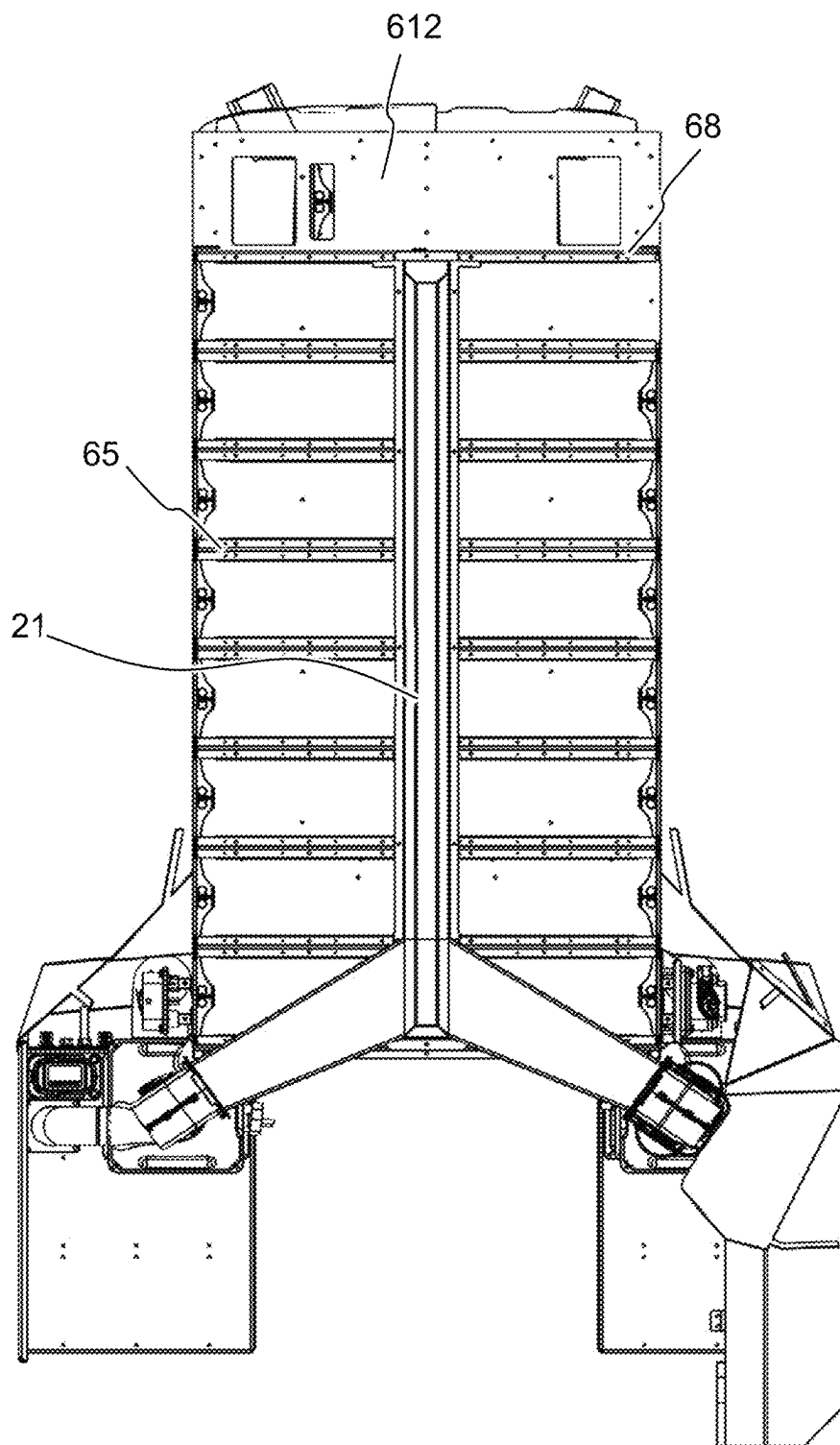


FIG. 16

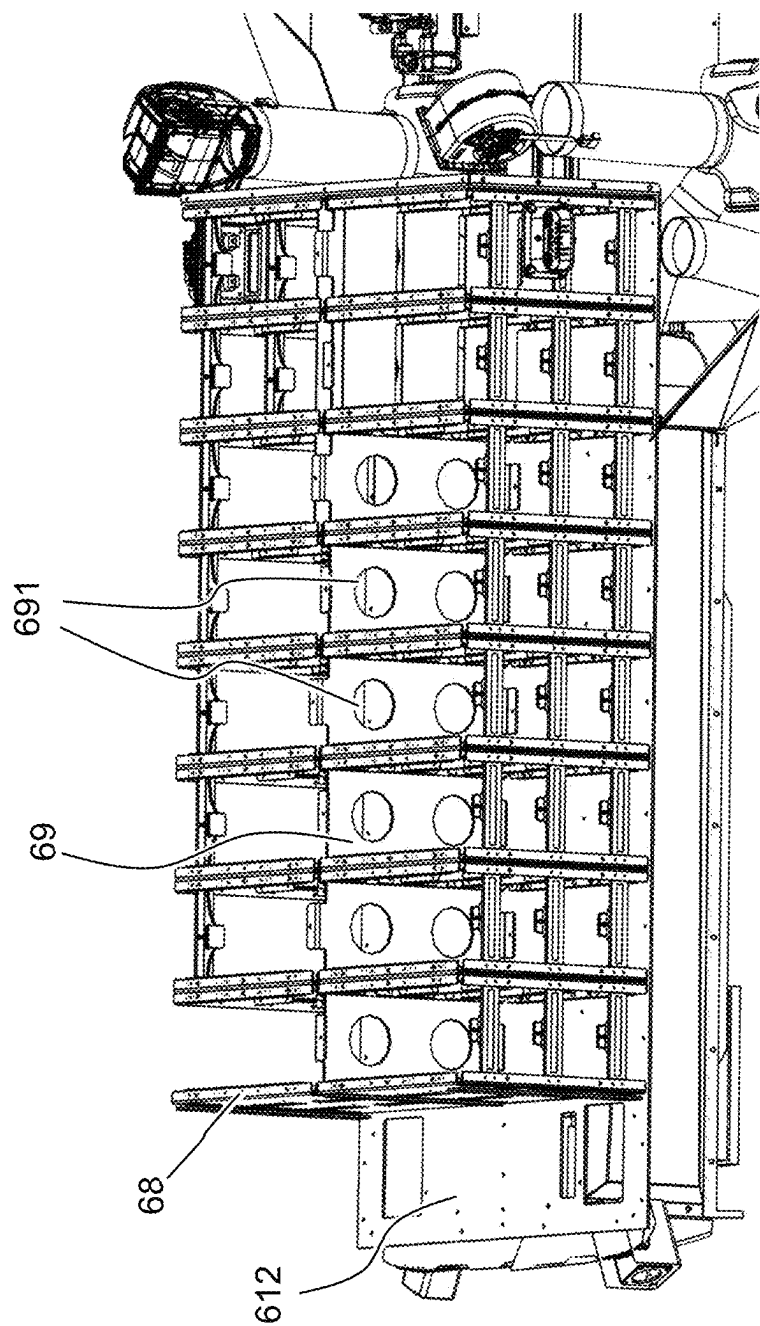


FIG. 17

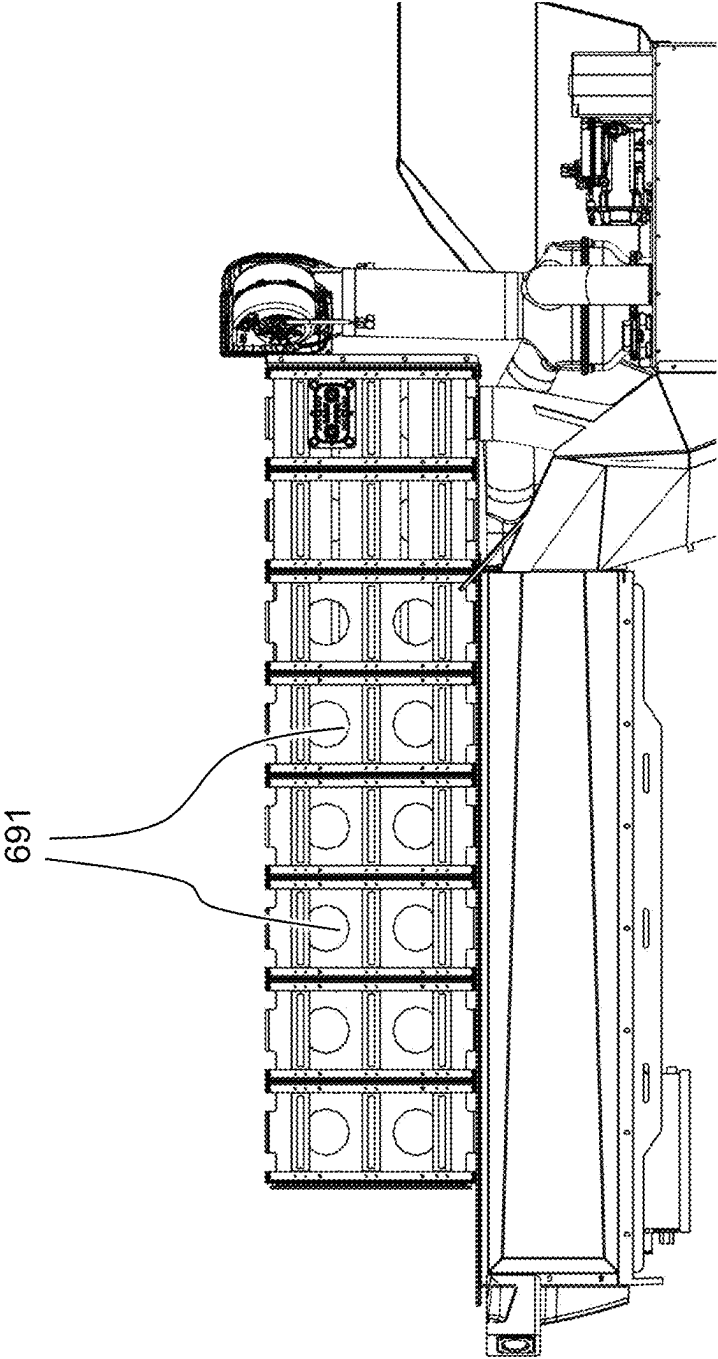


FIG. 18

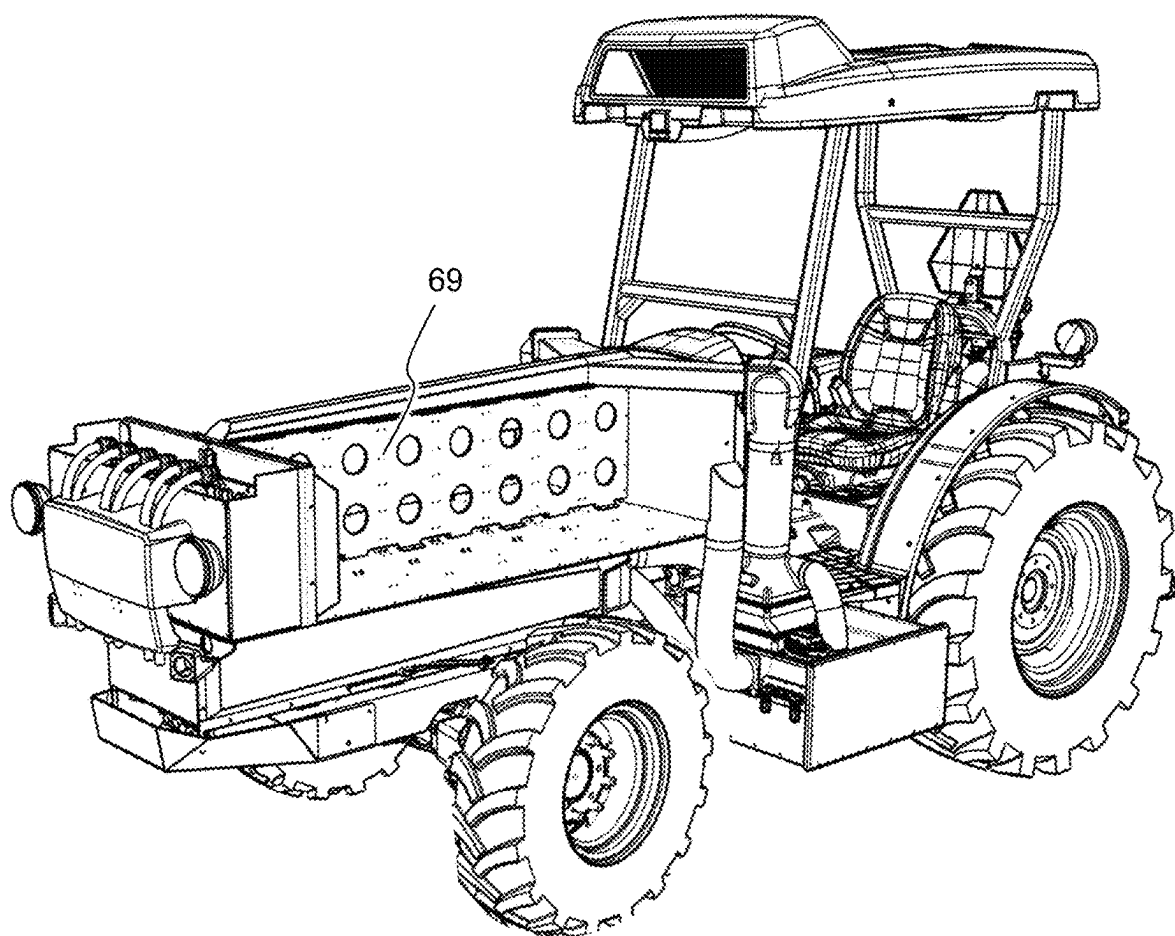


FIG. 19

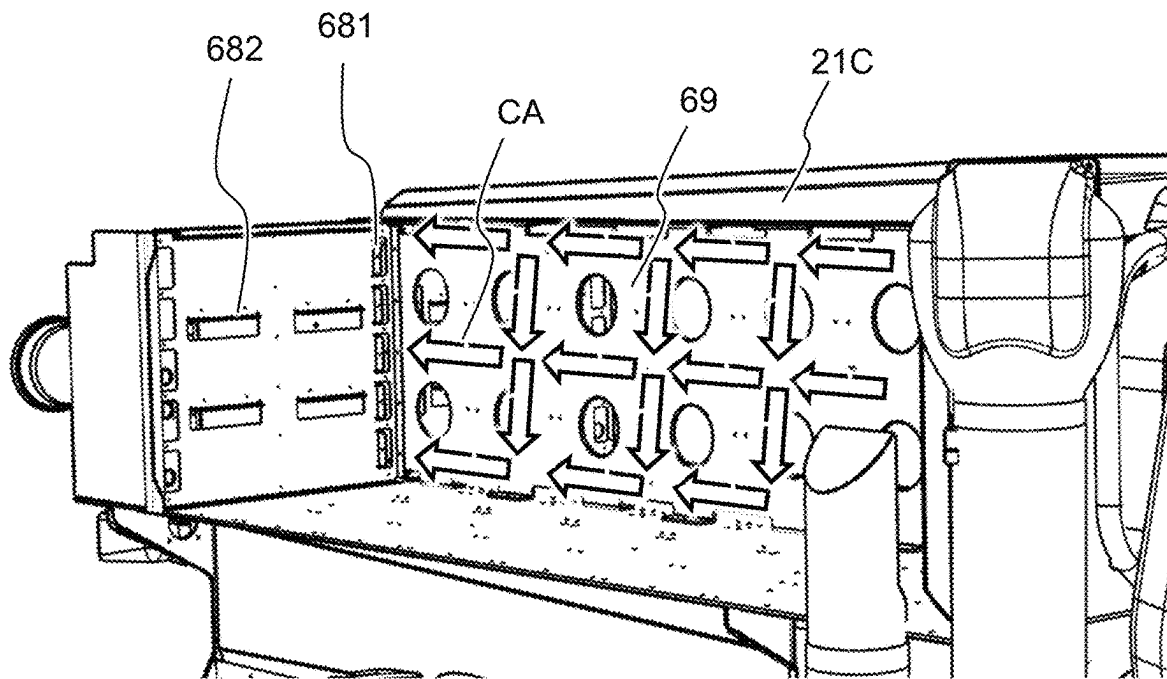


FIG. 20

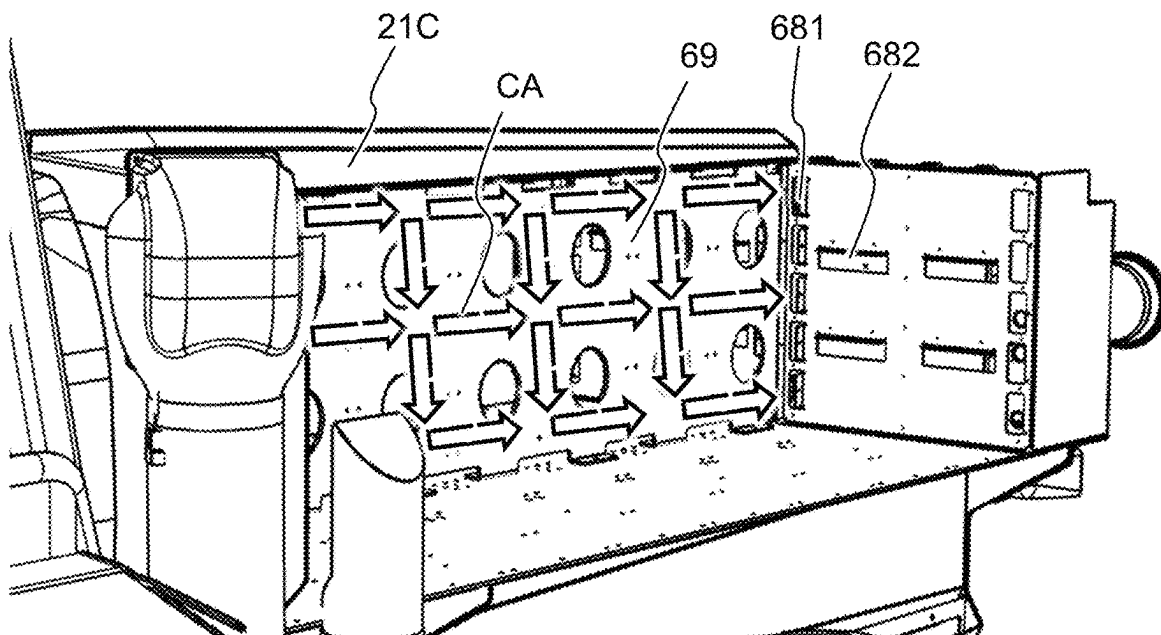


FIG. 21

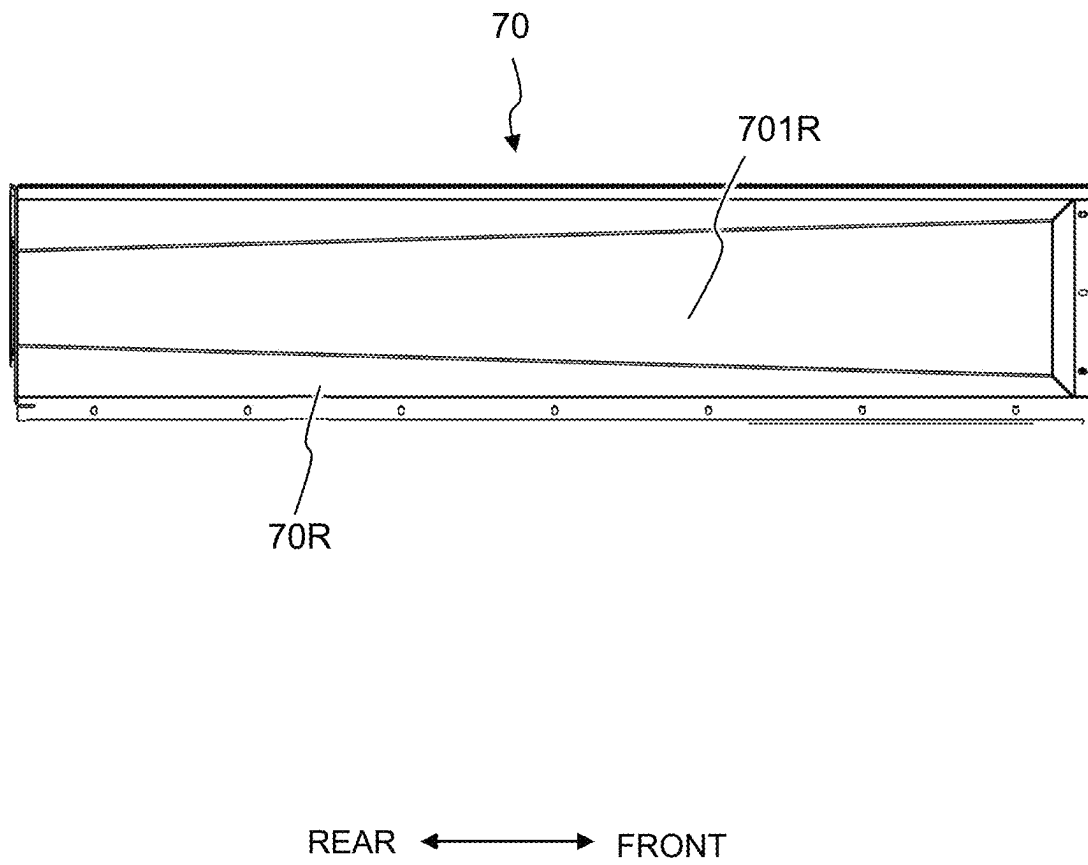


FIG. 22

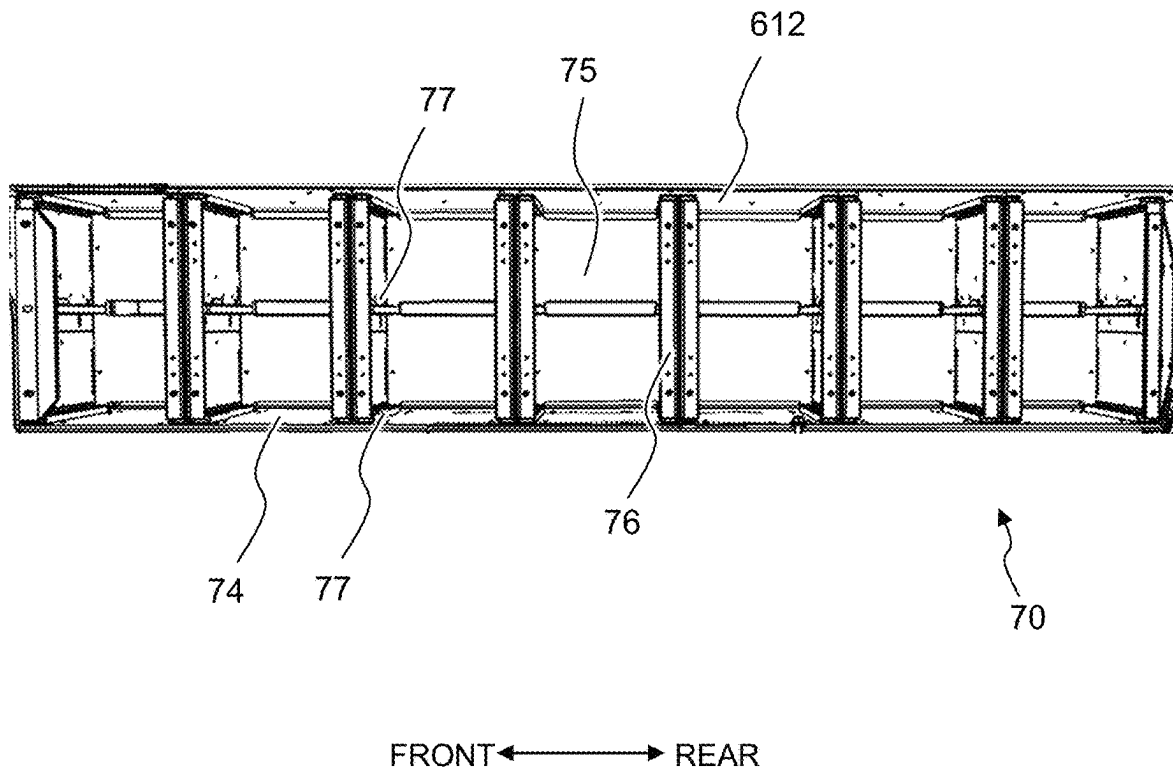


FIG. 23

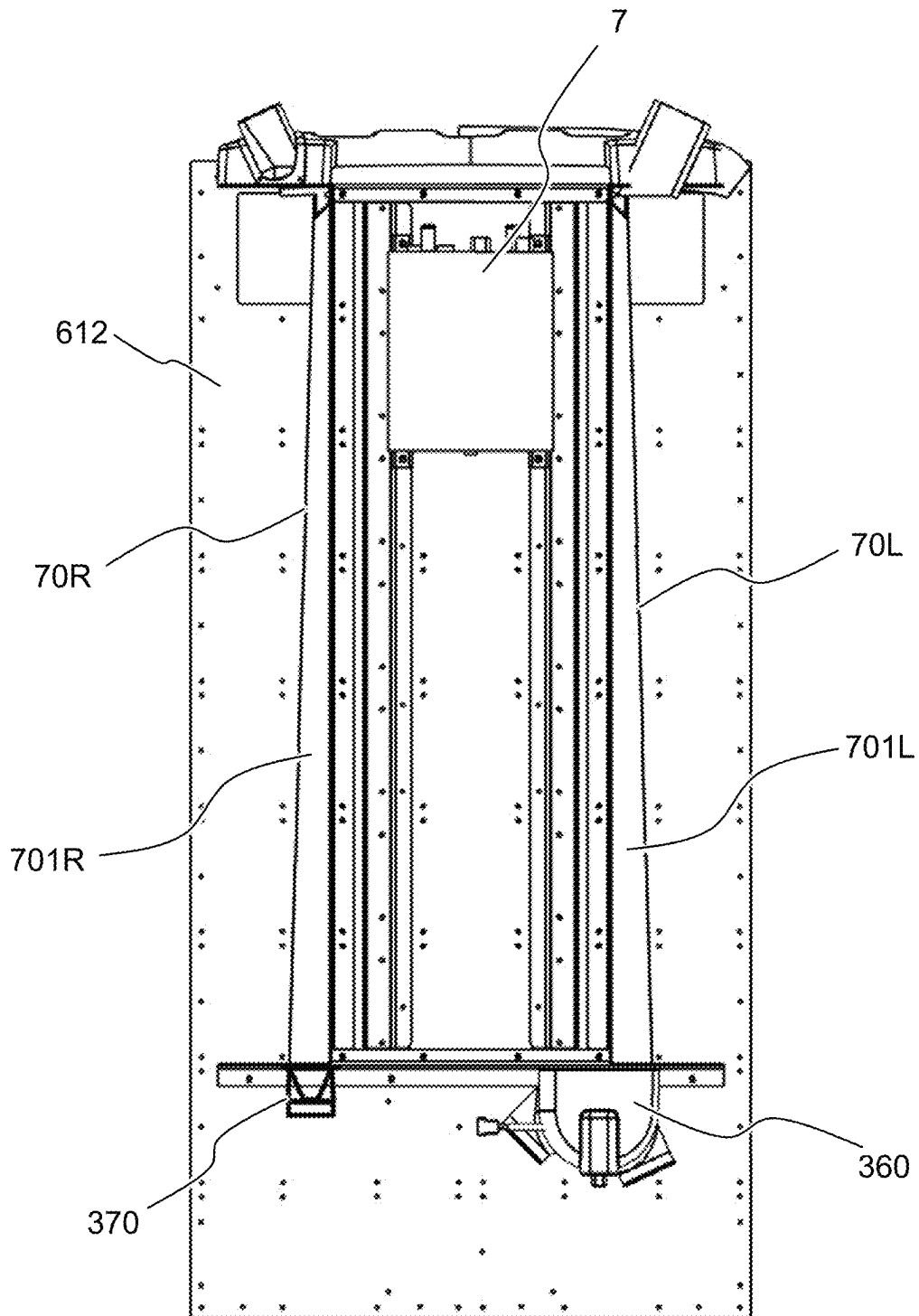


FIG. 24

FIG. 25

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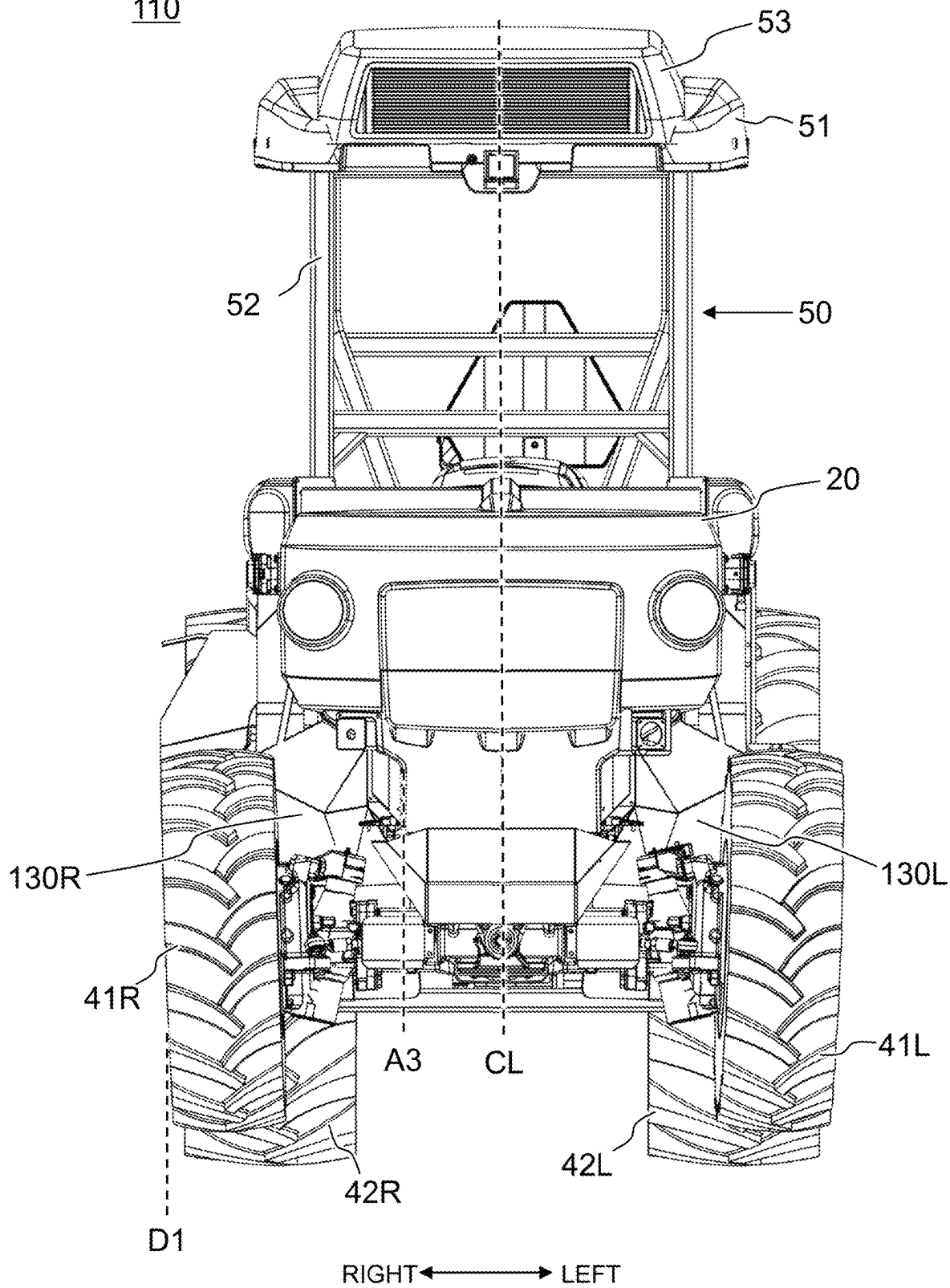


FIG. 26

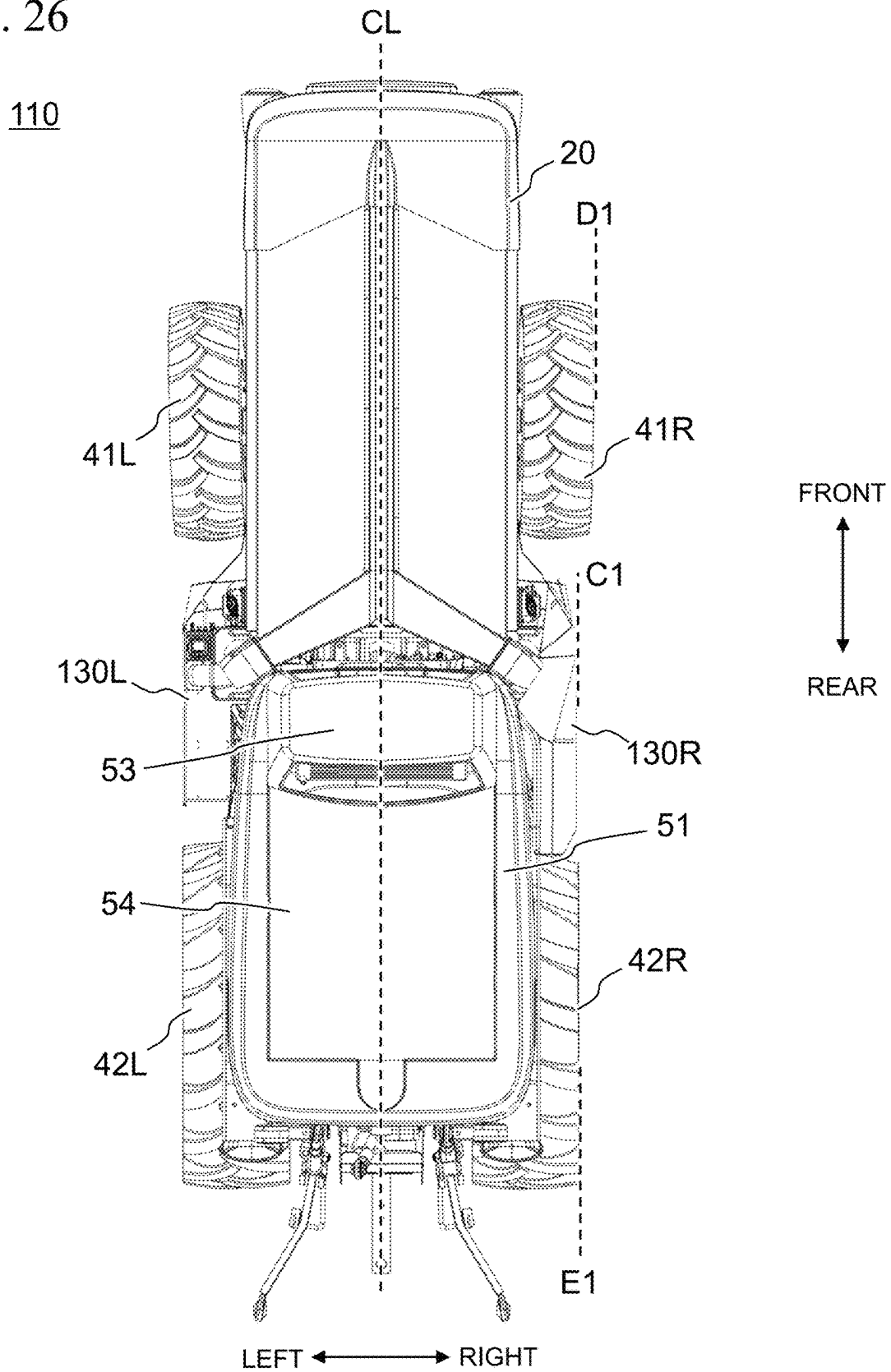


FIG. 27

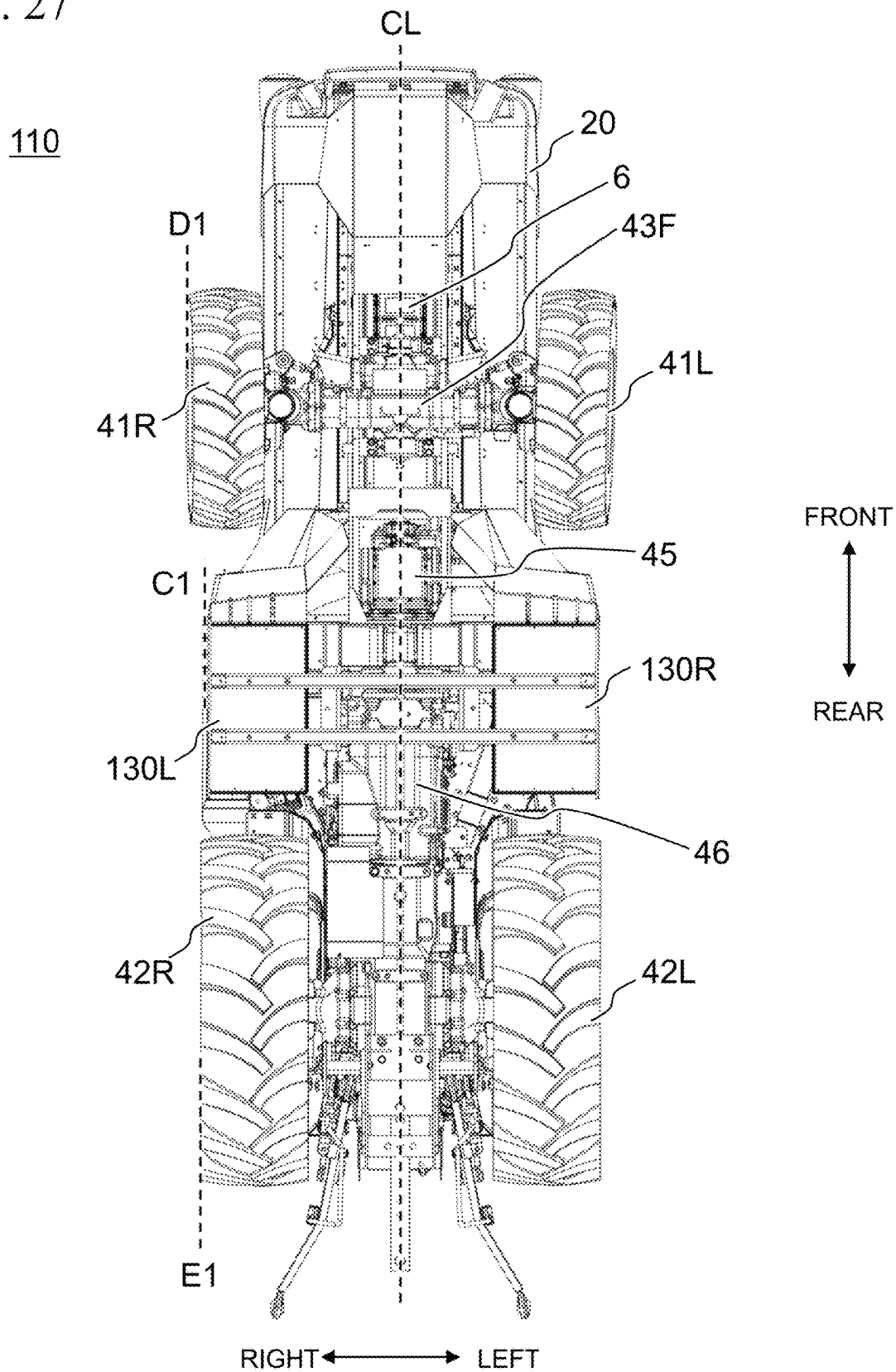
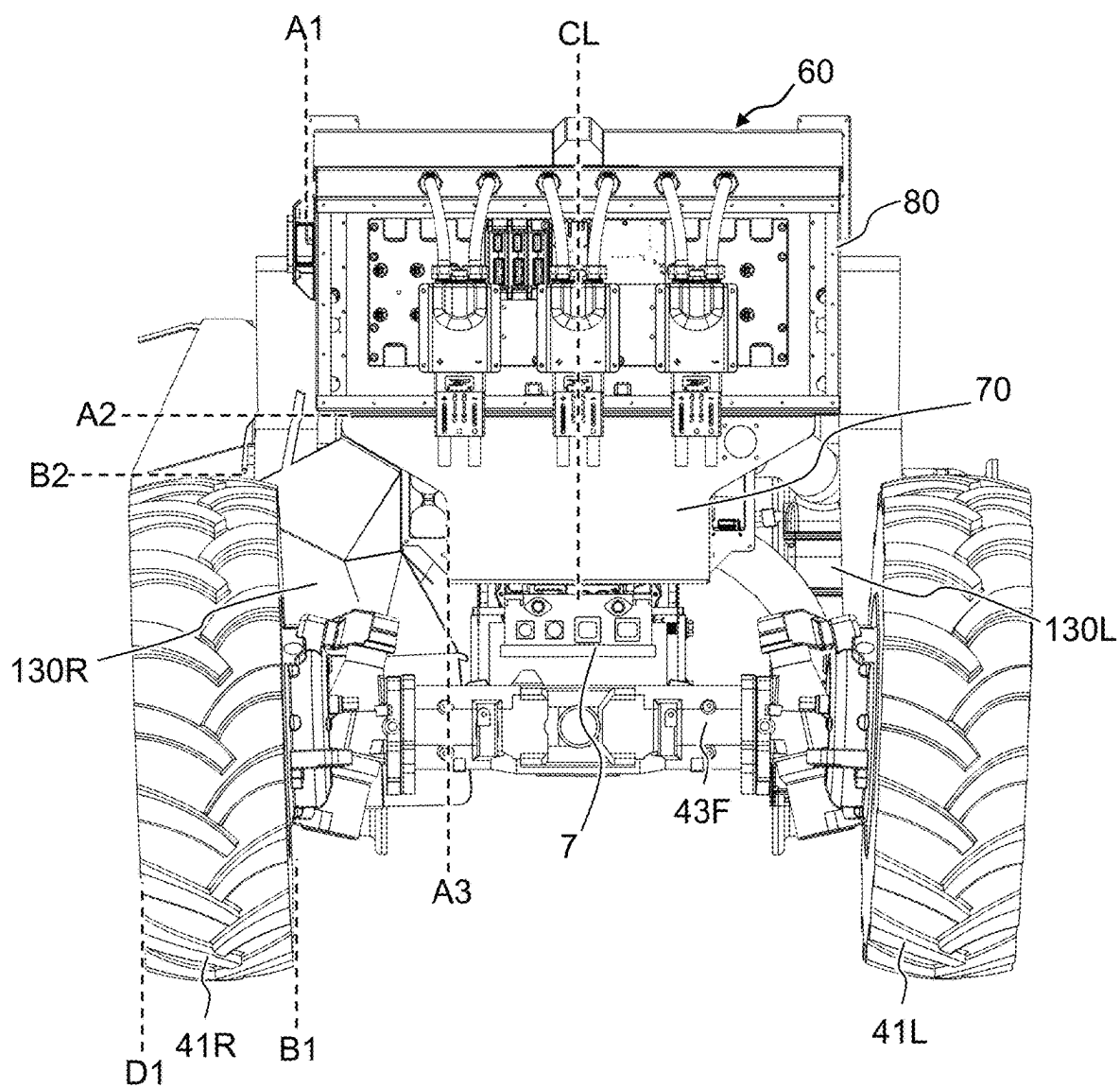


FIG. 28



RIGHT ← → LEFT

FIG. 29

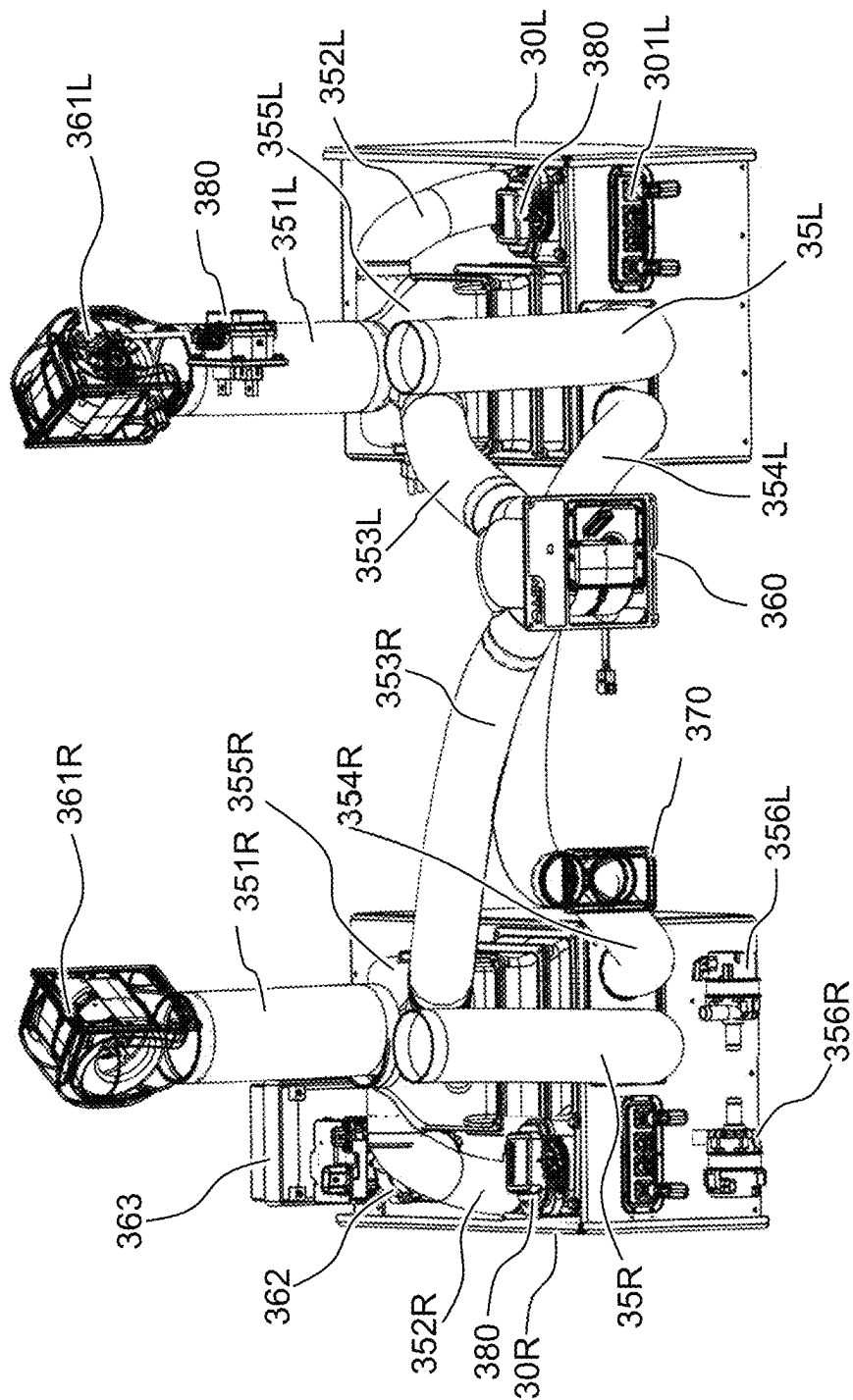


FIG. 30

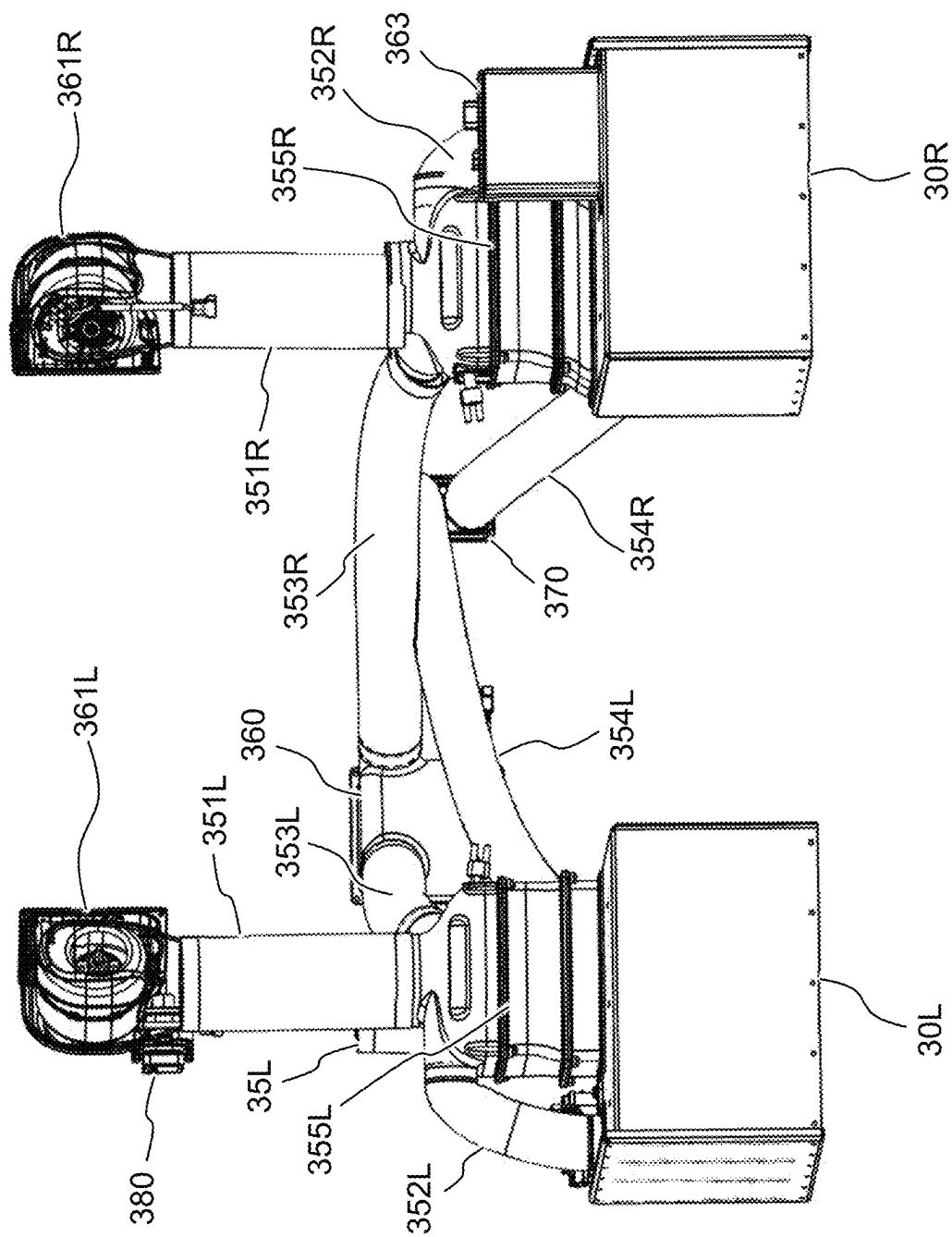


FIG. 31

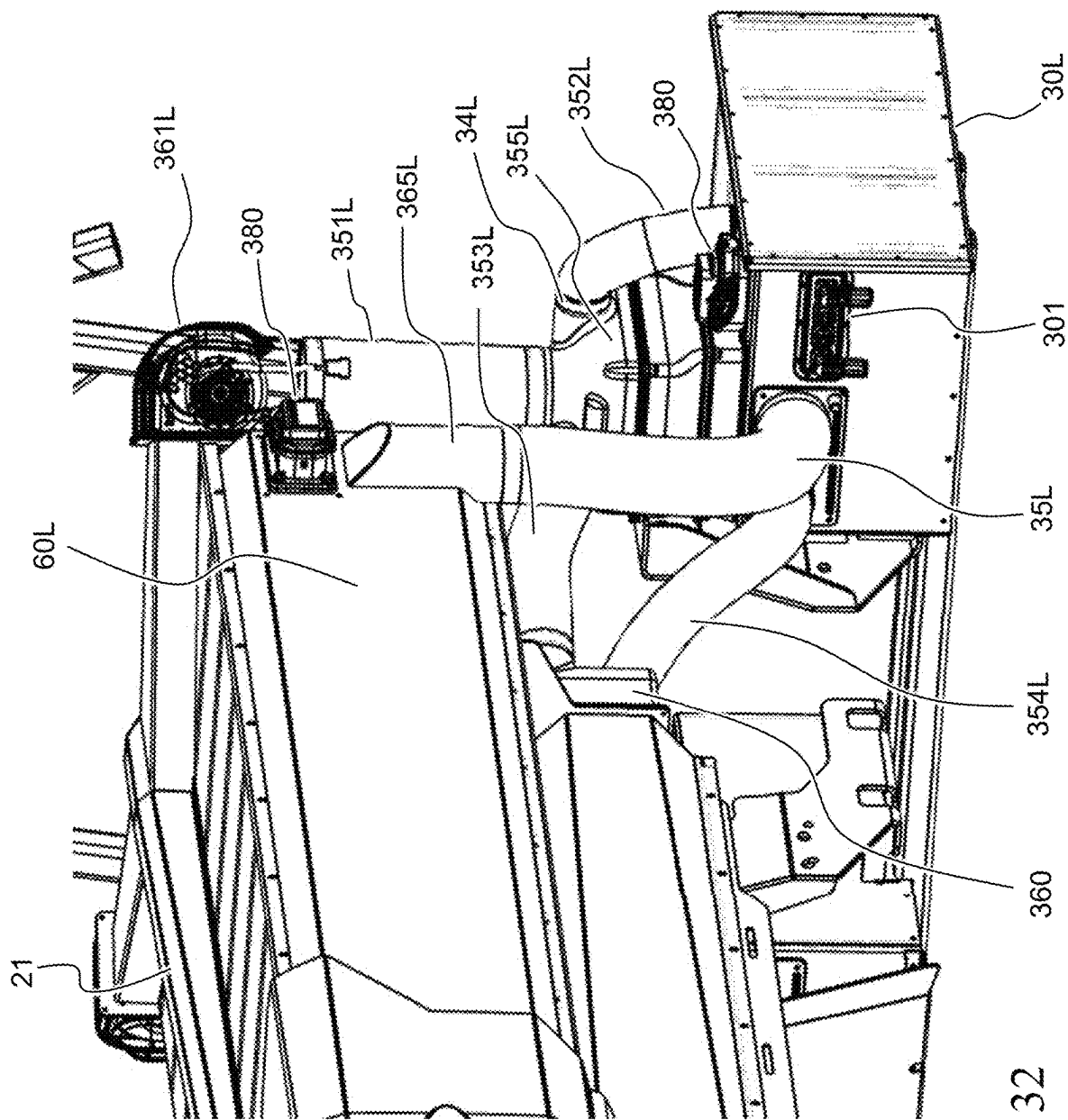


FIG. 32

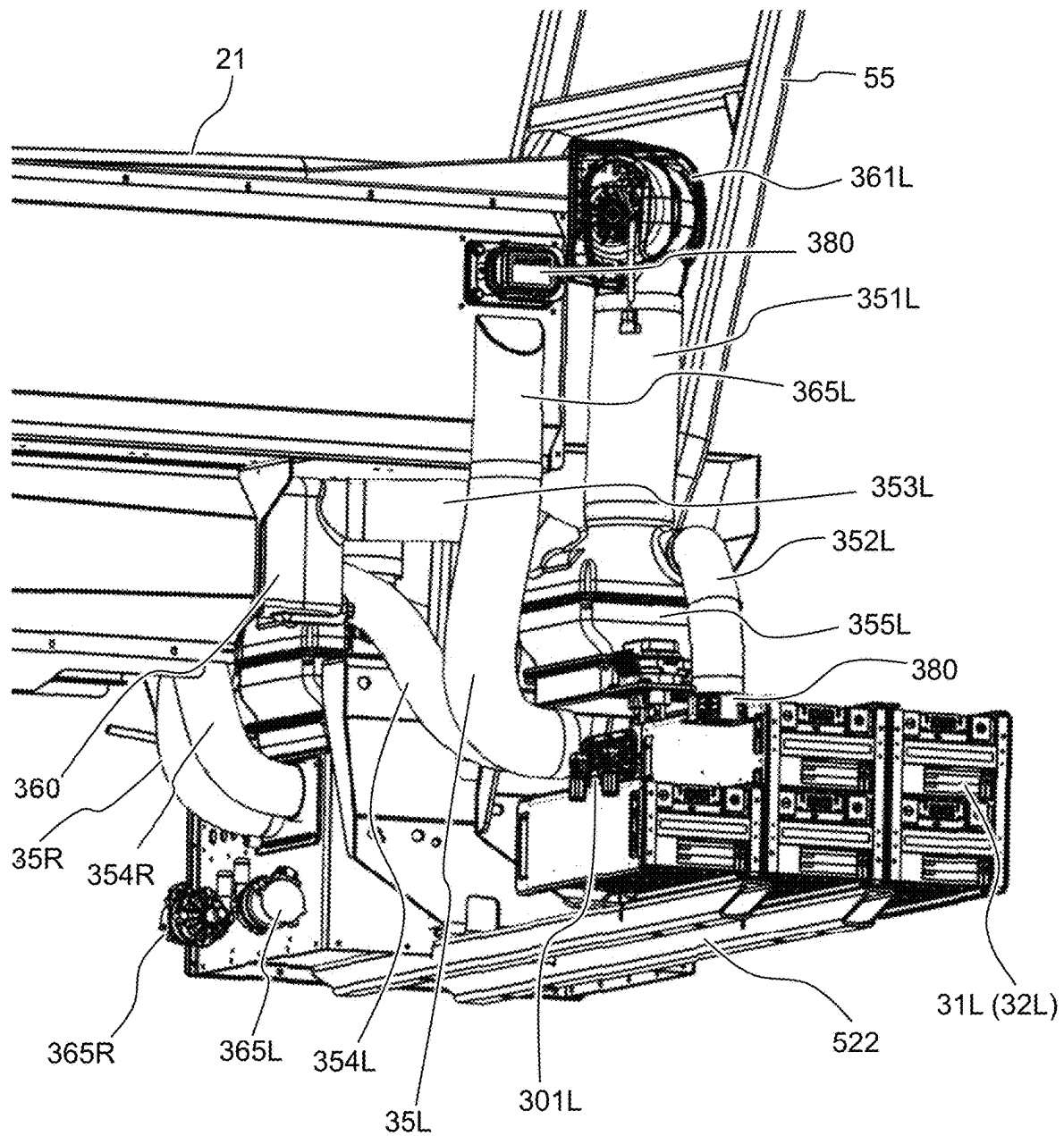


FIG. 33

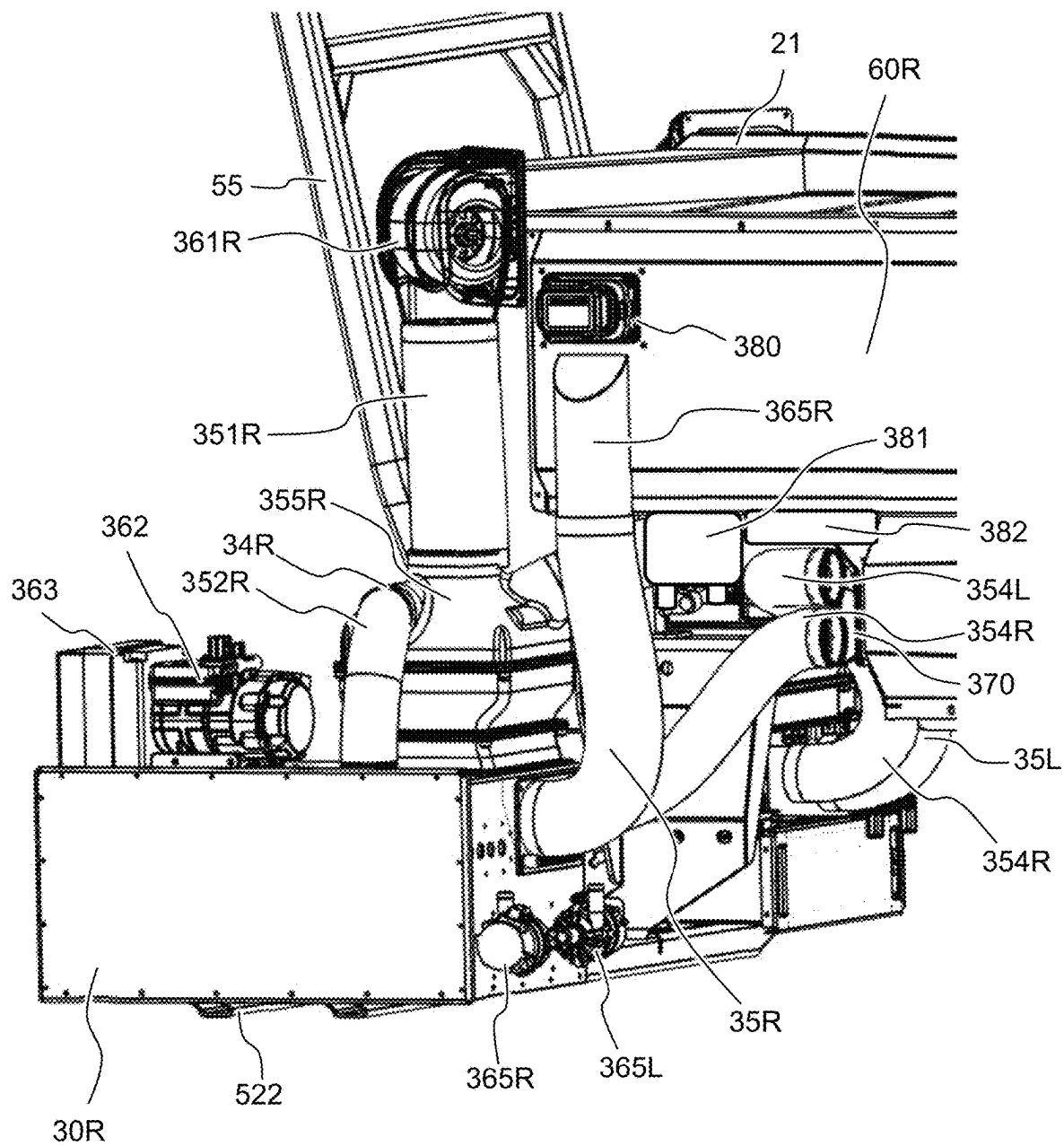


FIG. 34

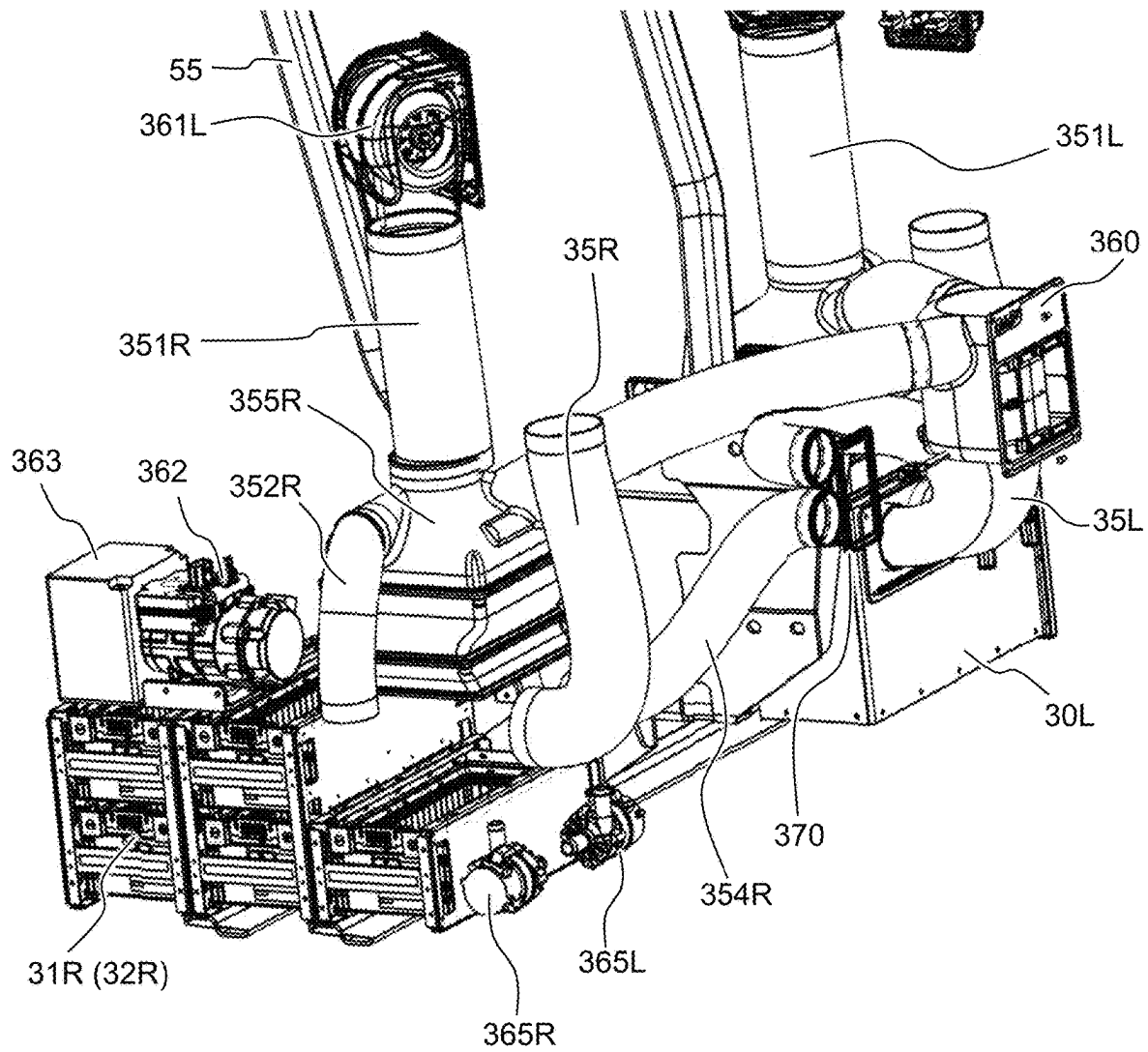


FIG. 35

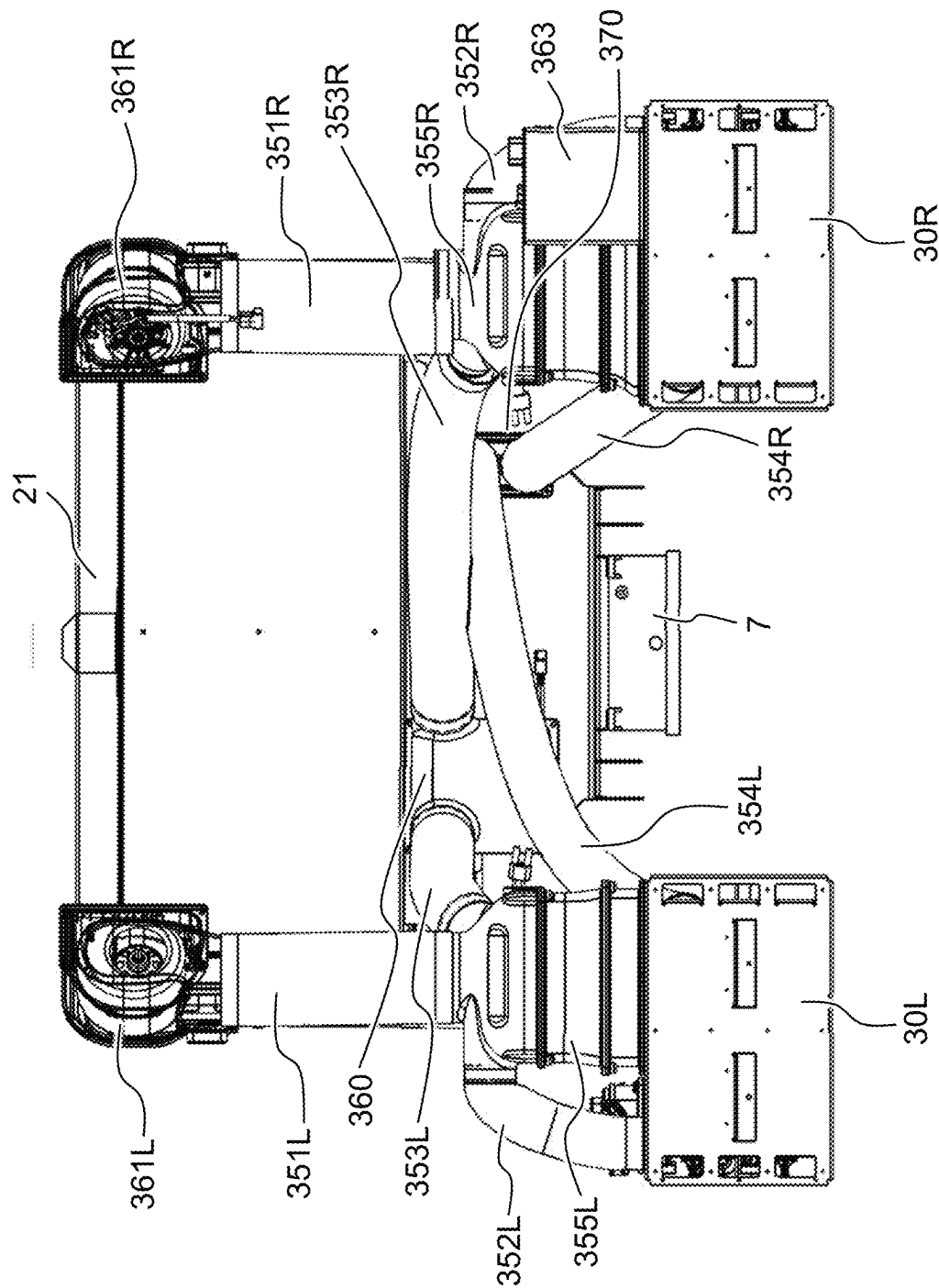


FIG. 36

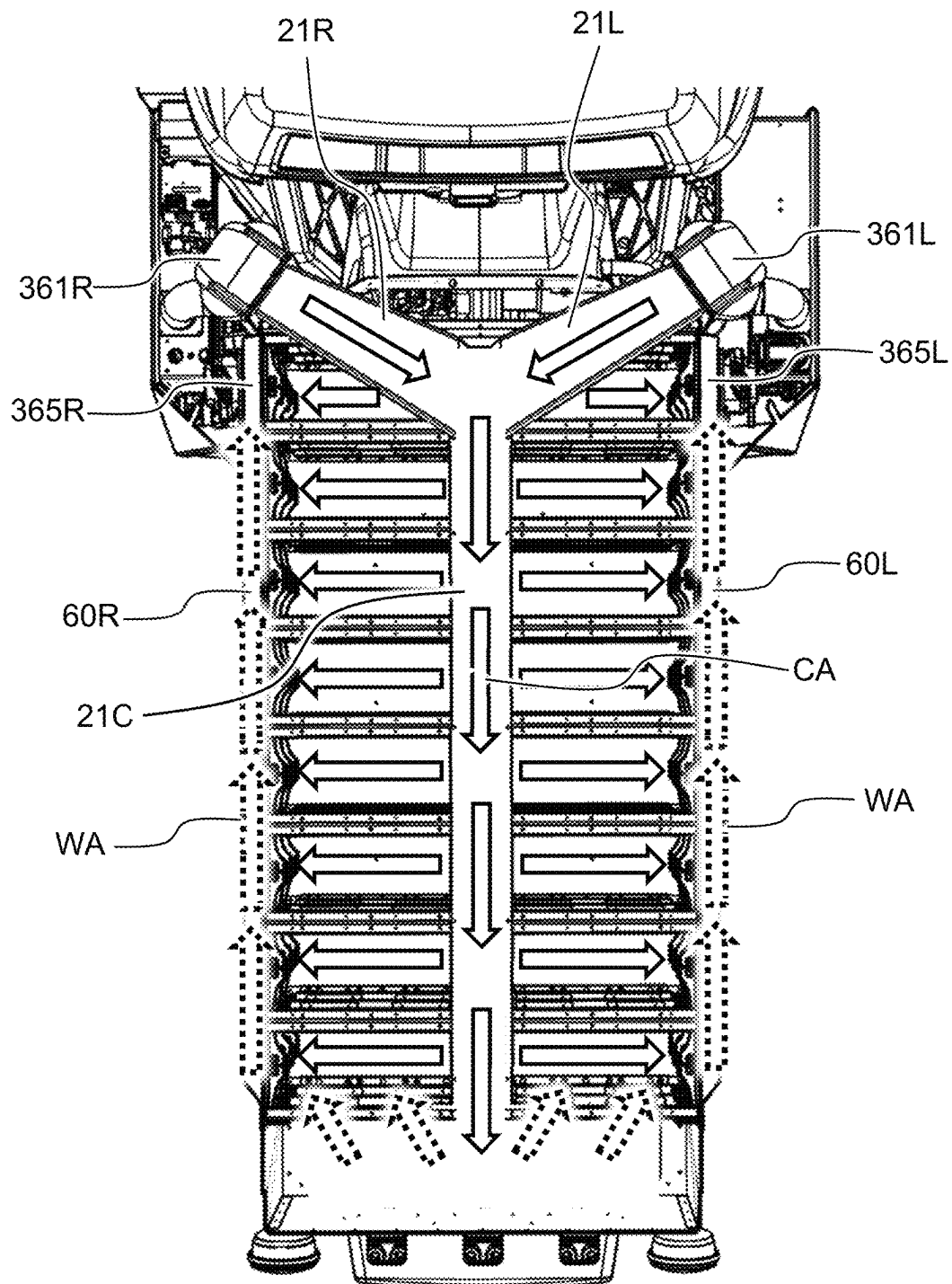


FIG. 37

FIG. 38A

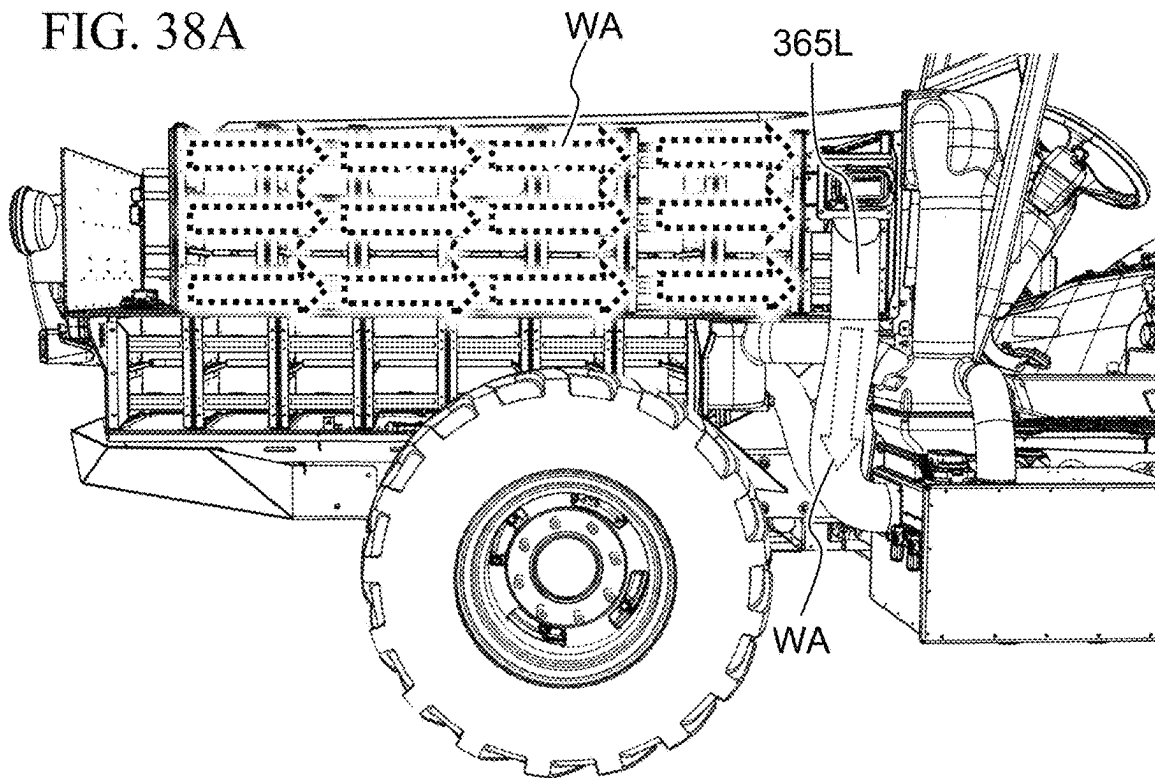
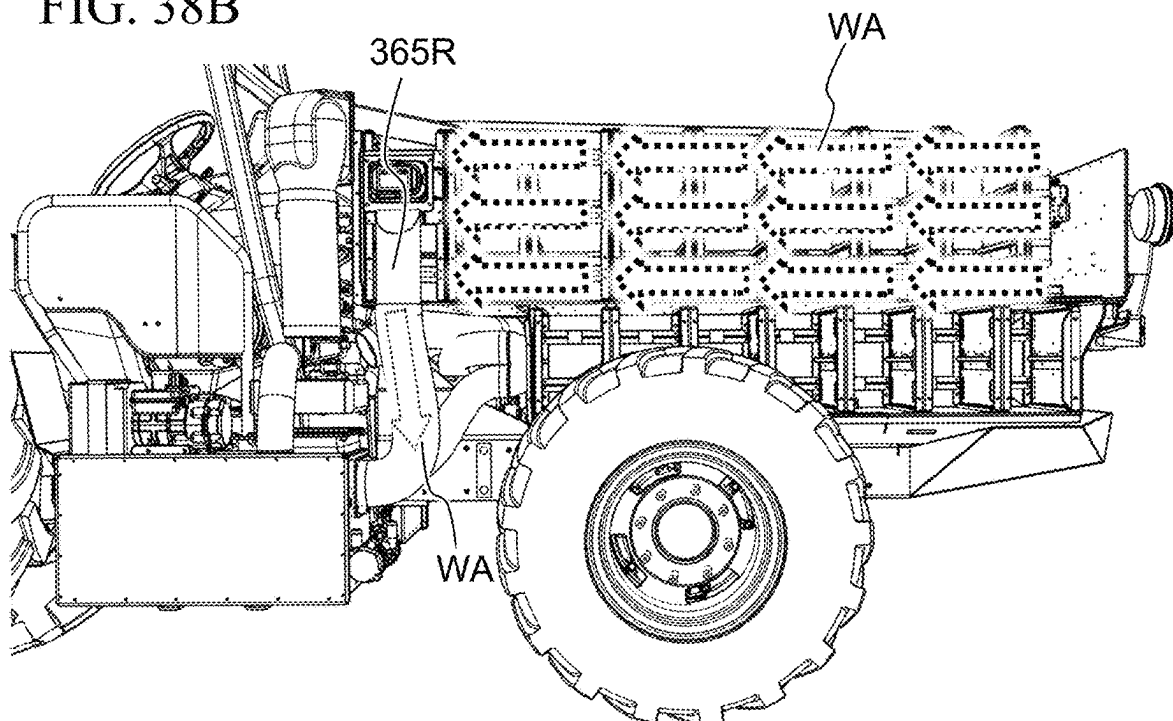


FIG. 38B



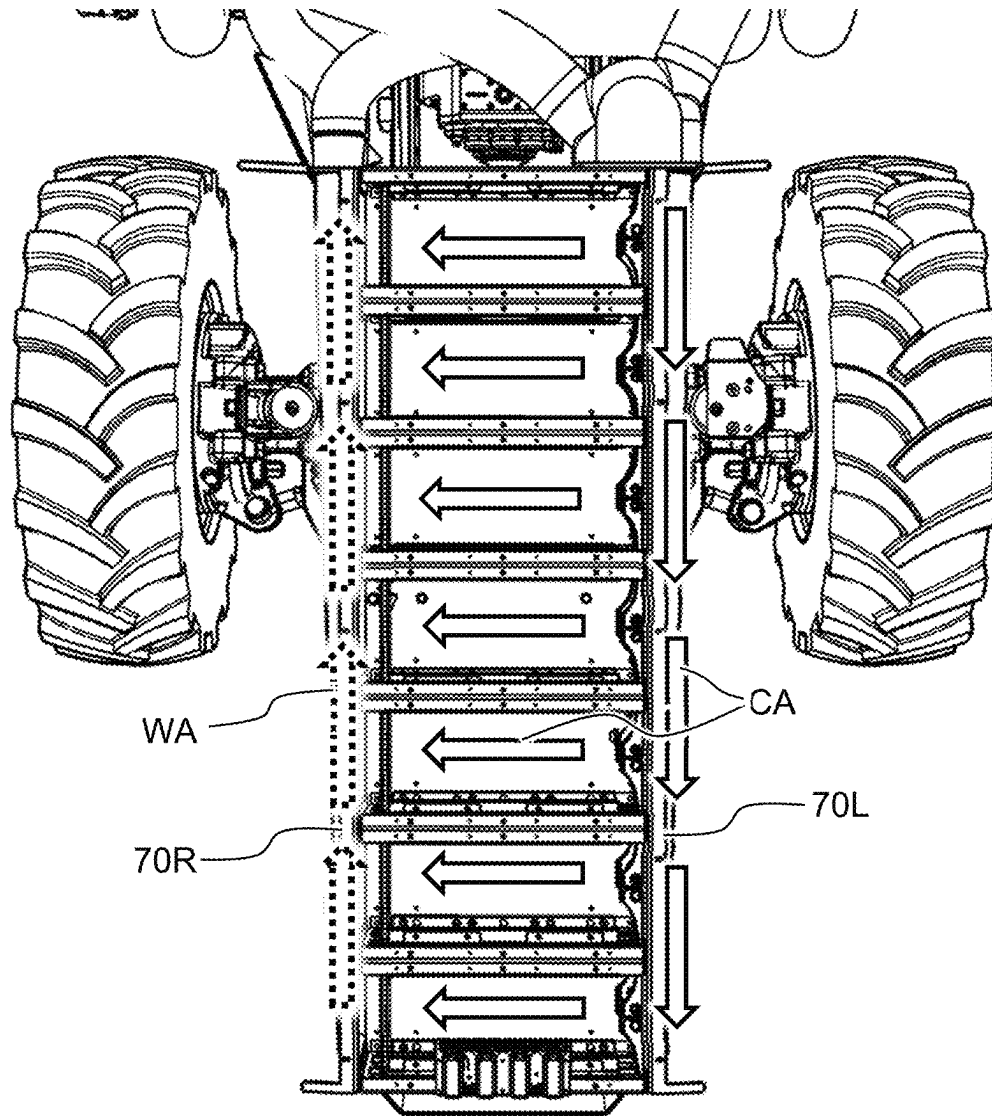


FIG. 39

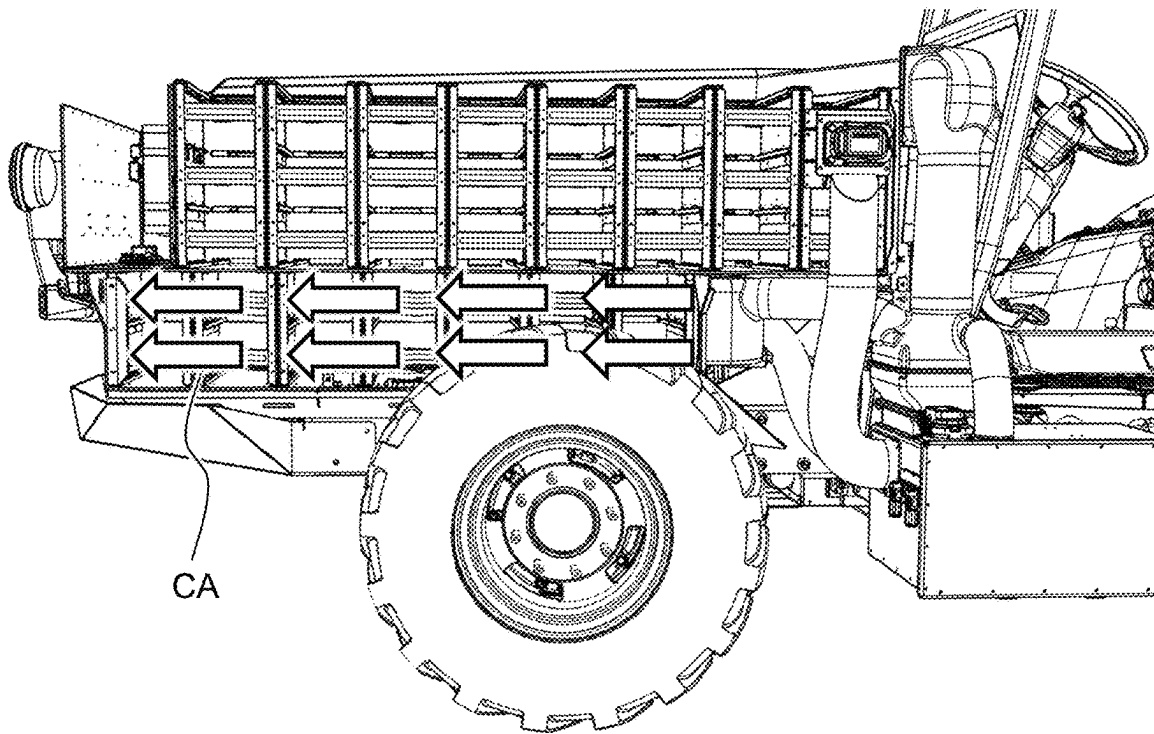


FIG. 40A

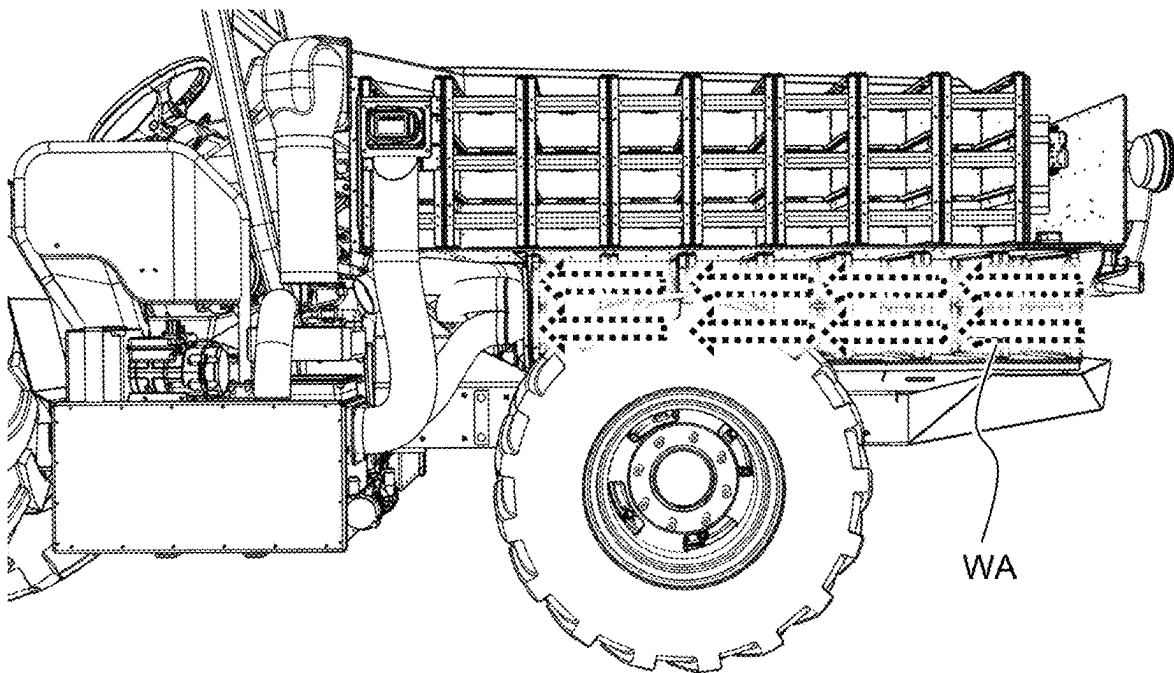


FIG. 40B

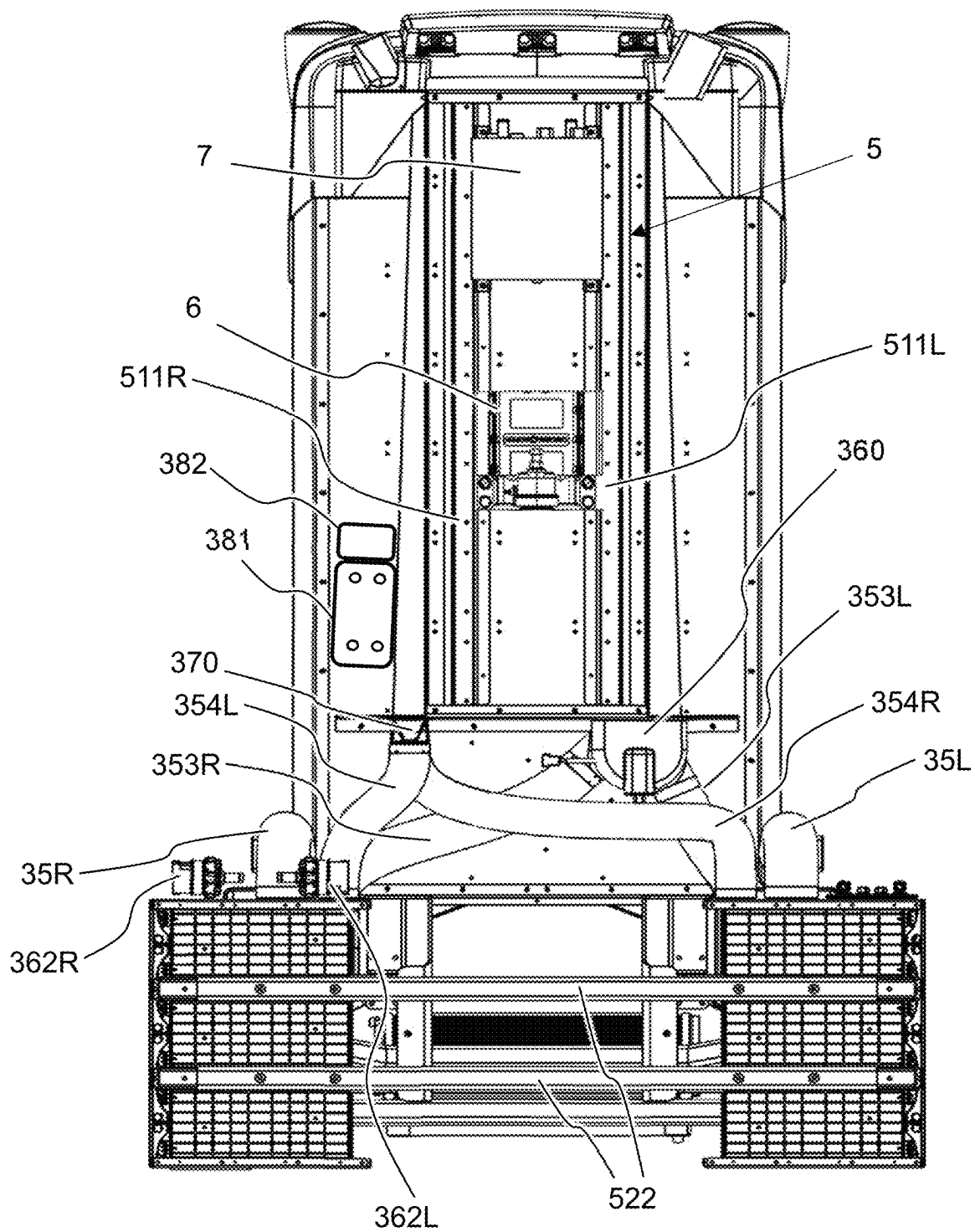


FIG. 41

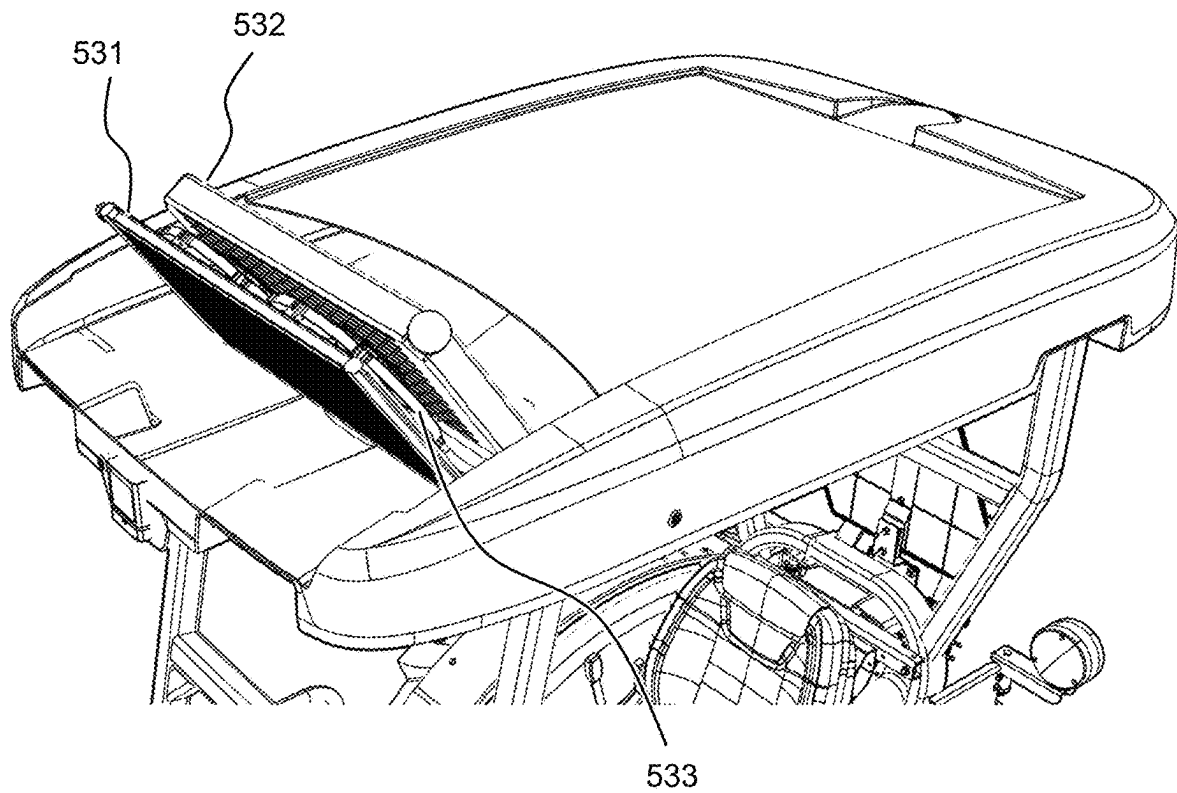


FIG. 42

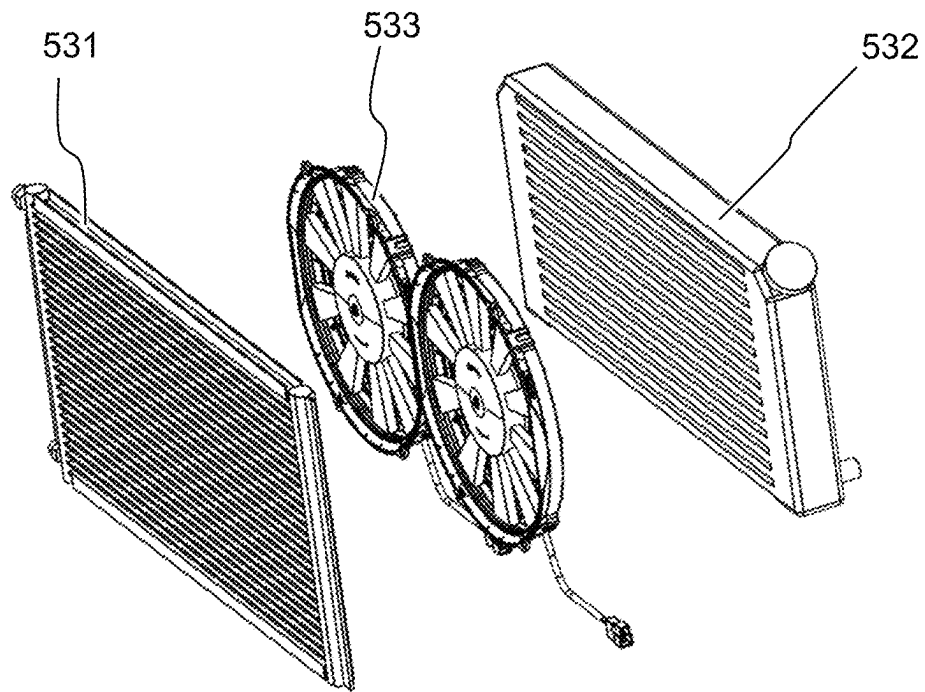


FIG. 43

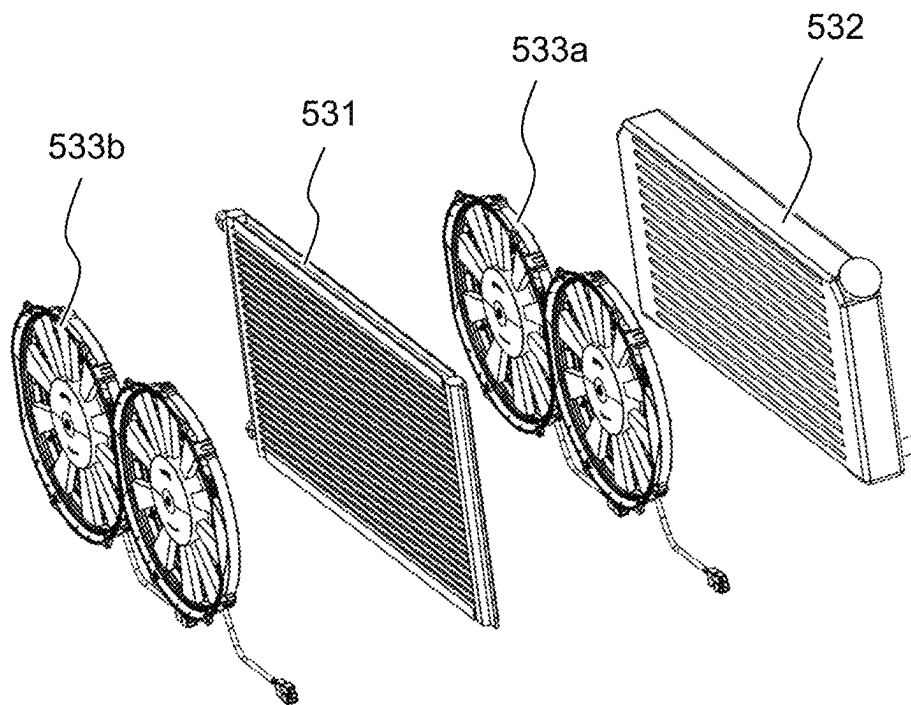


FIG. 44

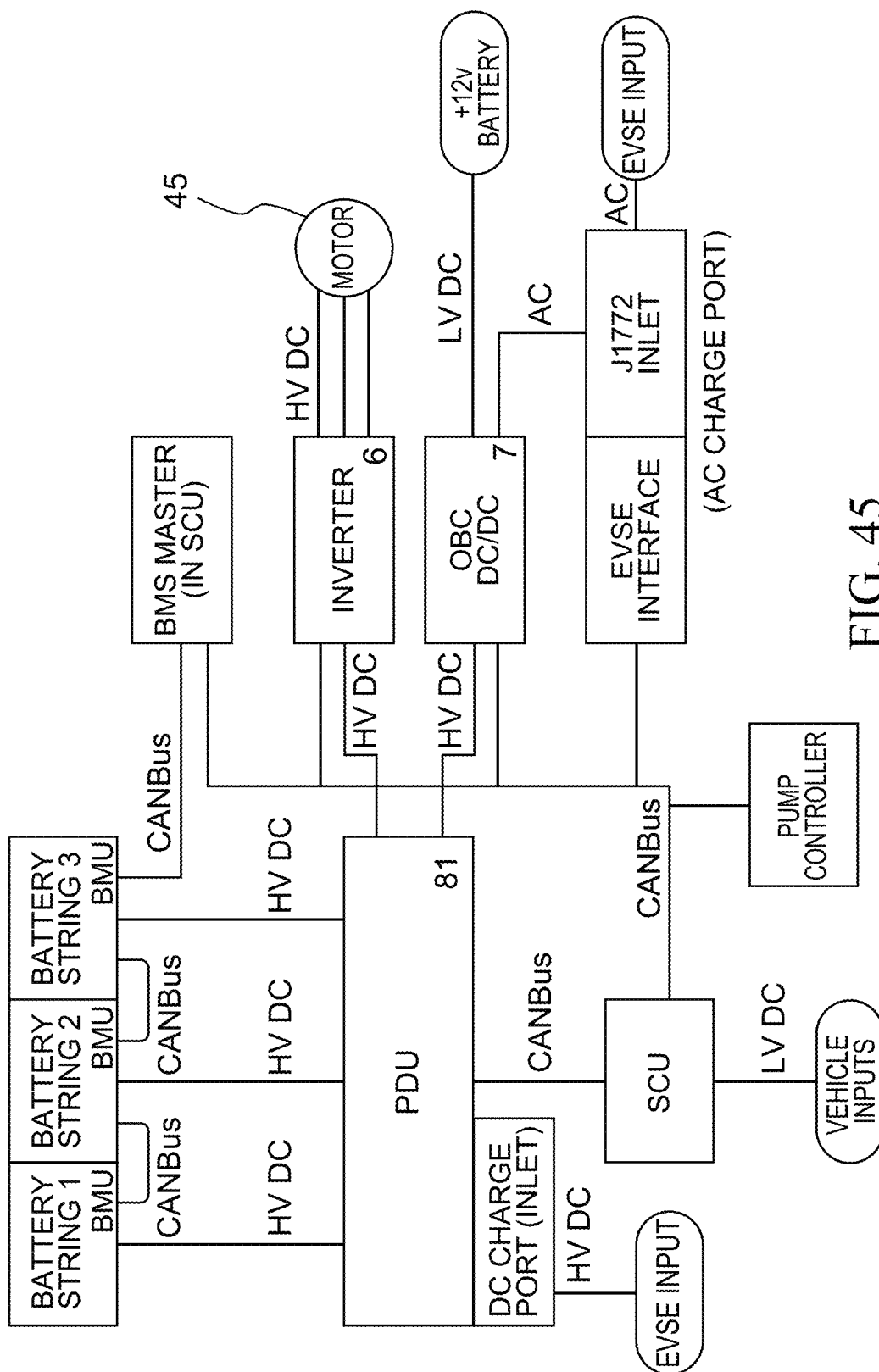


FIG. 45

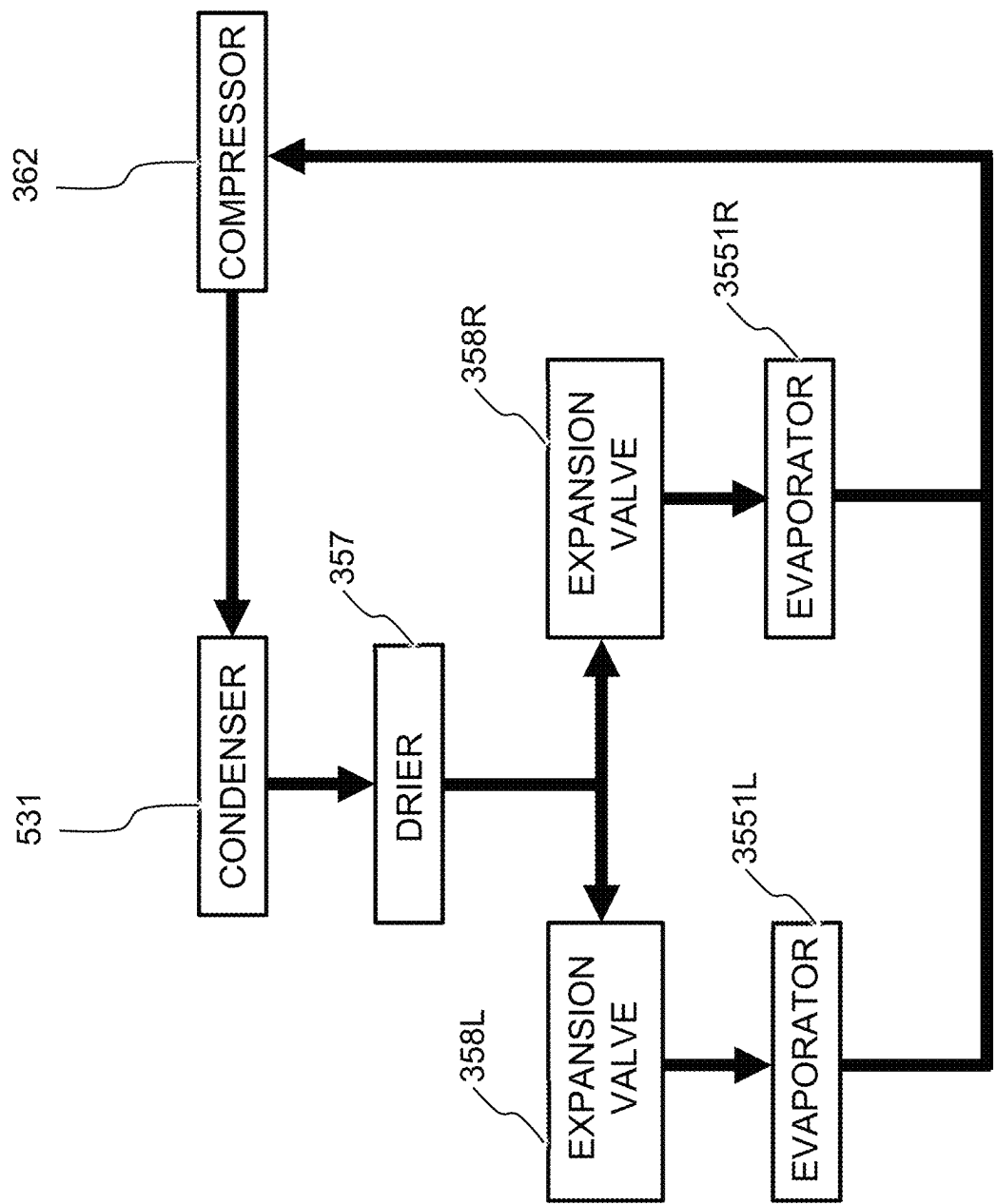


FIG. 46

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AIR COOLING SYSTEM FOR AN ELECTRIC VEHICLE

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to electric vehicles including structures and arrangements which can be used to cool components of the electric vehicle. More specifically, the present invention relates to electric vehicles including structures and arrangements that can be used to direct cooled air through components of an electric vehicle to thereby effectively cool components of the electric vehicle.

2. Description of the Related Art

Conventional structures and arrangements used to cool components of a vehicle typically only use liquid cooling for engine components and air cooling for operators of the vehicle. However, electric vehicles also require cooling for various additional components, which cooling cannot be provided using conventional liquid cooling and air cooling systems.

SUMMARY OF THE INVENTION

Preferred embodiments of the present invention provide electric vehicles that each include structures and arrangements which can be used to efficiently cool components of the electric vehicle.

A vehicle according to a preferred embodiment of the present invention includes a first battery housing to house a plurality of first battery modules, a second battery housing to house a plurality of second battery modules, and an air cooling system to cool the plurality of first battery modules and the plurality of second battery modules. The air cooling system includes a compressor, a condenser, a first evaporator coil that includes a first side at which warm air is received and a second side at which cool air exits after having passed through the first evaporator coil, a second evaporator coil that includes a first side at which warm air is received and a second side at which cool air exits after having passed through the second evaporator coil, a first blower to direct a first portion of the cool air from the second side of the first evaporator coil into the first battery housing, a second blower to direct a first portion of the cool air from the second side of the second evaporator coil into the first battery housing, and a third blower to direct a second portion of the cool air from the second side of the first evaporator coil and a second portion of the cool air from the second side of the second evaporator coil into the second battery housing.

In a preferred embodiment of the present invention, the vehicle further includes a first side housing located on a lateral side of the first battery housing, the first side housing to house a plurality of first side battery modules, and the first evaporator coil is supported by the first side housing.

In a preferred embodiment of the present invention, the vehicle further includes a first side blower to direct a third portion of the cool air from the second side of the first evaporator coil into the first side housing.

In a preferred embodiment of the present invention, the vehicle further includes a first side housing and a second side housing located on opposing lateral sides of the first battery housing, the first side housing to house a plurality of first side battery modules, and the second side housing to house a plurality of second side battery modules.

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In a preferred embodiment of the present invention, the compressor is mounted on the second side housing.

In a preferred embodiment of the present invention, the air cooling system further includes a drier, and the drier is attached to the first side housing.

In a preferred embodiment of the present invention, the vehicle further includes a battery, and the battery is mounted on the second side housing adjacent to the compressor.

In a preferred embodiment of the present invention, the vehicle further includes a first side blower that directs a third portion of the cool air from the second side of the first evaporator coil into the first side housing, and a second side blower that directs a third portion of the cool air from the second side of the second evaporator coil into the second side housing. The first side blower is connected to a first side outer duct through which the third portion of the cool air flows from the second side of the first evaporator coil into the first side housing, and the second side blower is connected to a second side outer duct through which the third portion of the cool air flows from the second side of the second evaporator coil into the second side housing.

In a preferred embodiment of the present invention, the first evaporator coil is supported by the first side housing, and the second evaporator coil is supported by the second side housing.

In a preferred embodiment of the present invention, the vehicle further includes a first evaporator air box that houses the first evaporator coil and is attached to the first side housing, and a second evaporator air box that houses the second evaporator coil and is attached to the second side housing. The air cooling system further includes a first expansion valve and a second expansion valve, the first expansion valve is housed in the first evaporator air box, and the second expansion valve is housed in the second evaporator air box.

In a preferred embodiment of the present invention, the first blower is connected to a first duct through which the first portion of the cool air from the second side of the first evaporator coil flows into the first battery housing, the second blower is connected to a second duct through which the first portion of the cool air from the second side of the second evaporator coil flows into the first battery housing, a bottom end of the first duct is connected to a first evaporator air box that houses the first evaporator coil, the first blower is connected to a top end of the first duct, a bottom end of the second duct is connected to a second evaporator air box that houses the second evaporator coil, and the second blower is connected to a top end of the second duct.

In a preferred embodiment of the present invention, the first battery housing includes a vertical partition wall that divides the first battery housing to include a first portion of the first battery housing and a second portion of the first battery housing, the first portion of the first battery housing houses a first battery module bank including a first portion of the plurality of first battery modules, the second portion of the first battery housing houses a second battery module bank including a second portion of the plurality of first battery modules, and the first portion of the cool air directed by the first blower and the first portion of the cool air directed by the second blower flow into the first battery housing at a central portion of the first battery housing.

In a preferred embodiment of the present invention, the cool air that flows into the first battery housing at the central portion of the first battery housing flows from the central portion of the first battery housing in a first lateral direction across the first battery module bank and in a second lateral direction across the second battery module bank.

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In a preferred embodiment of the present invention, the second portion of the cool air from the second side of the first evaporator coil and the second portion of the cool air from the second side of the second evaporator coil, which are directed by the third blower, flow into the second battery housing at a first side of the second battery housing in a left-right direction of the vehicle, and the third blower is attached to a rear surface of the second battery housing on the first side of the second battery housing.

In a preferred embodiment of the present invention, the third blower is connected to a first duct through which the second portion of the cool air from the second side of the first evaporator coil flows from the second side of the first evaporator coil into the second battery housing, and the third blower is connected to a second duct through which the second portion of the cool air from the second side of the second evaporator coil flows from the second side of the second evaporator coil into the second battery housing.

In a preferred embodiment of the present invention, a width of the first battery housing is greater than a width of the second battery housing in a width direction of the vehicle.

In a preferred embodiment of the present invention, the first battery housing and the second battery housing are offset in a front-rear direction of the vehicle, the offset defines a space in which an additional housing to house one or more electronic components is located, and at least one of the first portion of the cool air from the second side of the first evaporator coil directed by the first blower into the first battery housing and the first portion of the cool air from the second side of the second evaporator coil directed by the second blower into the first battery housing flows through the first battery housing and into the additional housing.

In a preferred embodiment of the present invention, the second battery housing is attached to a bottom surface of the first battery housing.

In a preferred embodiment of the present invention, the condenser is mounted on a roof of the vehicle.

In a preferred embodiment of the present invention, the vehicle further includes a cabin frame that is connected to a chassis of the vehicle and supports the roof of the vehicle, and a fluid connection between the compressor and the condenser extends along or through the cabin frame.

The above and other features, elements, steps, configurations, characteristics, and advantages of the present invention will become more apparent from the following detailed description of preferred embodiments of the present invention with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows a perspective right-front view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 2 shows a perspective left-rear view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 3 shows a front view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 4 shows a right side view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 5 shows a left side view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 6 shows a top view of an electric vehicle according to a preferred embodiment of the present invention.

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FIG. 7 shows a bottom view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 8 shows a partially assembled left-front perspective view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 9 shows a partially assembled right-front perspective view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 10 shows a partially assembled left side view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 11 shows a partially assembled front view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 12 shows a perspective view of a first battery housing and a second battery housing with their respective battery modules removed for clarity according to a preferred embodiment of the present invention.

FIG. 13 shows a top view of a first battery housing according to a preferred embodiment of the present invention.

FIG. 14 shows a partially assembled left-front perspective view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 15 shows a partially assembled left-front perspective view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 16 shows a partially assembled top view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 17 shows a partially assembled left perspective view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 18 shows a partially assembled left side view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 19 shows a partially assembled left-front perspective view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 20 shows a partially assembled left-rear perspective view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 21 shows a partially assembled right-rear perspective view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 22 is a right side view of a second battery housing according to a preferred embodiment of the present invention.

FIG. 23 is a left side view of a second battery housing with a first side cover and the second battery modules removed for clarity according to a preferred embodiment of the present invention.

FIG. 24 shows a partially assembled bottom view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 25 shows a front view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 26 shows a top view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 27 shows a bottom view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 28 shows a partially assembled front view of an electric vehicle according to a preferred embodiment of the present invention.

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FIG. 29 shows a schematic of an example of a liquid cooling system included in an electric vehicle according to a preferred embodiment of the present invention.

FIG. 30 shows components included in an air cooling system according to a preferred embodiment of the present invention.

FIG. 31 shows components included in an air cooling system according to a preferred embodiment of the present invention.

FIG. 32 shows components included in an air cooling system according to a preferred embodiment of the present invention.

FIG. 33 shows components included in an air cooling system according to a preferred embodiment of the present invention.

FIG. 34 shows components included in an air cooling system according to a preferred embodiment of the present invention.

FIG. 35 shows components included in an air cooling system according to a preferred embodiment of the present invention.

FIG. 36 shows components included in an air cooling system according to a preferred embodiment of the present invention.

FIG. 37 shows a plan view of an example of an air flow through a first battery housing according to a preferred embodiment of the present invention.

FIG. 38A shows a left side view of an example of an air flow through a first battery housing according to a preferred embodiment of the present invention.

FIG. 38B shows a right side view of an example of an air flow through a first battery housing according to a preferred embodiment of the present invention.

FIG. 39 shows a plan view of an example of an air flow through a second battery housing according to a preferred embodiment of the present invention.

FIG. 40A shows a left side view of an example of an air flow through a second battery housing according to a preferred embodiment of the present invention.

FIG. 40B shows a right side view of an example of the air flow through the second battery housing 70.

FIG. 41 shows a partially assembled bottom view of an electric vehicle according to a preferred embodiment of the present invention.

FIG. 42 shows a partially assembled view of a roof according to a preferred embodiment of the present invention.

FIG. 43 shows an exploded view of cooling components according to a preferred embodiment of the present invention.

FIG. 44 shows an exploded view of cooling components according to a preferred embodiment of the present invention.

FIG. 45 shows a block diagram of an electrical subsystem included in an electric vehicle according to a preferred embodiment of the present invention.

FIG. 46 is a schematic that shows a flow of refrigerant through portions of an air cooling system according to a preferred embodiment of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

FIGS. 1-7 show a vehicle 10 according to a preferred embodiment of the present invention. FIG. 1 shows an isometric view from a first (front) end of the vehicle 10, and FIG. 2 shows an isometric view from a second (rear) end of

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the vehicle 10. FIG. 3 shows a front view of the vehicle 10, and FIGS. 4 and 5 show side views of the vehicle 10. FIGS. 6 and 7 show top and bottom views of the vehicle 10, respectively.

In a preferred embodiment, the vehicle 10 includes a chassis 5, a left front wheel 41L and a right front wheel 41R connected by a front axle 43F, and a left rear wheel 42L and a right rear wheel 42R connected by a rear axle 43R. However, vehicles according to the preferred embodiments of the present invention are not specifically limited to four wheels and may include any appropriate number of wheels. For example, vehicles according to preferred embodiments may include only three wheels by removing one of the aforementioned four wheels, a fifth wheel (e.g., an additional wheel provided in line with a pair of one of the aforementioned four wheels, a wheel provided at a front or rear of the vehicle, etc.), six total wheels by including a pair of center wheels, and the like. According to other preferred embodiments of the present invention, vehicles can include one or more tracks or treads in place of one or more of the wheels.

In a preferred embodiment, the vehicle 10 can include a cabin 50 or the like with a roof 51 supported by a cabin frame that is supported by the chassis 5 and includes a rear cabin frame 52 and a forward cabin frame 55. The cabin 50 or the like can include various components, for example, a seat for an operator of the vehicle 10 and vehicle controls such as a steering wheel. However, the vehicle 10 is not limited to including a cabin 50, and can alternatively include a canopy or a Rollover Protection Structure (ROPS).

In a preferred embodiment of the present invention, the vehicle 10 includes a first battery housing 60 to house a plurality of first battery modules 61, and a second battery housing 70 to house a plurality of second battery modules 71. In a preferred embodiment, the first battery housing 60 and the second battery housing 70 can be supported by the chassis 5. As shown in FIGS. 8 and 9 in which certain structural features have been removed for illustrative purposes, the first battery housing 60 can include a plurality of openings 67 connected to a plurality of sub-housings that each receive a separate one of the plurality of first battery modules 61. Similarly, the second battery housing 70 can include a plurality of openings 75 connected to a plurality of sub-housings that each receive a separate one of the plurality of second battery modules 71.

In a preferred embodiment, the plurality of openings 67 of the first battery housing 60 and the plurality of openings 75 of the second battery housing 70 face a width/side (left-right) direction of the vehicle 10. The plurality of first battery modules 61 can be aligned, or substantially aligned, in rows and columns in both a vertical direction of the vehicle and the front-rear direction of the vehicle to provide compact storage of the plurality of first battery modules 61 and reduce a space required by the first battery housing 60. Similarly, the plurality of second battery modules 71 can be aligned, or substantially aligned, in rows and columns in both a vertical direction of the vehicle and the front-rear direction of the vehicle to provide compact storage of the plurality of second battery modules 71 and reduce a space required by the second battery housing 70. In a preferred embodiment, the terminal ends of each of the plurality of first battery modules 61 and the plurality of second battery modules 71 can face a width/side (left-right) direction of the vehicle 10 when housed in the respective first battery housing 60 and second battery housing 70, as shown in FIGS. 9 and 13, for example.

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In a preferred embodiment, the second battery housing 70 is attached to a bottom surface of the first battery housing 60, and the first battery housing 60 and the second battery housing 70 can be arranged in a “T” shape, as shown in FIG. 11, for example. In other words, the first battery housing 60 can have a width that is greater than a width of the second battery housing 70 in a width/side (left-right) direction of the vehicle, such that the first battery housing 60 extends beyond the second battery housing 70 in the width/side (left-right) direction of the vehicle. The structure of the first battery housing 60 overhanging/extending farther than the second battery housing 70 in the width direction of the vehicle can provide a space for the left front wheel 41L and the right front wheel 41R to operate and turn.

As shown in FIGS. 3 and 11, an outer edge A1 of the first battery housing 60 can be located inside of an inner edge B1 of each of the front wheels 41R and 41L in the width direction of the vehicle. Furthermore, a lower edge A2 of the first battery housing 60 can be spaced away by a predetermined distance from an upper edge B2 of each of the front wheels 41R and 41L. A line connecting the upper edge B2 of the left front wheel 41L and the upper edge B2 of the right front wheel 41R can pass through a center portion of the second battery housing 70. In other words, in a vertical direction of the vehicle 10, the line connecting the upper edge B2 of each of the front wheels 41R and 41L may be located between an upper surface of the second battery housing 70 and a lower surface of the second battery housing 70 in the vertical direction of the vehicle.

Alternatively, the outer edge A1 of the first battery housing 60 can be aligned or substantially aligned with the inner edge B1 of each of the front wheels 41R and 41L, or the outer edge A1 of the first battery housing 60 can be located outside of the inner edge B1 of each of the front wheels 41R and 41L in the width direction of the vehicle.

In a preferred embodiment of the present invention, the first battery housing 60 and the second battery housing 70 can be offset in a direction from the first (front) end of the vehicle to the second (rear) end of the vehicle, as shown in FIG. 10, for example. For example, the second battery housing 70 can extend closer to the first (front) end of the vehicle 10 than the first battery housing 60. As an example, the second battery housing 70 can include at least one battery module 71A that is not overlapped by the first battery housing 60 in a vertical direction. A power distribution unit (PDU) housing 80 (e.g., an additional housing) to house one or more electrical components including a power distribution unit (PDU) 81 can be provided in a space defined by the forward offset of the second battery housing 70 with respect to the first battery housing 60, as shown in FIGS. 8-11, for example. As shown in FIG. 10, for example, the power distribution unit (PDU) 81 is housed on the power distribution unit (PDU) housing 80 and is attached to a front surface 68 of the first battery housing 60.

In a preferred embodiment, the first battery housing 60 can extend closer to the second (rear) end of the vehicle than the second battery housing 70. As an example, the first battery housing 60 can include at least one battery module 61A that is not overlapped by the second battery housing 70 in a vertical direction, as shown in FIG. 10, for example. Portions of an air cooling system discussed in more detail below can be provided in a space defined by the rearward offset of the first battery housing 60 with respect to the second battery housing 70.

FIG. 12 shows a perspective view of the first battery housing 60 and the second battery housing 70 with their respective battery modules 61 and 71 removed for clarity. A

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portion of the PDU housing 80 is also removed from the view shown in FIG. 12. FIG. 13 shows a top view of the first battery housing 60. As shown in FIGS. 12 and 13, the first battery housing 60 can include a first portion 60A (e.g., a left portion) and a second portion 60B (e.g., a right portion). For example, the first battery housing 60 can be divided into the first portion 60A and the second portion 60B by a vertical partition wall 69. The vertical partition wall 69 is shown, for example, in FIGS. 19-21, in which portions of the first portion 60A and the second portion 60B of the first battery housing 60 have been removed to better illustrate the vertical partition wall 69.

In a preferred embodiment, as shown in FIGS. 8 and 9, the first portion 60A of the first battery housing 60 can house a first battery module bank 62 including a first portion of the plurality of first battery modules 61, and the second portion 60B of the first battery housing 60 can house a second battery module bank 63 including a second portion of the plurality of first battery modules 61. In a preferred embodiment, each of the first battery module bank 62 and the second battery module bank 63 can be provided in rows and columns. As an example, each of the first battery module bank 62 and the second battery module bank 63 can include three rows and eight columns of the plurality of first battery modules 61, for a total of twenty-four of the first battery modules 61 provided in the first battery module bank 62 housed in the first portion 60A, and twenty-four of the first battery modules 61 provided in the second battery module bank 63 housed in the second portion 60B.

As shown, for example, in FIGS. 1 and 2, the first battery housing 60 preferably includes a first side cover 60L (a left side cover) which covers a first side of the first battery housing 60, a second side cover 60R (a right side cover) which covers a second side of the first battery housing 60, and an upper cover 64 which covers an upper side of the first battery housing 60. As shown in FIGS. 15 and 16, the first battery housing 60 further includes cross walls 65 that delineate the vertical columns in which the first battery modules 61 are arrayed, mounting bars 66 that support the first battery modules 61, the openings 67 through which the first battery modules 61 are inserted, a front surface 68 that defines a forwardmost portion of the first battery housing 60, and a base plate 612.

As shown in FIGS. 15, 20, and 21, the front surface 68 preferably includes inner ventilation holes 681 and intermediate ventilation holes 682 that permit a flow of air between the first battery housing 60 and the PDU housing 80, discussed in more detail below. As shown in FIG. 15, the cross walls 65 preferably include side ventilation holes 651 that permit a flow of air between adjacent first battery modules 61 arrayed in a front to rear direction of the vehicle, also discussed in more detail below. As shown in FIGS. 17-21, the vertical partition wall 69 preferably includes center partition holes 691.

In a preferred embodiment, the second battery housing 70 houses a third battery module bank 72 including the plurality of second battery modules 71, as shown in FIG. 8, for example. The third battery module bank 72 can be provided in rows and columns. As an example, the third battery module bank 72 can include two rows and seven columns of the second battery modules 71, for a total of fourteen of the second battery modules 71 provided in the third battery module bank 72 housed in the second battery housing 70.

In a preferred embodiment of the present invention, the second battery housing 70 includes a first side cover 70L (a left side cover), a second side cover 70R (a right side cover), and a base plate 74, as shown in FIGS. 1, 5, and 22-24, for

example. In a preferred embodiment, the base plate **612** of the first battery housing **60** defines and functions as an upper cover of the second battery housing **70**.

FIG. **22** is a right side view of the second battery housing **70**, and FIG. **23** is a left side view of the second battery housing **70** with the first side cover **70L** and the second battery modules **71** removed. As shown in FIG. **23**, for example, the second battery housing **70** preferably includes cross walls **76** that delineate the vertical columns in which the second battery modules **71** are arrayed, mounting bars **77** that are used to support individual ones of the second battery modules **71**, and the openings **75** through which the battery modules **71** are inserted. In a preferred embodiment, the first side cover **70L** includes a first sloped channel **701L**, and the second side cover **70R** includes a second sloped channel **701R**, as shown in FIG. **24**, for example. As discussed in more detail below, the first sloped channel **701L** and the second sloped channel **701R** guide a flow of air through the second battery housing **70**.

In a preferred embodiment of the present invention, the vehicle **10** can also include one or more side housings **30**. The one or more side housings **30** can be spaced away from the first battery housing **60** and the second battery housing **70**. For example, as shown in FIGS. **1-7**, the vehicle **10** can include a first (left) side housing **30L** and a second (right) side housing **30R**. The first side housing **30L** can be located at least partially between the left front wheel **41L** and the left rear wheel **42L**, and the second side housing **30R** can be located at least partially between the right front wheel **41R** and the right rear wheel **42R**.

The first side housing **30L** can store a plurality of first side battery modules **31L**, and the second side housing **30R** can store a plurality of second side battery modules **31R**. The first side housing **30L** can house a fourth battery module bank **32L** including the plurality of the first side battery modules **31L** provided in rows and columns, and the second side housing **30R** can house a fifth battery module bank **32R** including the plurality of second side battery modules **31R** provided in rows and columns. As examples, the fourth battery module bank **32L** can include one row of two battery modules **31L** and one row of three battery modules **31L**, and the fifth battery module bank **32R** can include one row of two battery modules **31R** and one row of three battery modules **31R**. Accordingly, a total of five of the first side battery modules **31L** can be provided in the fourth battery module bank **32L**, and a total of five of the second side battery modules **31R** can be provided in the fifth battery module bank **32R**. In a preferred embodiment, each of the fourth battery module bank **32L** and the fifth battery module bank **32R** can be provided in series with the third battery module bank **72** housed in the second battery housing **70**.

In a preferred embodiment, the first side housing **30L** and the second side housing **30R** preferably include service plugs **380** which can be used to disconnect the first side battery modules **31L** and the second side battery modules **31R** from main power chains of the vehicle so that the first side battery modules **31L** and the second side battery modules **31R** may be serviced. The first side housing **30L** and the second side housing **30R** also preferably respectively include saddle power interface connections **301L** and **301R** which are used to respectively connect the first side battery modules **31L** and the second side battery modules **31R** with the third battery module bank **72** housed in the second battery housing **70**.

In a preferred embodiment of the present invention, the first battery modules **61**, the second battery modules **71**, the first side battery modules **31L**, and the second side battery

modules **31R** can be charged and discharged in groups. For example, one or two group(s) of battery modules can be discharged to provide electrical power to operate the vehicle **10** while other battery group(s) remain idle or are charged. For example, the first battery module bank **62** can define a first battery module group, the second battery module bank **63** can define a second battery module group, and the third battery module bank **72** can define a third battery module group. If the fourth battery module bank **32L** and/or the fifth battery module bank **32R** are included in the vehicle **10**, the fourth battery module bank **32L** and the fifth battery module bank **32R** can be included in the third battery module group. Accordingly, each of the first battery module group, the second battery module group, and the third battery module group can include a total of twenty-four battery modules. However, other groupings of battery modules can be provided, and are included within the preferred embodiments of the present invention. For example, the first battery module group can include a greater number of battery modules than the third battery module group, and the second battery module group can include a greater number of battery modules than the third battery module group.

In a preferred embodiment of the present invention, the first battery modules **61**, the second battery modules **71**, the first side battery modules **31L**, and the second side battery modules **31R** can be lithium titanium oxide (LTO) battery modules. For example, each of the battery modules **61**, **71**, **31L**, and **31R** can have a nominal voltage of about 27.6 V, which provides a nominal voltage of about 662.4 V for each of the first battery module group, the second battery module group, and the third battery module group. In addition, each of the first battery module group, the second battery module group, and the third battery module group can have a minimum voltage of about 432 V and a maximum voltage of about 778 V, for example. Each of the battery modules **61**, **71**, **31L**, and **31R** can store about 90 kWh of energy, for example. The battery modules **61**, **71**, **31L**, and **31R** can be sealed within their corresponding housings (the first battery housing **60**, the second battery housing **70**, the first side housing **30L**, and the second side housing **30R**), and each of the battery modules **61**, **71**, **31L**, **31R** can be removed and replaced upon failure.

As other examples, the first battery modules **61**, the second battery modules **71**, the first side battery modules **31L**, and the second side battery modules **31R** can be lithium ferrophosphate (LFP) battery modules or nickel manganese cobalt (NMC) battery modules, for example. LFP battery modules or NMC battery modules can be provided as smaller and/or lighter battery modules than LTO battery modules.

In a preferred embodiment, each of the first battery modules **61**, the second battery modules **71**, the first side battery modules **31L**, and the second side battery modules **31R** can be a removable battery module that can be individually installed and removed from the corresponding first battery housing **60**, second battery housing **70**, first side housing **30L**, and second side housing **30R**. However, the battery modules **61**, **71**, **31L**, **31R** can instead be sealed and secured within their corresponding housing (first battery housing **60**, second battery housing **70**, first side housing **30L**, or second side housing **30R**) to insulate the battery modules and/or prevent corrosion of battery terminals. As an example, terminals of the battery modules **31L**, **31R**, **61**, and **71** can be oriented towards an interior of the vehicle **10**. As described above in an example, one or two group(s) of battery modules can be discharged to provide electrical power to operate the vehicle **10** while other battery group(s)

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remain idle or are charged. Accordingly, for example, battery modules included in the idle battery group(s) can be removed or replaced while the vehicle 10 is supplied with electrical power from the non-idle battery group(s).

In a preferred embodiment, communication among the first battery modules 61, the second battery modules 71, the first side battery modules 31L, and the second side battery modules 31R can be provided by a battery management unit (BMU) or the like over a controller area network (CAN) bus. The BMU, or a plurality of BMUs, can be provided in or on one or more of the first battery housing 60, the second battery housing 70, the first side housing 30L, and the second side housing 30R, or can be provided at other location(s) in or on the vehicle 10. The vehicle 10 can also include a battery management system (BMS) that aggregates information from each of the battery module groups. The BMS can be implemented by a supervisory control unit (SCU) that can be connected to a PDU housed in the PDU housing 80. The BMS can control charging and discharging of the battery modules, and preferably maintain each of the first battery module group, the second battery module group, and the third battery module group in a state of charge between about 20% and about 80%, for example, to reduce battery module degradation. The state of charge can be maintained on a cell level within each battery module, a battery module level, a battery module bank level, or on a battery module group level.

FIGS. 25-28 show a vehicle 110 according to another preferred embodiment of the present invention. As shown in the plan views of FIGS. 26 and 27, the vehicle 110 includes a first (left) side housing 130L and a second (right) side housing 130R that are located closer to a center line CL of the vehicle 110 in a left-right direction of the vehicle 110 than the first side housing 30L and the second side housing 30R of the vehicle 10. For example, an outer edge C1 of the second side housing 130R can be located closer to the center line CL of the vehicle 110 in the left-right direction than each of an outer edge D1 of the right front wheel 41R and an outer edge E1 of the right rear wheel 42R. Similarly, an outer edge of the first side housing 130L can be located closer to the center line CL of the vehicle 110 in the left-right direction of the vehicle 110 than each of an outer edge of the left front wheel 41L and an outer edge of the left rear wheel 42L. Furthermore, a line extending from the outer edge C1 of the second side housing 130R can pass through one or both of the right front wheel 41R and the right rear wheel 42R in a width direction of the right wheels 41R and 42R. Similarly, a line extending from the outer edge of the first side housing 130L can pass through one or both of the left front wheel 41L and the left rear wheel 42L in a width direction of the right wheels 41L and 42L.

However, the outer edge C1 of the second side housing 130R can instead be located closer to the center line CL of the vehicle 110 in the left-right direction than only one of the outer edge D1 of the right front wheel 41R and the outer edge E1 of the right rear wheel 42R. Similarly, the outer edge of the first side housing 130L can be located closer to the center line CL of the vehicle 110 in the left-right direction than only one of the outer edge of the left front wheel 41L and the outer edge of the left rear wheel 42L. For example, if a size of the rear wheels 42R and 42L is increased to provide improved traction, the outer edge C1 of the second side housing 130R can be located, in the left-right direction, between the outer edge D1 of the right front wheel 41R and the outer edge E1 of the right rear wheel 42R, and the outer edge of the first side housing 130L can also be

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located, in the left-right direction, between the outer edge of the left front wheel 41L and the outer edge of the left rear wheel 42L.

As another alternative, the outer edges of the first and second side housings 130L and 130R can also be located in line with, or substantially in line with, any of the wheels 41R, 41L, 42R, and 42L. The first and second side housings 130L and 130R can also be individually adjusted with respect to the center line CL of the vehicle 110.

Other elements of the vehicle 110 are the same as, or similar to, the vehicle 10 described above, and repeated description thereof is omitted.

FIG. 45 shows a block diagram of an electrical sub-system of the electric vehicle 10. As shown in FIG. 45, the electrical sub-system includes a power distribution unit (PDU) that connects to three battery strings (e.g., the first battery module group, the second battery module group, and the third battery module group) via high-voltage (HV) DC lines. Each battery string is controlled by a battery management unit (BMU) that communicates with other BMUs and a battery management system (BMS) Master via RJ45 CANBus lines, for example. A BMS is responsible to aggregate the information from all the battery strings and respond to the appropriate BMU if the situation requires. In a preferred embodiment, the function of the BMS may be implemented in a Supervisory Control Unit (SCU). The power distribution unit (PDU) is further connected to an inverter 6 via HV DC lines to drive the electric motor 45 of the vehicle, and an onboard charge controller (OBC) 7 which may include a DC/DC converter in the present preferred embodiment as shown. The OBC 7 is connected to a 12 v battery via low-voltage (LV) DC line, and to the AC Charge port, which may be a J1772 inlet. As shown in FIG. 45, in a preferred embodiment of the present invention, two Electric Vehicle Supply Equipment (EVSE) systems are provided. One is a standard SAE J1772 system for Level 2 charging at 240 VAC via an on-board charger (OBC DC/DC). The second EVSE is a custom implementation for DC fast charging. Both EVSEs supply power to the PDU where it is distributed to charge each battery string. The DC Fast Charger connects to the Battery Strings 1-3 of the battery pack through the PDU. A Supervisory Control Unit (SCU) is connected to the PDU via CANBus. The SCU connects to vehicle inputs or sensors via low-voltage (LV) DC lines. A pump controller, which controls one or more of a radiator 532, a blower fan 533, a first coolant pump 356L, a second coolant pump 356R discussed in more detail below, is connected to the SCU via a CANBus.

In a preferred embodiment of the present invention, electrical power can be supplied to the electric motor 45 via the inverter 6. As shown in FIG. 7, for example, the electric motor 45 can be located between the first side housing 30L and the second side housing 30R in a width/side (left-right) direction of the vehicle 10, and the electric motor 45 can be located forward of the first side housing 30L and the second side housing 30R in a front-rear direction of the vehicle 10. In a preferred embodiment, the inverter 6 is located forward of the electric motor 45 in a front-rear direction of the vehicle 10. The vehicle 10 can include a transmission 46 to drive each of the left front wheel 41L, the right front wheel 41R, the left rear wheel 42L, and the right rear wheel 42R in a four-wheel drive arrangement. However, the electric motor 45 and the transmission 46 can also be implemented in two-wheel drive arrangements.

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In a preferred embodiment, the Onboard Battery Charger with DCDC converter (OBC/DCDC) 7 can be attached to a bottom surface of the second battery housing 70, as shown in FIGS. 8-11, for example.

In a preferred embodiment of the present invention, the vehicle 10 can include a liquid cooling system that is able to sufficiently cool heat generating elements of the vehicle 10 while also being efficiently arranged at different locations around the vehicle 10, as discussed in detail below.

FIG. 29 shows a schematic of an example of a liquid cooling system 1 for a vehicle according to a preferred embodiment of the present invention. The liquid cooling system 1 preferably includes a radiator 532, a blower fan 533, a first coolant pump 356L, a second coolant pump 356R, a heat exchanger 381, the inverter 6, the electric motor 45, the Onboard Battery Charger with DCDC converter (OBC/DCDC) 7, and a secondary DC to DC Converter (DCDC) 382. Preferably, the liquid cooling system 1 includes a pair of parallel cooling path circuits (e.g., a first cooling path circuit and a second cooling path circuit) which are provided between a first cooling path coupling 11 and a second cooling path coupling 12. The first cooling path coupling 11 preferably includes a Y connector, and the second cooling path coupling 12 preferably includes a T connector, for example. As shown in FIG. 29, the first cooling path circuit preferably includes the first coolant pump 356L, the inverter 6, and the electric motor 45. The second cooling path circuit preferably includes the second coolant pump 356R, the heat exchanger 381, the OBC/DCDC 7, and the secondary DC to DC Converter (DCDC) 382. The first cooling path circuit and the second cooling path circuit are preferably parallel portions of a cooling loop defined by the liquid cooling system 1. In a preferred embodiment, the first coolant pump 356L and the second coolant pump 356R are preferably Coolant Water Aggregate (CWA) type pumps and have a flow rate of 25 l/min @ 0.53 bar and 35 l/min @ 0.25 bar at an operating voltage of 8 to 16V, for example.

In a preferred embodiment, the first cooling path circuit receives cold coolant (e.g., water, a water/glycol mixture, or a water/ethylene glycol mixture) from the radiator 532, and the first coolant pump 356L is used to circulate the coolant through the inverter 6 and the motor 45, before the coolant returns to the radiator 532. The coolant absorbs heat from the inverter 6, thereby cooling the inverter 6, and then continues into a cooling channel defined within the motor 45. The coolant absorbs heat from the motor 45 while the coolant is within the cooling channel defined within the motor 45, and then continues to the radiator 532. In the radiator 532, the coolant which has been heated by passing through the inverter 6 and the motor 45, is collected and cooled.

In a preferred embodiment, the second cooling path circuit receives cold coolant from the radiator 532, and the second coolant pump 356R is used to circulate the coolant through the heat exchanger 381, the OBC/DCDC 7, and the secondary DC to DC Converter (DCDC) 382, before the coolant returns to the radiator 532. The coolant absorbs heat from the heat exchanger 381, thereby cooling the heat exchanger 381, and then continues into a cooling circuit within the OBC/DCDC 7 to absorb heat from the OBC/DCDC 7. Once the coolant has passed through the cooling circuit within the OBC/DCDC 7, the coolant continues to a cooling circuit within the secondary DC to DC Converter (DCDC) 382 to absorb heat from the secondary DC to DC Converter (DCDC) 382, and then continues to the radiator 532. In the radiator 532, the coolant which has been heated

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by passing through the heat exchanger 381, the OBC/DCDC 7, and the secondary DC to DC Converter (DCDC) 382, is collected and cooled.

In a preferred embodiment of the present invention, the heat exchanger 381 is connected to a hydraulic pump 481 such that oil (e.g., hydraulic fluid), which is used by the hydraulic pump 481 to operate hydraulic components of the vehicle (e.g., a three point lift for accessories of the vehicle, a braking system, a power steering, etc.), can be cooled. For example, the oil and the coolant are each circulated through the heat exchanger 381 such that heat from the oil can be transferred to the coolant, thereby cooling the heat exchanger 381 and the oil that is circulated through the heat exchanger 381. The heat exchanger is preferably rated at 125,000 BTU/Hr, but any other desirable type of heat exchanger could be used. In a preferred embodiment, the hydraulic pump 481 is attached to a rear frame of the vehicle adjacent to the rear right wheel 42R, as shown in FIG. 7, for example. In a preferred embodiment, the hydraulic pump 481 is located rearward of the heat exchanger 381 in a front-rear direction of the vehicle 10.

As shown, in FIG. 30 the first coolant pump 356L and the second coolant pump 356R are preferably mounted on a lower front surface of the second side housing 30R. The radiator 532 is preferably housed in a radiator and condenser module 53 affixed to a roof 51 of a cabin 50 of the vehicle 10, as shown in FIG. 1, for example. In a preferred embodiment, a fluid connection between the radiator 532 and the first coolant pump 356L and a fluid connection (e.g., tubing or piping) between the radiator 532 and the second coolant pump 356R extend through internal portions of the cabin frame or along an outside of the cabin frame.

In a preferred embodiment, the heat exchanger 381 is preferably fixed to a portion of the vehicle adjacent to the second side housing 30R. For example, in a preferred embodiment of the present invention, the heat exchanger 381 is attached to a right side of the second battery housing 70, as shown in FIGS. 34 and 41, for example. The location of the heat exchanger 381 makes it possible for a connection between the second coolant pump 356R and the heat exchanger 381 to be as short as possible while also permitting an easy connection to the hydraulic system, the hydraulic system preferably being located at a right side of the vehicle.

As discussed above, the radiator 532 is attached to the roof 51 of the cabin 50 of the vehicle 10. Thus, a connection (e.g., tubing or piping) between the radiator 532 and the first coolant pump 356L and the second coolant pump 356R extends up to the roof 51. The connection between the radiator 532 and the first coolant pump 356L and the second coolant pump 356R can include tubing or piping that extends along an outside or inside of the rear cabin frame 52 or the forward cabin frame 55. Providing tubing alongside or inside of the rear cabin frame 52 or the forward cabin frame 55 provides an efficient way of routing the connection between the first and second coolant pumps 356L and 356R and the radiator 532. Further, by providing the radiator 532 on the roof 51, it is possible to space the radiator 532 far away from other heat generating components of the vehicle to thereby increase its thermal cooling capacity. Additionally, by providing the radiator 532 on the roof 51, it is possible to reduce the front-rear length of the vehicle 10 compared to if the radiator 532 was attached to a front of the vehicle 10.

FIG. 41 shows a partially assembled bottom view of a vehicle including components of the liquid cooling system according to a preferred embodiment of the present inven-

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tion. The vehicle preferably includes a chassis **5** which supports a lower surface of the second battery housing **70**. The chassis **5** preferably further includes a pair of frame rails **511L** and **511R** which support the inverter **6** and the OBC/DCDC **7** below the second battery housing **70**. In a preferred embodiment, the inverter **6** and the OBC/DCDC **7** are also mounted under the first battery housing **60** in an up-down direction of the vehicle. The OBC/DCDC **7** is preferably arranged forward of the inverter **6** in a front-rear direction of the vehicle. The chassis **5** also preferably includes a pair of saddle support beams **522** which are affixed to bottom portions of the first side housing **30L** and the second side housing **30R** to firmly support the first side housing **30L** and the second side housing **30R** with respect to the chassis **5** either indirectly or through a direct connection.

FIG. **42** shows a partially assembled view of the roof **51** with a cover of the radiator and condenser module **53** removed for illustrative purposes. FIGS. **43** and **44** exploded views of a vehicle and cooling components according to preferred embodiments of the present invention. In a preferred embodiment shown in FIGS. **42** and **43**, the blower fan **533** is located between the radiator **532** and a condenser **531** (discussed in more detail below) and to provide cooling airflow to both the radiator **532** and the condenser **531** at the same time. In an alternative preferred embodiment shown in FIG. **44**, the vehicle **10** includes a first blower fan **533a** located between the condenser **531** and the radiator **532** and a second blower fan **533b** located in front of the condenser **531**.

In a preferred embodiment of the present invention, the vehicle **10** preferably further includes an air cooling system that is structured to cool the first battery modules **61**, the second battery modules **71**, the first side battery modules **31L**, and the second side battery modules **31R**. FIG. **46** is a schematic that shows a flow of refrigerant **R** through portions of the air cooling system. For example, FIG. **46** shows a flow of refrigerant through an AC compressor **362**, a condenser **531**, a drier **357**, a first expansion valve **358L**, a second expansion valve **358R**, a first evaporator coil **3551L**, and a second evaporator coil **3551R**. In a preferred embodiment, the AC compressor **362** can include an integrated motor and compressor which is powered by one or more of the first battery modules **61**, the second battery modules **71**, the first side battery modules **31L**, and the second side battery modules **31R**. The AC compressor **362** is operable to increase the pressure of the refrigerant **R** and sends hot, high pressure refrigerant **R** (gas) to the condenser **531**. The condenser **521** reduces the temperature of the refrigerant **R** but maintains the refrigerant **R** at high pressure. The reduced-temperature, high pressure refrigerant **R** (liquid) is sent from the condenser **521** to the drier **357**, in which water/moisture is removed from the refrigerant **R**, before the refrigerant **R** (liquid) is sent to the first expansion valve **358L** and the second expansion valve **358R**. The first expansion valve **358L** and the second expansion valve **358R** reduce the pressure of the refrigerant **R**, which reduces the temperature of the refrigerant **R**. The cold refrigerant **R** (liquid) from the first expansion valve **358L** is sent to the first evaporator coil **3551L** to cool the first evaporator coil **3551L**, and the cold refrigerant **R** (liquid) from the second expansion valve **358R** is sent to the second evaporator coil **3551R** to cool the second evaporator coil **3551R**. After the refrigerant **R** passes through the first evaporator coil **3551L** and the second evaporator coil **3551R**, the low temperature, low pressure refrigerant **R** (gas) is sent back to the compressor **362**.

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In a preferred embodiment, the condenser **531** is mounted on the roof **51** and a fluid connection between the compressor **362** and the condenser **531** extends along or through the rear cabin frame **52** or the forward cabin frame **55**. As discussed above and as shown in FIG. **12**, for example, the condenser **531** is preferably located adjacent to the radiator **532** on the roof **51** with the blower fan **533** located between the condenser **531** and the radiator **532** such that both of the condenser **531** and the radiator **532** are located far away from other heat generating components of the vehicle to thereby increase thermal cooling capacity.

In a preferred embodiment, the air cooling system includes, attached to the first side housing **30L**, a first side return duct **35L**, a first side outer duct **352L**, a first lower return duct **354L**, and a first evaporator air box **355L**, as shown in FIGS. **30-36**, for example. In a preferred embodiment, the drier **357** is also attached to the first side housing **30L**, as shown in FIG. **1**, for example. The first evaporator air box **355L** is connected to the first side outer duct **352L**, a first side chimney duct **351L**, and a first lower duct **353L**. The air cooling system also preferably includes, attached to the second side housing **30R**, a second side return duct **35R**, a second side outer duct **352R**, a second lower return duct **354R**, the compressor **362**, a battery **363**, and a second evaporator air box **355R**. The second evaporator air box **355R** is connected to the second side outer duct **352R**, a second side chimney duct **351R**, and a second lower duct **353R**. In a preferred embodiment, the first expansion valve **358L** is housed in the first evaporator air box **355L** and the second expansion valve **358R** is housed in the second evaporator air box **355R**.

In a preferred embodiment, the first side chimney duct **351L** is connected between the first evaporator air box **355L** and a first chimney duct blower **361L**, and the second side chimney duct **351R** is connected between the second evaporator air box **355R** and a second chimney duct blower **361R**. The first lower duct **353L** is connected between the first evaporator air box **355L** and a lower blower **360**, and the second lower duct **353R** is connected between the second evaporator air box **355R** and the lower blower **360**. The first side outer duct **352L** is connected between the first evaporator air box **355L** and the first side housing **30L**, and the second side outer duct **352R** is connected between the second evaporator air box **355R** and the second side housing **30R**. As shown in FIGS. **32** and **34**, for example, the first side return duct **35L** is connected between the first side housing **30L** and a first return housing **365L** attached to the left side cover **60L** of the first battery housing **60**, and the second side return duct **35R** is connected between the second side housing **30R** and a second return housing **365R** attached to the right side cover **60R** of the first battery housing **60**. The first lower return duct **354L** is connected between the first side housing **30L** and a return duct opening **370** attached to a right rear of the second battery housing **70**, and the second lower return duct **354R** is connected between the second side housing **30R** and the return duct opening **370**.

Next, a flow of air through the air cooling system according to a preferred embodiment of the present disclosure will be described. The first evaporator air box **355L** and the second evaporator air box **355R** each cool air circulating through the air cooling system. For example, the first evaporator air box **355L** includes first evaporator coil **3551L** that cool air circulating through the air cooling system, and the second evaporator air box **355R** includes second evaporator coil **3551R** that cool air circulating through the air cooling system. Each of the first evaporator air box **355L** and

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the second evaporator air box 355R includes a first side (e.g., a bottom side) at which warmed air enters the evaporator air box and a second side (e.g., a top side) at which cool air exits the evaporator air box after passing through the evaporator coil included in the respective evaporator air box.

In a preferred embodiment, the cool air that exits the second side of the first evaporator air box 355L flows to the first chimney duct 351L, the first side outer duct 352L, and the first lower duct 353L. As discussed above, the first chimney duct 351L is connected to a first chimney duct blower 361L that is provided at an upper end of the first chimney duct 351L. The first chimney duct blower 361L preferably includes a blower motor and fan which directs cooled air from the first evaporator air box 355L through the first chimney duct 351L and into a first portion 21L (a left side portion) of an upper cooling duct 21, discussed in more detail below. In a preferred embodiment, the first side outer duct 352L is connected to a first side blower 34L that directs cool air from the first evaporator air box 355L into the first side housing 30L to cool the first side battery modules 31L, as shown in FIG. 32, for example. The first lower duct 353L is connected to a lower blower 360 that directs cool air from the first evaporator air box 355L into the second battery housing 70, also discussed in more detail below.

Warmed air returns to the first evaporator air box 355L (i.e., to the first side of the first evaporator air box 355L) through an inside of the first side housing 30L, which receives cool air through the first side outer duct 352L, as discussed above. The inside of the first side housing 30L also receives air from the first side return ducting 35L and the first lower return duct 354L, which can also cool the first side battery modules 31L.

In a preferred embodiment, the cool air that exits the second side of the second evaporator air box 355R flows to the second chimney duct 351R, the second side outer duct 352R, and the second lower duct 353R. As discussed above, the second chimney duct 351R is connected to a second chimney duct blower 361R that is provided at an upper end of the second chimney duct 351R. The second chimney duct blower 361R preferably includes a blower motor and fan that directs cooled air from the second evaporator air box 355R through the second chimney duct 351R into a second portion 21R (a right side portion) of the upper cooling duct 21, discussed in more detail below. The second side outer duct 352R is connected to a second side blower 34R that directs cool air from the second evaporator air box 355R into the second side housing 30R to cool the second side battery modules 31R, as shown in FIG. 34, for example. The second lower duct 353R is connected to the lower blower 360 which directs cool air from the second evaporator air box 355R into the second battery housing 70, also discussed in more detail below.

Warmed air returns to the second evaporator air box 355R (i.e., to the first side of the second evaporator air box 355R) through an inside of the second side housing 30R, which receives cool air through the second side outer duct 352R, as discussed above. The inside of the second side housing 30R also receives air from a second side return duct 35R and a second lower return duct 354R, which can also cool the second side battery modules 31R.

As discussed above, the first chimney duct blower 361L directs cooled air from the first evaporator air box 355L through the first chimney duct 351L and into the first portion/duct 21L (a left side portion) of the upper cooling duct 21, and the second chimney duct blower 361R directs cooled air from the second evaporator air box 355R through the second chimney duct 351R into the second portion/duct

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21R (a right side portion) of the upper cooling duct 21. The cooled air from the first portion 21L and the second portion 21R flows into a center portion 21C of the upper cooling duct 21, from which the cooled air flows into a central portion of the first battery housing 60 along the vertical partition wall 69 in a front-rear direction and an up-down direction. For example, the solid arrows in FIGS. 20 and 21 show a flow of cool air CA along the vertical partition wall 69 in the front-rear direction and the up-down direction.

In a preferred embodiment of the present invention, the cool air CA that flows along the vertical partition wall 69 then flows in lateral directions (width directions of the vehicle) across the first battery modules 61 included in the first battery module bank 62 housed in the first portion 60A of the first battery housing 60 and across the first battery modules 61 included in the second battery module bank 63 housed in the second portion 60B of the first battery housing 60. For example, FIG. 37 shows a plan view of an example of the air flow through the first battery housing 60 in which the solid arrows show a flow of cool air CA and the dashed arrows show a flow of warm air WA. FIG. 38A shows a left side view of an example of the air flow through the first battery housing 60, and FIG. 38B shows a right side view of an example of the air flow through the first battery housing 60.

As shown in FIG. 37, the cool air CA flows from the center portion 21C of the upper cooling duct 21/the vertical partition wall 69 in a first lateral direction (a leftward direction) across the first battery modules 61 included in the first battery module bank 62 housed in the first portion 60A of the first battery housing 60 before reaching the first side cover 60L as warm air (warm air that has been used to cool the first battery modules 61 included in the first battery module bank 62). In a preferred embodiment, the cool air CA that flows in the first lateral direction flows across top and/or bottom surfaces of the first battery modules 61 housed in the first portion 60A of the first battery housing 60. The warm air that reaches the first side cover 60L then flows rearwardly along the first side cover 60L before flowing into the first return housing 365L. When the warm air flows rearwardly along the first side cover 60L, the warm air flows across terminal ends of the first battery modules 61 included in the first battery module bank 62.

In a preferred embodiment, the first return housing 365L is connected to an upper end of the first side return duct 35L and guides the warm air flowing along the first side cover 60L into the first side return duct 35L. The warm air that flows into the first side return duct 35L then flows into the first side housing 30L to which the first side return duct 35L is connected.

As shown in FIG. 37, the cool air CA also flows from the center portion 21C of the upper cooling duct 21/the vertical partition wall 69 in a second lateral direction (a right direction) across the first battery modules 61 included in the second battery module bank 63 housed in the second portion 60B of the first battery housing 60 before reaching the second side cover 60R as warm air (warm air that has been used to cool the first battery modules 61 included in the second battery module bank 63). In a preferred embodiment, the cool air CA that flows in the second lateral direction flows across top and/or bottom surfaces of the first battery modules 61 housed in the second portion 60B of the first battery housing 60. The warm air that reaches the second side cover 60R then travels rearwardly along the second side cover 60R before flowing into the second return housing 365R. When the warm air flows rearwardly along the second

side cover **60R**, the warm air flows across terminal ends of the first battery modules **61** included in the second battery module bank **63**.

In a preferred embodiment, the second return housing **365R** is connected to an upper end of the second side return duct **35R** and guides the warm air flowing along the second side cover **60R** into the second side return duct **35R**. The warm air that flows into the second side return duct **35R** then flows into the second side housing **30R** to which the second side return duct **35R** is connected.

In a preferred embodiment of the present invention, the cool air that flows along the vertical partition wall **69** in a front-rear direction can also flow through the inner ventilation holes **681** and into the PDU housing **80**, as shown in FIGS. **15** and **37**, for example. The cool air that flows into the PDU housing **80** can cool the components housed in the PDU housing **80** before exiting the PDU housing **80** as warm air through the intermediate ventilation holes **682**. The warm air that leaves the PDU housing **80** through the intermediate ventilation holes **682** can flow rearwardly along the first side cover **60L** toward the first return housing **365L** and rearwardly along the second side cover **60R** toward the second return housing **365R**.

As discussed above, the lower blower **360** directs cool air from the first evaporator air box **355L** through the first lower ducts **353L** and into the second battery housing **70**, and the lower blower **360** directs cool air from the second evaporator air box **355R** through the second lower duct **353R** and into the second battery housing **70**. In a preferred embodiment, the cool air from the first evaporator air box **355L** and the cool air from the second evaporator air box **355R** flows into the second battery housing **70** at a first side (e.g., a left side) of the second battery housing **70** in a left-right direction of the vehicle. In a preferred embodiment, the lower blower **360** is attached to a rear surface of the second battery housing **70** and on the first side of the second battery housing **70** in the left-right direction of the vehicle.

In a preferred embodiment of the present invention, the cool air CA directed by the lower blower **360** flows along the first side cover **70L** in a front-rear direction and also flows in a lateral direction (a width directions of the vehicle) across the second battery modules **71** included in the third battery module bank **72** housed in the second battery housing **70**. For example, FIG. **39** shows a plan view of an example of the air flow through the second battery housing **70** in which the solid arrows show a flow of cool air CA and the dashed arrows show a flow of warm air WA. FIG. **40A** shows a left side view of an example of the air flow through the second battery housing **70**, and FIG. **40B** shows a right side view of an example of the air flow through the second battery housing **70**.

As shown in FIG. **39**, the cool air CA flows from the first side cover **70L** in a first lateral direction (a rightward direction) across the second battery modules **71** included in the third battery module bank **72** housed in the second battery housing **70** before reaching the second side cover **70R** as warm air (warm air that has been used to cool the second battery modules **71** included in the third battery module bank **72**). In a preferred embodiment, the cool air CA that flows in the first lateral direction flows across top and/or bottom surfaces of the second battery modules **71** housed in the second battery housing **70**. The warm air that reaches the second side cover **70R** then travels rearwardly along the second side cover **70R** before flowing into the return duct opening **370**. In a preferred embodiment, when the warm air flows rearwardly along the second side cover

70R, the warm air flows across terminal ends of the second battery modules **71** included in the third battery module bank **72**.

In a preferred embodiment, the return duct opening **370** is connected to a forward end of the first lower return duct **354L** and a forward end of the second lower return duct **354R** and guides the warm air flowing along the second side cover **70R** into the first lower return duct **354L** and the second lower return duct **354R**. In a preferred embodiment, a first portion of the warm air that flows into the return duct opening **370** flows through the first lower return duct **354L** into the first side housing **30L**, and a second portion of the warm air that flows into the return duct opening **370** flows through the second lower return duct **354R** into the second side housing **30R**.

In a preferred embodiment of the present invention, a volume of the first battery housing **60** is larger than a volume of the second battery housing **70**. Therefore, in a preferred embodiment, a combined volume of the cool air from the second side of the first evaporator coil **3551L** directed by the first chimney duct blower **361L** (e.g., a first portion of the cool air from the second side of the first evaporator coil **3551L**) and the cool air from the second side of the second evaporator coil **3551R** directed by the second chimney duct blower **361R** (e.g., a first portion of the cool air from the second side of the second evaporator coil **3551R**) is greater than a combined volume of the cool air from the second side of the first evaporator coil **3551L** (e.g., a second portion of the cool air from the second side of the first evaporator coil **3551L**) and the cool air from the second side of the second evaporator coil **3551R** (e.g., a second portion of the cool air from the second side of the second evaporator coil **3551R**) directed by the lower blower **360**. In this way, the first battery housing **60**, which has a volume larger than a volume of the second battery housing **70**, receives a larger volume of cool air from the second side of the first evaporator coil **3551L** and the second side of the second evaporator coil **3551R** than does the second battery housing **70**.

In a preferred embodiment of the present invention, a volume of the first side housing **30L** is less than the volume of the second battery housing **70**, and a volume of the second side housing **30R** is less than the volume of the second battery housing **70**. Therefore, in a preferred embodiment, a volume of the cool air from the second side of the first evaporator coil **3551L** directed by the first side blower **34L** (e.g., a third portion of the cool air from the second side of the first evaporator coil **3551L**) is less than the combined volume of the cool air from the second side of the first evaporator coil **3551L** and the cool air from the second side of the second evaporator coil **3551R** directed by the lower blower **360**, and a volume of the cool air from the second side of the second evaporator coil **3551R** directed by the second side blower **34R** (e.g., a third portion of the cool air from the second side of the second evaporator coil **3551R**) is less than the combined volume of the cool air from the second side of the first evaporator coil **3551L** and the cool air from the second side of the second evaporator coil **3551R** directed by the lower blower **360**. In this way, the second battery housing **70**, which has a volume larger than each of the first side housing **30L** and the second side housing **30R**, receives a larger volume of cool air from the second side of the first evaporator coil **3551L** and the second side of the second evaporator coil **3551R** than does the first side housing **30L** and the second side housing **30R**.

In a preferred embodiment of the present invention, one or more temperature sensors are provided to measure a temperature of each of the first battery modules **61**, the

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second battery modules **71**, the first side battery modules **31L**, and the second side battery modules **31R**. For example, in a preferred embodiment, each of the battery modules included in the first battery modules **61**, the second battery modules **71**, the first side battery modules **31L**, and the second side battery modules **31R** can include a temperature sensor to measure the temperature of the respective battery module.

In a preferred embodiment, the first chimney duct blower **361L**, the second chimney duct blower **361R**, the lower blower **360**, the first side blower **34L**, and the second side blower **34R** can be operated in accordance with the temperatures detected by the one or more temperature sensors. For example, when one or more of the temperatures sensors measure a temperature above a predetermined temperature threshold, the first chimney duct blower **361L**, the second chimney duct blower **361R**, the lower blower **360**, the first side blower **34L**, and the second side blower **34R** can be turned ON (e.g., controlled to direct cool air). For example, in a preferred embodiment of the present invention, the first chimney duct blower **361L**, the second chimney duct blower **361R**, the lower blower **360**, the first side blower **34L**, and the second side blower **34R** can be turned ON when one or more of (at least one of) the temperature sensors measures a temperature above a first predetermined temperature threshold (e.g., a turn ON target temperature) and then turn back OFF when each of the one or more of the temperatures sensors measures a temperature below a second predetermined temperature threshold (e.g., a turn OFF target temperature), for example. In a preferred embodiment, the first predetermined temperature threshold (e.g., the turn ON target temperature) is greater than the second predetermined temperature threshold (e.g., the turn OFF target temperature).

In another preferred embodiment, it is possible for the cooling system to control certain ones of the first chimney duct blower **361L**, the second chimney duct blower **361R**, the lower blower **360**, the first side blower **34L**, and the second side blower **34R** based on the temperatures detected by the one or more temperature sensors. For example, in a preferred embodiment, it is possible to turn ON only the first chimney duct blower **361L** and the second chimney duct blower **361R** based on the temperatures detected by the one or more temperature sensors included in the first battery modules **61**. Similarly, in a preferred embodiment, it is possible to turn ON only the lower blower **360** based on the temperatures detected by the one or more temperature sensors included in the second battery modules **71**. Similarly, in a preferred embodiment, it is possible to turn ON only the first side blower **34L** and the second side blower **34R** based on the temperatures detected by the one or more temperature sensors included in the first side battery modules **31L** and the second side battery modules **31R**.

In a preferred embodiment of the present invention, the secondary DC-to-DC converter provides power to one or more accessories of the vehicle including the first chimney duct blower **361L**, the second chimney duct blower **361R**, the lower blower **360**, the first side blower **34L**, the second side blower **34R**, the first coolant pump **356L**, and the second coolant pump **356R**, for example. The secondary DCDC **382** and the OBC/DCDC **7** preferably output power to a common 12V bus. The common 12V bus is preferably buffered by the battery **363** which is supported on an upper surface of the second side housing **30R**.

It should be understood that the foregoing description is only illustrative of the present invention. Various alternatives and modifications can be devised by those skilled in the

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art without departing from the present invention. Accordingly, the present invention is intended to embrace all such alternatives, modifications, and variances that fall within the scope of the appended claims.

What is claimed is:

1. A vehicle comprising:

a first battery housing to house a plurality of first battery modules;

a second battery housing to house a plurality of second battery modules;

an air cooling system to cool the plurality of first battery modules and the plurality of second battery modules, the air cooling system including:

a compressor;

a condenser;

a first evaporator coil that includes a first side at which warm air is received and a second side at which cool air exits after having passed through the first evaporator coil;

a second evaporator coil that includes a first side at which warm air is received and a second side at which cool air exits after having passed through the second evaporator coil;

a first blower to direct a first portion of the cool air from the second side of the first evaporator coil into the first battery housing;

a second blower to direct a first portion of the cool air from the second side of the second evaporator coil into the first battery housing; and

a third blower to direct a second portion of the cool air from the second side of the first evaporator coil and a second portion of the cool air from the second side of the second evaporator coil into the second battery housing.

2. The vehicle according to claim 1, further comprising:

a first side housing located on a lateral side of the first battery housing, the first side housing to house a plurality of first side battery modules; wherein

the first evaporator coil is supported by the first side housing.

3. The vehicle according to claim 2, further comprising:

a first side blower to direct a third portion of the cool air from the second side of the first evaporator coil into the first side housing.

4. The vehicle according to claim 1, further comprising:

a first side housing and a second side housing located on opposing lateral sides of the first battery housing, the first side housing to house a plurality of first side battery modules, and the second side housing to house a plurality of second side battery modules.

5. The vehicle according to claim 4, wherein the compressor is mounted on the second side housing.

6. The vehicle according to claim 5, wherein the air cooling system further includes a drier; and the drier is attached to the first side housing.

7. The vehicle according to claim 5, further comprising: a battery; wherein

the battery is mounted on the second side housing adjacent to the compressor.

8. The vehicle according to claim 4, further comprising:

a first side blower that directs a third portion of the cool air from the second side of the first evaporator coil into the first side housing; and

a second side blower that directs a third portion of the cool air from the second side of the second evaporator coil into the second side housing; wherein

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the first side blower is connected to a first side outer duct through which the third portion of the cool air flows from the second side of the first evaporator coil into the first side housing; and

the second side blower is connected to a second side outer duct through which the third portion of the cool air flows from the second side of the second evaporator coil into the second side housing.

9. The vehicle according to claim 4, wherein the first evaporator coil is supported by the first side housing; and the second evaporator coil is supported by the second side housing.

10. The vehicle according to claim 9, further comprising: a first evaporator air box that houses the first evaporator coil and is attached to the first side housing; a second evaporator air box that houses the second evaporator coil and is attached to the second side housing; wherein the air cooling system further includes a first expansion valve and a second expansion valve; the first expansion valve is housed in the first evaporator air box; and the second expansion valve is housed in the second evaporator air box.

11. The vehicle according to claim 1, wherein the first blower is connected to a first duct through which the first portion of the cool air from the second side of the first evaporator coil flows into the first battery housing; the second blower is connected to a second duct through which the first portion of the cool air from the second side of the second evaporator coil flows into the first battery housing; a bottom end of the first duct is connected to a first evaporator air box that houses the first evaporator coil; the first blower is connected to a top end of the first duct; a bottom end of the second duct is connected to a second evaporator air box that houses the second evaporator coil; and the second blower is connected to a top end of the second duct.

12. The vehicle according to claim 1, wherein the first battery housing includes a vertical partition wall that divides the first battery housing to include a first portion of the first battery housing and a second portion of the first battery housing; the first portion of the first battery housing houses a first battery module bank including a first portion of the plurality of first battery modules; the second portion of the first battery housing houses a second battery module bank including a second portion of the plurality of first battery modules; and the first portion of the cool air directed by the first blower and the first portion of the cool air directed by the second blower flow into the first battery housing at a central portion of the first battery housing.

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13. The vehicle according to claim 12, the cool air that flows into the first battery housing at the central portion of the first battery housing flows from the central portion of the first battery housing in a first lateral direction across the first battery module bank and in a second lateral direction across the second battery module bank.

14. The vehicle according to claim 1, wherein the second portion of the cool air from the second side of the first evaporator coil and the second portion of the cool air from the second side of the second evaporator coil, which are directed by the third blower, flow into the second battery housing at a first side of the second battery housing in a left-right direction of the vehicle; and the third blower is attached to a rear surface of the second battery housing on the first side of the second battery housing.

15. The vehicle according to claim 14, wherein the third blower is connected to a first duct through which the second portion of the cool air from the second side of the first evaporator coil flows from the second side of the first evaporator coil into the second battery housing; and the third blower is connected to a second duct through which the second portion of the cool air from the second side of the second evaporator coil flows from the second side of the second evaporator coil into the second battery housing.

16. The vehicle according to claim 1, a width of the first battery housing is greater than a width of the second battery housing in a width direction of the vehicle.

17. The vehicle according to claim 1, the first battery housing and the second battery housing are offset in a front-rear direction of the vehicle; the offset defines a space in which an additional housing to house one or more electronic components is located; and at least one of the first portion of the cool air from the second side of the first evaporator coil directed by the first blower into the first battery housing and the first portion of the cool air from the second side of the second evaporator coil directed by the second blower into the first battery housing flows through the first battery housing and into the additional housing.

18. The vehicle according to claim 1, the second battery housing is attached to a bottom surface of the first battery housing.

19. The vehicle according to claim 1, wherein the condenser is mounted on a roof of the vehicle.

20. The vehicle according to claim 19, further comprising: a cabin frame that is connected to a chassis of the vehicle and supports the roof of the vehicle; wherein a fluid connection between the compressor and the condenser extends along or through the cabin frame.

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